Hyper-**N**etwork for **e**lectro**Mo**bility

NeMo D1.1. Consolidated Version of the Use Cases and Actors' Requirements

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Glossary of terms

Term	Description
Electromobility or eMobility	The use of electric-powered drivetrains for road vehicles designed to shift vehicle design away from the use of fossil fuels and carbon gas emissions.
e-roaming or eRoaming	A market model in electro mobility whereby EV drivers may charge their vehicles at all charging stations, regardless of any contracts concluded with operators. The billing then takes place via the customer's own contractual partner.
Hyper-Network	A distributed environment with open architecture based on standardised interfaces, in which actors (physical or digital) can connect and interact seamlessly, exchange data and provide integrated and interoperable ICT services.
Actor	All physical or digital entities of the electromobility domain, which may interact with the Hyper-Network. They can be drivers and owners of EVs, vehicle manufacturers, Charge point owners or operators, energy suppliers, DSOs, service providers, public authorities, policy makers, municipalities, regulatory bodies, customers' associations.
Business Component	A Business Component is part of a system that has the potential to operate autonomously. It does not necessarily identify physical systems.
Business Object	A Business Object describes a basic data structure that is used within a Business Scenario.
Business Rule	A Business Rule is an expression of a business policy. As such, a business rule may: • Provide a set of conditions that govern business behaviour. • Provide the criteria for when an action is successfully or unsuccessfully completed. • A business rule is a policy, constraint, or required operation that applies to a specific set of business conditions or dependencies.
Business Scenario	A Business Scenario describes on a business level the scenarios under which actors will perform specific functions and will interact with each other.
Inter-Roaming	Inter-Roaming refers to a market model in electromobility where eRoaming-Platforms connect to each other to enable their charge point operators and electromobility service providers to offer or access infrastructure of their opposite platform no matter where they are originally connected to.

Term	Description
Use case	A Use Case outlines a Business Scenario in more detail. • Use cases provide a structure to express functional requirements within the context of business and system processes. • The summary of all Use Cases are the Use Case Model, which describes the functional requirements of the system under development. • Scenarios of a Use Case could be illustrated by a workflow diagram.

List of abbreviations and acronyms

Abbreviation	Meaning
API	Application Programming Interface
B2B	Business to Business
B2B2C	Business to Business to Customer
B2C	Business to Customer
BMS	Battery Management System
BS	Business Scenario
CDR	Charge Detail Record
CP	Charge Point
CPM	Charge Point Manufacturer
CPO	Charge Point Operator
DMD	Data Management Domain
DSO	Distribution System Operator
eMI3	eMobility ICT Interoperability Innovation group, open group of significant actors from the global Electric Vehicles market to harmonize the standards (ICT data definitions, formats, interfaces etc.) for Electric Vehicles.
EMIP	Electro-Mobility Interoperation Protocol
EMP	Electromobility service Provider
EoC	End of Charge
EUCAR	European Council for Automotive Research & Development
EURELECTRIC	Union of the Electricity Industry
EV	Electric Vehicle

Abbreviation	Meaning
EVSE	Electric Vehicle Supply Equipment
HyER	European Association for Hydrogen and fuel cells and Electromobility in European Regions
ICE	Internal Combustion Engine
ICT	Information and Communication technologies
IoT	Internet of Things
ITS	Intelligent Transport Systems
KPI	Key Performance Indicator
NSP	Routing and Navigation Service Provider
OA	Open Access
OCHP	Open Clearing House Protocol
OCPI	Open Charge Point Interface
OCPP	Open Charge Point Protocol
OEM	Original Equipment Manufacturer, refers to the Vehicle Manufacturer
OICP	Open InterCharge Protocol (InterCharge protocol by Hubject)
OSCP	Open Smart Charging Protocol
OWL-S	Semantic Markup for Web Services
PaaS	Platform as a Service
PEST	Political, Economic, Social and Technological
POD	Point Of Delivery
RES	Renewable Energy Sources
SOA	Service-oriented architecture

Abbreviation	Meaning
SoC	State of Charge
SOH	State-Of-Health
SPV	Special Purpose Vehicle
TCO	Total Cost of Ownership
TSO	Transmission System Operator
UITP	International Association of Public Transport
UML	Unified Modelling Language
UC	Use Case
V2G	Vehicle to Grid
VM	Vehicle Manufacturer

Executive Summary

The Deliverable describes the Actors involved in the electromobility sector, grouped into macro-areas. The Actors have been selected based on their relation with the NeMo Hyper-Network. The interrelations between macro-areas and actors are displayed via UML (Unified Modelling Language) diagrams which describe the behaviour and the structure of a system. Moreover, the Deliverable presents a selection of Business Scenarios that will be tested in five European test sites located in: France, Austria, Germany, Italy and Spain. The identification of the Business Scenarios is the result of the discussion carried out among project participants since the very beginning of NeMo project. The Business Scenarios consist of a set of technologies and services related to the electromobility sector. In order to avoid that the same NeMo tool or technology is tested in different scenarios, a list of standard services was initially identified. Based on the list of standard services, seven main Business Scenarios have been selected. Business Scenarios consist of a high level description of the services that will be tested in the test sites. A more detailed description of the functionalities proposed within each Business Scenario is provided by the Use Cases, which additionally include the description of the requirements. A table matches the requirements from the complete list that are relevant for each use Case. Finally, the Deliverable provides a tentative description and overview of the test sites, including the cross-country pilot.

The Deliverable is considered as the basis for future activities, in particular it represents a first attempt to decide what will be tested in each test site and how the NeMo tools and technologies will look. Based on the requirements, WP2 will extract specifications, will describe the solutions to the actors 'needs and requirements and will define the Hyper-Network reference architecture.

More specifically, Task 2.2 will define the NeMo Hyper-Network sub-system based on the actors' definition and Business Scenarios. Task 2.3, based on the requirements identified in T1.4, will describe the NeMo Hyper-Network basic functionalities. Finally, Task 2.4 will define the standards and the protocols based on the actors identified in T1.1.

The activities to be carried out in the context of WP4, are based on the Business Scenarios and Use Cases described in D1.1 and identified in T1.3. Specifically, T4.1 is based on the functional and non-functional requirements described in T1.4 in order to define a low level design for the NeMo Hyper-Network. WP6 will validate the requirements defined in T1.4. Finally, WP7 will define business models and roll out process of Business Scenarios described in T1.3.

1. Introduction

This Deliverable reports the work carried out in the first six months of the project duration. These activities relate to three Tasks: Task 1.1 which was dedicated to the description of the Actors needs and of their interrelations, Task 1.3 which focused on the description of the Business scenarios and Use Cases and T1.4 which identified the requirements in relation to the identified Use Cases.

The Deliverable is divided in four main sections based on the results of the three Tasks (T1.1, T1.3, and T1.4). Chapter 2 consists of a detailed description of the Actors involved in the electromobility sector and their interrelations. A new approach to graphically present the interrelations is employed and a definition of each actor and the decision about the acronyms to be used are presented.

Chapter 3 is dedicated to the high level description of the Business Scenarios which include the new NeMo services, including the existing ones, which will be tested in the project test sites. A more detailed description of the specific Use Cases that form each Business Scenario is reported in Chapter 4. For each Use Case, the objectives, the technical characteristics of the functionality and the description of the requirements are provided. Chapter 5 is a first tentative selection of `the cross-country pilot.

Finally, Chapter 6 consists of the full list of requirements, which are mapped with each Use Case.

2. Actors' description and interrelations

2.1 Description of the methodology

The objective of this work was to identify, classify and describe all the actors and stakeholders that may interact with the NeMo Hyper-Network and may use its services. To this aim a methodology was adopted based on the following steps:

1. To define a list of actors involved in the electro-mobility domain (based on market analysis, related literature and previous projects review);
2. To group the identified actors in so-called macro-areas, based on common objectives and/or features;
3. To describe the role and main activities of each actor;
4. To define a common dictionary for the key concepts and definitions used in the actors' description;
5. To collect actors' needs in the NeMo Hyper-Network;
6. To define the relationships between actors by means of a relationship matrix, i.e., a matrix showing if a relation between each actor pair exists: this relation is defined starting from the needs collected in the previous step, i.e., the relation is established to satisfy an actor's need;
7. To develop UML diagrams that specify the actors and macro-areas interrelations, starting from the relationship matrix.

2.2 Actors' description

The identified actors were grouped in 9 macro-areas per common features and/or needs, as follows: Industry, IT Suppliers, Charging Infrastructure, Decision Makers, Public Authorities, Energy, European and International Organizations and Networks, End Users and Complementary Services.

Firstly, a short description is provided for each macro-area; then a description is provided for each actor belonging to that macro-area, mainly referring to the existing standards, e.g. ISO 15118" Vehicle-to-Grid Communication Interface" specification and previous projects' findings.

2.2.1 Industry

The Industry macro-area groups all the actors mainly related to the production of goods. The list of actors belonging to the Industry macro-area is:

- **Vehicle Manufacturer (VM)**: this actor develops, produces and sells EVs to the market. According to ISO 15118, it can be also defined as a company which builds Electric Vehicles that are different to all other products made by other companies;
- **Charge Point Manufacturer (CPM)**: it develops, produces and sells poles / charge points for charge, i.e., a company which builds Charge Points that are different to all other products made by other companies;

- **Electromobility Service Provider (EMP)**: each customer, to access the Hyper-Network, must pledge a contract with an EMP for all the services related to the electromobility operations such as subscription models for charging and billing, direct-payment or navigation services to the nearest available charge point, tariff schemes for end and business customers, B2C billing and invoicing.

2.2.2 IT Suppliers

The IT Suppliers actors are companies providing IT services to other actors interacting in the electro-mobility Hyper-Network.

- **IT Service Developer**: this actor represents software companies providing back-end application or services, for example, to enable the billing and roaming or to forecast the electricity demand;
- **IT Cloud Platform Provider**: it includes companies specialized in developing cloud platforms belong to this class. The cloud platform is provided in the form of Infrastructure-as-a-Service (IaaS) or Platform-as-a-Service (PaaS) to the IT Service Developers;
- **eRoaming Platform Provider**: this actor is a legal entity providing IT solutions for eRoaming services;
- **Map Service Provider**: it is a legal entity providing dynamic map database and content to IT developers;
- **Routing and Navigation Service Provider**: this public or private entity makes use of map data to provide routing and navigation services, based also on vehicle and driver information;
- **Travel and Traffic Information Service Provider**: this public or private entity provides a service to other actors by collecting and aggregating travel and traffic information.

2.2.3 Charging Infrastructure

This macro-area includes actors which constitute the framework necessary for the EVs to complete the charging process.

- **Charge Point Operator (CPO)**: this class includes companies responsible for the management and servicing of charge points and charging stations. Its main responsibilities concern the management of the charge points by means of an IT system, the billing and invoicing;
- **Parking and Service Stations Provider**: this legal entity provides parking spots equipped with charging services.

2.2.4 Decision Makers

Decision Makers are actors involved in the preparation and enforcement of regulations and legislation concerning the electro-mobility Hyper-Network.

- **Policy Maker**: this class includes member of governments responsible for preparing and enforcing rules and laws and public investment decisions;
- **Regulatory Body**: this actor is a public authority or government agency responsible for exercising autonomous authority in a regulatory or supervisory capacity.
- **Energy Regulatory Body**: it is an organization created by governments with the purpose of protecting the interests of users and consumers, promoting competition and ensuring efficient, cost-effective and profitable nationwide services with satisfactory quality levels in the electricity sectors.

2.2.5 Public Authorities

This macro-area includes public authorities which are indirectly involved with the electromobility Hyper-Network and market.

- **Municipality**: it is an urban administrative division with corporate status and powers of self-government or jurisdiction. For the purposes of NeMo, this term will also be used to cover other public authorities (below national or state level) with similar roles/powers, such as counties, provinces or metropolitan regions;
- **Tax Authority**: it is a government entity authorized by the law to assess, levy and collect taxes;
- **Control Authority**: it is a government entity authorized to guarantee the observance of the law (can be Police, financial police, courts and legal entities);
- **Road Authority**: this actor is the body responsible for the administration of public roads within a given jurisdiction;
- **Road Operator**: this actor monitors, maintains and manages the road traffic and road side infrastructure systems (this may be the same as the Road Authority or it may be a separate company owning or operating the road for a Road Authority).

2.2.6 Energy

This macro-area groups all the actors involved in the management of the energy.

- **Transmission System Operator (TSO)**: this entity is responsible for operating the transmission system which transports electricity on the high-voltage. The TSO guarantees the maintenance and, eventually, the development of the transmission system, also providing system services as balancing services, reserve capacity and power quality. It helps reducing or increasing the energy load;
- **Distribution System Operator (DSO)**: this entity is responsible for the voltage stability in the distribution grid;
- **Facility Manager**: this entity is responsible for coordinating demand and supply of facilities and services within public and private organizations. It monitors and manages building energy usage;
- **Energy Retailer**: this actor sells the electricity to customers;

- **Aggregator**: it can be called also Energy Trader and acts on behalf of groups of customers by using small volume inputs and creating saleable portfolios to be sold on different electricity markets.

2.2.7 European and International Organizations and Networks

This macro-area groups actors that cooperate and provide opinions and ideas useful for the Hyper-Network to satisfy all kinds of users and customers.

- **Association of Electric Vehicles Market Stakeholders**: this actor is a representative body of stakeholders in the development of an Electric Vehicle market. Some examples of such associations include: e-violin, AMP, UK EVSE, AFIREV, AEDIV;
- **Stakeholders Interest Group**: this actor is an association of entities interested in the development of the electromobility sector and collects data and experiences of electric mobility past and present projects. Some examples include: eMobility ICT Interoperability Innovation Group (eMI3), Hydrogen Fuel Cells and electromobility in European Regions (HyER);
- **NeMo Alliance**: this association will be established within WP7 of NeMo project and will grant access to the NeMo Hyper-Network to legal entities.

2.2.8 End Users

End Users are the final users of the NeMo Hyper-Network.

- **Driver**: a person using the vehicle and providing information about driving needs and consequently influencing charging patterns;
- **Owner of EVs**: a person or legal entity owning the vehicle;
- **Fleet Operator**: a person or legal entity operating a fleet of EVs, eventually having contracts with EMPs;
- **Logistic Company**: a company that offers logistic services using EVs.

2.2.9 Complementary Services

This macro-area groups actors offering additional services fundamental for the functioning of the NeMo Hyper-Network.

- **Payment Service Provider (PSP)**: a PSP offers online services for accepting electronic payments by a variety of payment methods (e.g., credit card, direct debit, bank transfer, online banking). It manages the payment and reimbursement procedure for EV users and other operators;
- **Insurance Operator**: it is a company providing coverage in the form of compensation resulting from loss, damages, injury, treatment or hardship for EV owners and drivers;
- **Charge Card Operator**: this company provides cashless services to EV users.

2.3 Actors' needs

In this section the actors' needs with respect to the NeMo Hyper-Network are described in detail.

The **Vehicle Manufacturer** needs:

- To identify and satisfy End Users needs in order to offer better services and increase its EV market penetration;
- To access in-vehicle data from EV Users in order to increase EV performance (subject to EV driver/owner acceptance)
- To ensure easy and seamless e-mobility through easy access to charge point;
- To be informed on available EV services and possibly to interact with them;
- To exchange data in order to develop additional EV services functionalities;
- To monitor and increase EV performances;
- To design and develop the HMI application for EV User;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Charge Point Manufacturer** needs:

- To identify and satisfy CPOs' needs to offer better services and to increase sale of charge points and charge points;
- To access charge points data from CPOs in order to increase charge point performance;
- To influence regulations and standards, and advice on new policies;
- To share data on technical compatibility of vehicles and CPs (interoperability standards);
- To comply with the needs of the road operator or parking/service station operator regarding the desired installation;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Electro Mobility Service Provider** and **eRoaming Platform Provider** need:

- To manage local and remote authorization requests;
- To get Charge-Detail-Records from the charge point;
- To exchange information related to an in-progress charge session;
- To get static and dynamic information about charge points;
- To get information related to the End-User's mobility need and vehicle status;
- To send the charge need to the CPO;
- To get information from the CPO related to B2B tariff and prices changes;
- To receive charge points reservations from end users;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **IT Service Developer** needs:

- To access End Users data;
- To display results from IT Service to end user;
- To access data from the Map Service Provider;

- To access static and dynamic charging data from CPOs;
- To receive rules about standardization and interoperability from Decision Makers;
- To access in-vehicle data from connected car backend of VM;
- To use "Service Creation" and "Service Execution" and "Service Delivery" domains of the NeMo Open Cloud Marketplace to develop a new IT Service;
- To offer IT services in the NeMo "Service Delivery" domain;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **IT Cloud Platform Provider** needs:

- To receive rules about standardization and interoperability from Decision Makers;
- To provide "Service Creation", "Service Delivery" and "Service Execution" domains;
- To provide bill for using the cloud platform;
- To access the "NeMo Network";
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Map Service Provider** needs**:**

- To access information about the locations and types of charging facilities from CPOs;
- To access current location information from Drivers / End Users;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Travel and Traffic Information Service Provider** and **Routing and Navigation Service Provider** need:

- To access current location information from Drivers / End Users;
- To access and aggregate traffic information from Drivers and Road Operators;
- To access map data from a map service provider;
- To access static vehicle information (dimensions, charge usage profile, etc.) provided by a VM or a Driver;
- To access dynamic vehicle information (charge status, speed, temperature, etc.) provided by a standard interface (only Routing and Navigation Service Provider);
- To access driver profile information (preferred charging locations, expected destinations, driving profile, etc.) (only Routing and Navigation Service Provider);
- To access dynamic environment information (e.g., current and expected weather);
- To access dynamic charge point estimated B2B and B2C pricing;
- To access charge point information (e.g., charge rate, plug types, etc.);
- To access dynamic charge point availability;
- To access the Drivers' account or token information with his eRoaming Platform Provider to perform authentication, booking and payment actions;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Charge Point Operator** needs**:**

- To manage local and remote authorisation requests;
- To send Charge-Detail-Records;
- To provide charging load schedule;
- To receive indications about the installation of new charge points in order to ensure good quality of the grid from DSO;

- To exchange information related to an in-progress charge-session;
- To upload static and dynamic information about charge points;
- To upload information related to B2B tariff and prices changes;
- To access traffic and usage data;
- To access data on network status;
- To foster electromobility deployment in urban areas;
- To provide CP real-time status;
- To handle charging profiles;
- To perform predictive smart charging;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Parking** and **Service Stations Provider** needs**:**

- To access data regarding parking requests from End Users;
- To receive deployment rules from Regulatory Bodies;
- To provide status information about availability of parking and charge point places;
- To inform about plans for the installation of new charge points within the parking area;
- To provide billing information for the parking;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Policy Maker** needs**:**

- To access traffic and usage data;
- To access data about EV users/registrations;
- To access charge point usage data;
- To communicate and promote mobility policies;
- To have detailed information about the End Users' mobility needs and the transport network constraints.

The **Regulatory Body** needs**:**

- To access traffic and usage data;
- To access charge point usage data;
- To create regulations looking to improve the mobility according to the policies set by the Policy Maker;
- To receive feedback about the effects of the regulations.

The **Energy Regulatory Body** needs**:**

- To access data on EV charging network status, in order to evaluate the quality of service (DSO);
- To access data on exchanged energy and applied tariffs for EV market evaluation.

The **Municipality** needs**:**

- To identify and satisfy End Users' needs to offer better services and increase its EV market penetration;
- To access in-vehicle data from EV users to increase EV performance;
- To access evaluation data tools to define continuous improvement strategies;
- To access data about EV users/registrations;
- To implement policies and ensure the compliance with current regulations;
- To receive rules about standardization and interoperability from Decision Makers;

- To comply with the laws of the regulatory bodies.

The **Tax Authority** needs**:**

- To access data about charge point usage;
- To access data about EV users/registrations;
- To access data about EV usage and status;
- To obtain energy and exploitation costs of the charging network in order to define accurate taxes according to the policies;
- To set taxation policies for the service and the network access.

The **Control Authority** needs**:**

- To access traffic and usage data;
- To access a vehicle identification system;
- To receive rules about standardization and interoperability from Decision Makers;
- To communicate issues (penalties, incidents, etc.).

The **Road Authority** needs**:**

- To access and communicate traffic data;
- To access charge point usage and status;
- To access EV usage data;
- To promote road usage regulations in conjunction with the Control Authority;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Road Operator** needs**:**

- To access data exchange between EV and infrastructure;
- To access data exchange between infrastructures;
- To provide traffic data to authorities and traffic information to drivers;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **TSO** needs**:**

- To access CP usage data;
- To access DSO needs;
- To transport energy to the distribution systems;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **DSO** needs**:**

- To maintain the stability of the electric grid and/or for future MV/LV dispatching purposes;
- To connect charge points to the grid;
- To access data about traffic and usage to perform load forecast for network operation purposes;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Energy Retailer** needs**:**

- To access grid status;
- To decide retail price;

- To perform energy retailing;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Aggregator** needs:

- To access data on critical DSO network balance requirements: The Aggregator will use its portfolio of clients (CPO, End Users, retailers…) to respond to DSO needs in a demand-response scheme;
- To access data about traffic and usage, to be sent to DSO/TSO for forecasting purposes;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **EV market stakeholders** and **Stakeholder Interest Groups** may (according to their sector and objectives) need:

- To access data about traffic and usage;
- To access data about charge point usage;
- To access data on technical compatibility of vehicles and CPs (interoperability standards);
- To access data on costs and payment options for different CPs;
- To influence regulations or advice on new policy;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **NeMo Alliance** needs:

- To receive requests from stakeholder/legal entities to access the NeMo Hyper-Network;
- To grant access to the NeMo Hyper-Network after validating access needs from stakeholder/legal entities;
- To access data about traffic and usage;
- To access data about charge point usage;
- To access data on prices and clearing of payments between operators;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Driver** and **Owner of EVs** need:

- To communicate driving needs, such as range and time of availability, necessary to achieve the most appropriate energy transfer scenario;
- To find a suitable charge point to charge his/her vehicle when necessary;
- To access data about the location and real-time availability of charging infrastructures;
- To reserve charge points/stations and to modify or cancel prior reservations;
- To receive indications on how to reach a suitable charge point;
- To set up and manage an account which gives access to charge points across different operators in different locations (including cancelling accounts, moving to a different account provider or moving an account to a different vehicle);
- To receive data concerning additional services;
- To access the EV HMI and to easily get useful information;
- To be informed on EV services availability;
- To choose between various payment options (subscription-based vs. ad-hoc authorisation, pay per charge, flat rate per month, etc.);

- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Fleet Operator** and **Logistic Company** need:

- To charge a vehicle when necessary according to building power capacity constraint;
- To set up and manage corporate accounts covering the fleet;
- To set up and manage an account which gives access to charging across different operators in different locations (including deleting accounts, moving to a different account provider or moving an account to a different vehicle) (only **Logistic Company**);
- To be informed of the location and real-time status of charging infrastructure by the CPOs (only **Logistic Company**);
- To receive indications on how to reach a suitable charge point (only **Logistic Company**);
- In the event of an account for several vehicles, to identify which vehicle has used which facilities and the energy transfer involved (in case the cost is passed on directly to final clients, for example rental vehicles);
- To receive data concerning additional services;
- To get information to manage fleet: mileage, SoC, autonomy by every team shift;
- To propose solutions for home charging and to refund consumed energy (only **Fleet Operator**);
- To access global EV statistics and usage (only **Fleet Operator**);
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Payment Service Provider** needs:

- To access charge points data;
- To access data about sold energy and incentives;
- To provide the payment service;
- To access charging payment data from CPOs;
- To deposit the due amount to the CPO's account;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

The **Insurance Operator** needs:

- To access traffic and usage data from End Users;
- To access data from Control Authority;
- To provide the insurance services;
- To comply with the laws of the regulatory bodies.

The **Charge Card Operator** needs:

- To access charging payment data from CPOs;
- To deposit the due amount to the CPO's account;
- To access data analytics about transactions per EV user, per CPO, per region.
- To provide charge cards and related services;
- To receive rules about standardization and interoperability from Decision Makers;
- To comply with the laws of the regulatory bodies.

2.4 Macro-area interrelations

Based on the previous analysis of needs, this section presents the identified macro-areas containing the electro-mobility actors and their interrelations. Each macro-area and their interrelations are represented by using the UML package diagram (see Fig. 1).

In the UML Package Diagram, packages are used to group elements of any kind. It allows making up a model of a large-scale system and it is used to depict dependencies among major elements in the system. In NeMo, packages represent macro-areas and are used to group classes; each class represents one of the actors of the electro-mobility network. In the diagram, packages are represented with boxes with the name of the package on the top and the name of the contained classes inside the box.

The Package Diagram connects different macro-areas: an arrow connecting one package to another depicts one or more actors belonging to the first macro-area sending data to one or more actors of the second macro-area. Each arrow also contains a brief description of the exchanged data.

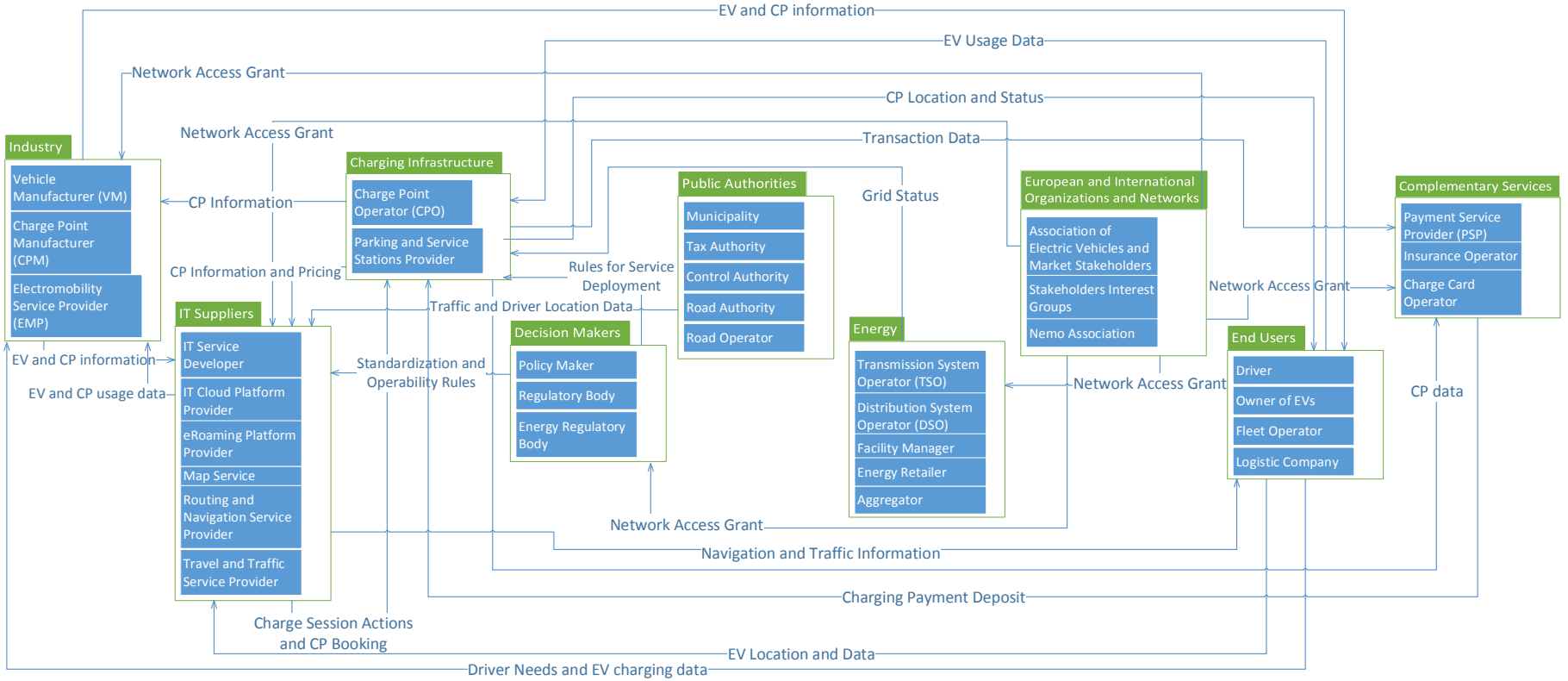


Figure 1: The Package Diagram for the Macro-areas interrelations

2.5 Actors' interrelations

This section presents and models the actors' interrelations by means of a relationship matrix and UML Class Diagrams. The relationship matrix indicates if a relation between each actor pair exists, which can be unidirectional or bidirectional (see Figure 3).

In the UML, a class diagram is a type of static structure diagram that describes the structure of a system by showing the system's classes, operations (or methods), and the relationships among objects. In NeMo, the Class Diagram is used to represent the interrelations between actors and to specify their main functions in the network. For each macro-area, a Class Diagram is provided.

In the diagrams, classes are represented with boxes that contain two compartments: the top compartment contains the name of the class and the bottom one contains the methods or operations that the class, i.e. the actor, can execute. Moreover, each arrow contains the address of the senders/recipients' actors and the exchanged data.

In particular, the Class Diagram of each macro-area is provided. The Figure 2 shows how to read the interrelations in the Class Diagrams illustrated in Figures 4-12.

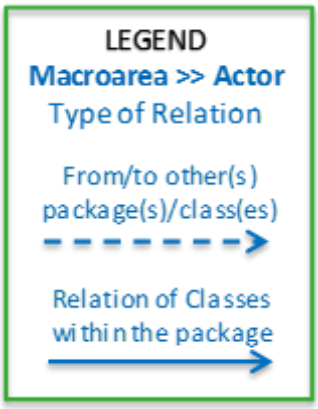


Figure 2: The Class Diagram legend

			a.1	a.2	a.3	b.1	b.2	b.3	b.4	b.5	b.6	c.1	c.2	d.1	d.2	d.3	e.1	e.2	e.3	e.4	e.5	f.1	f.2	f.3	f.4	f.5	g.1	g.2	g.3	h.1	h.2	h.3	h.4	i.1	i.2	i.3
Industry	Vehicle Manufacturer (VM)	a.1	-		X			X																			X	X		X	X	X	X			
	Charge Point Manufacturer (CPM)	a.2		-										X	X	X											X	X	X								
	Electro-Mobility Service Provider (EMP)	a.3	X		-			X				X					X													X	X	X	X			
IT Suppliers	IT Service Developer	b.1				-											X													X	X	X	X			
	IT Cloud Platform Provider	b.2				X	-										X																			
	eRoaming Platform Provider	b.3			X			-				X					X																			
	Map Service Provider	b.4							-	X	X						X																			
	Routing and Navigation Service Provider	b.5								-							X													X						
	Travel and Traffic Information Service Provider	b.6							X	X	-						X						X							X						
Charging Infrastructure	Charge Point Operator (CPO)	c.1		X	X	X		X	X		X	-	X	X	X	X		X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Parking and Service Stations Provider	c.2								X			-														X	X		X	X	X	X			
Decision Makers	Policy Maker	d.1		X		X	X							-		X	X	X	X	X	X							X								
	Regulatory Body	d.2		X		X	X						X		-		X	X	X	X	X							X								
	Energy Regulatory Body	d.3		X		X	X								X	-	X	X	X	X	X							X								
Public Authorities	Municipality	e.1												X	X	X	-	X									X	X	X							
	Tax Authotity	e.2												X	X	X	X	-	X								X	X	X	X	X	X	X			
	Control Authority	e.3												X	X	X			-								X	X	X	X	X	X	X		X	
	Road Authority	e.4												X	X	X			X	-							X	X	X							
	Road Operator	e.5		X							X			X	X	X	X			X	-						X	X	X							
Energy	Transmission System Operator (TSO)	f.1																				-	X													
	Distribution System Operator (DSO)	f.2										X										X	-	X	X	X										
	Facility Manager	f.3																						-												
	Energy Retailer	f.4										X	X			X	X								-	X				X	X	X	X	X		
	Aggregator	f.5																					X			-				X	X	X	X			
European and International Organizations and Networs	Association of EV and Market Stakeholders	g.1												X	X	X											-									
	Stakeholders Interest Groups	g.2	X	X								X	X	X	X	X												-									
	Nemo Association	g.3	X	X	X	X	X	X	X	X	X			X	X	X						X	X	X	X	X			-					X	X	X
End Users	Driver	h.1	X		X	X			X			X		X	X	X	X	X	X	X	X	X	X	X	X	X				-		X		X	X	X
	EV owner	h.2										X					X														-			X	X	X
	Fleet Operator	h.3			X							X					X													X		-		X	X	X
	Logistic Company	h.4			X							X					X																-	X	X	X
Complementary Services	Payment Service Provider (PSP)	i.1										X															X	X	X	X	X	X	X	-		
	Insurance Operator	i.2										X																							-	
	Charge Card Operator	i.3										X																		X	X	X	X			-

Figure 3: The actors' relationships matrix

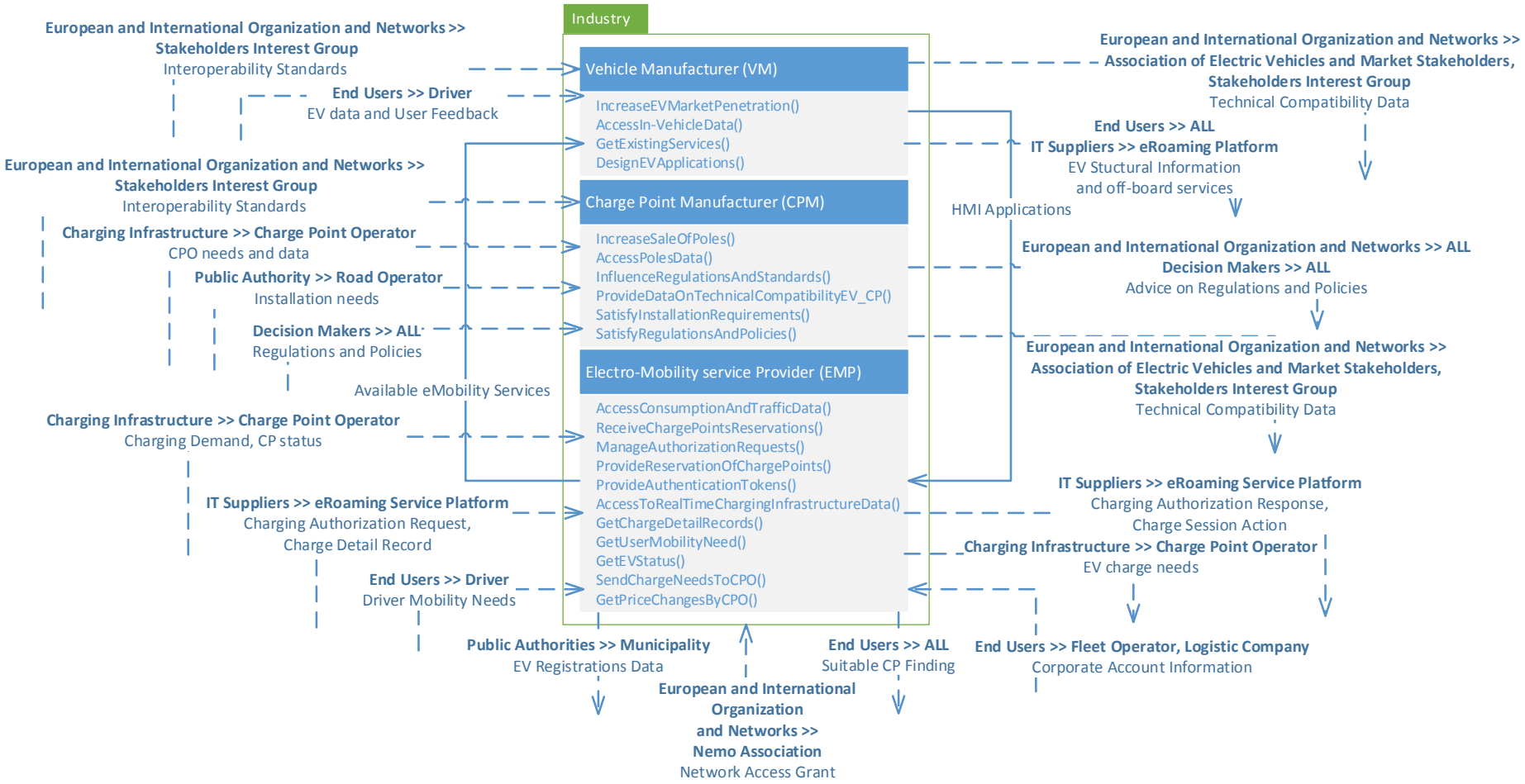


Figure 4: The Class Diagram of the Industry actors

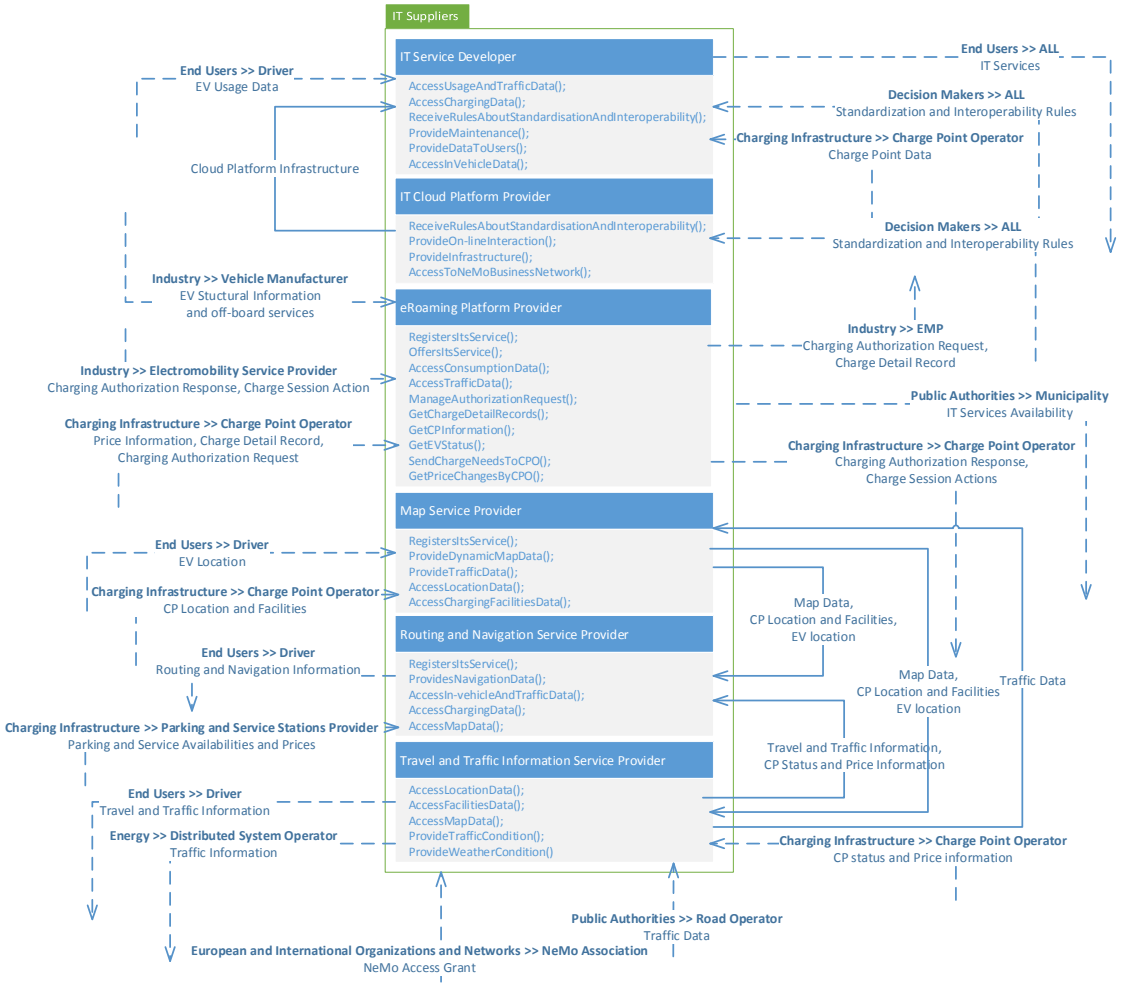


Figure 5: The Class Diagram of the IT Suppliers actors

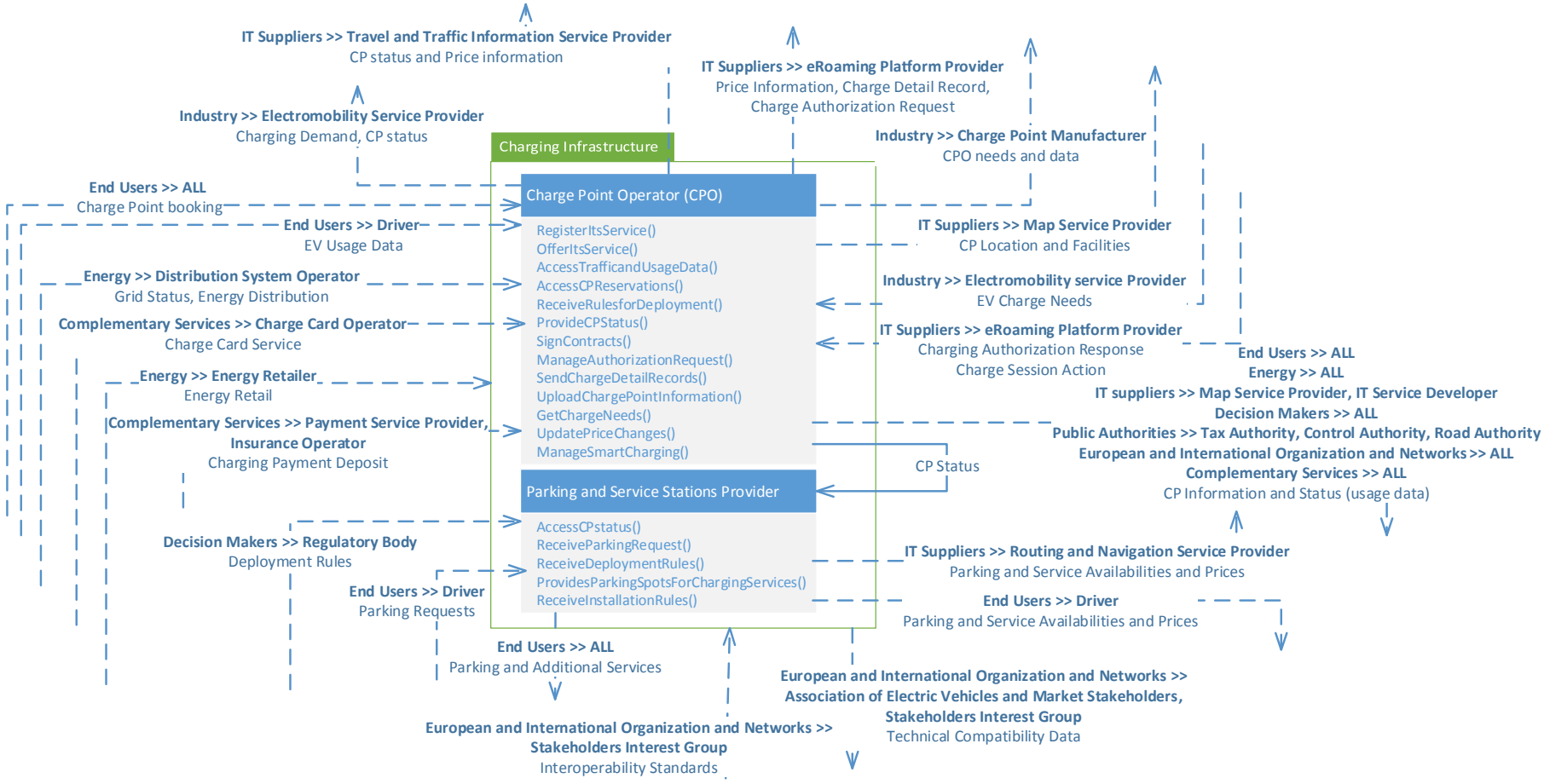


Figure 6: The Class Diagram of the Charging Infrastructure actors

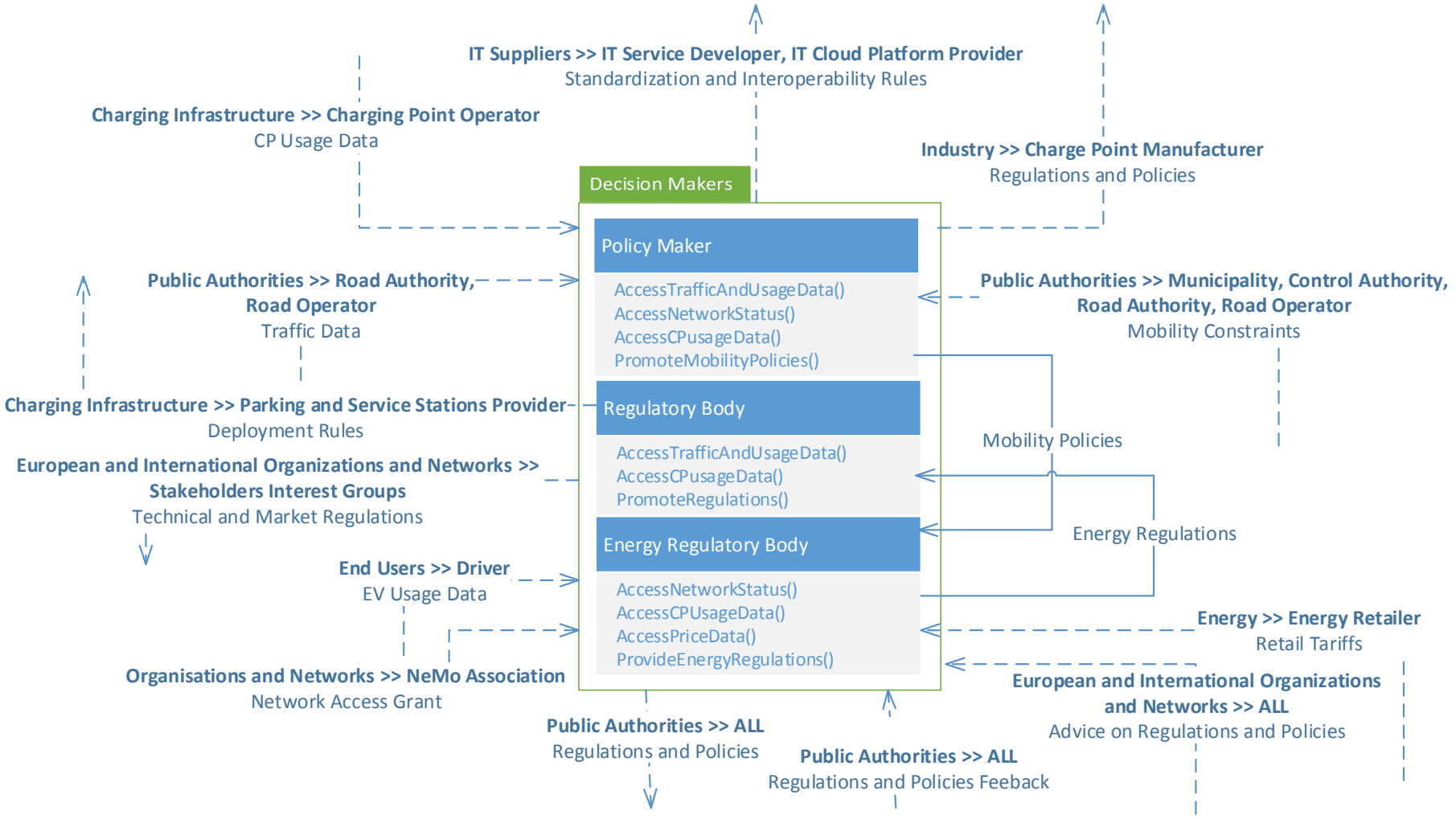


Figure 7: The Class Diagram of the Decision Makers actors

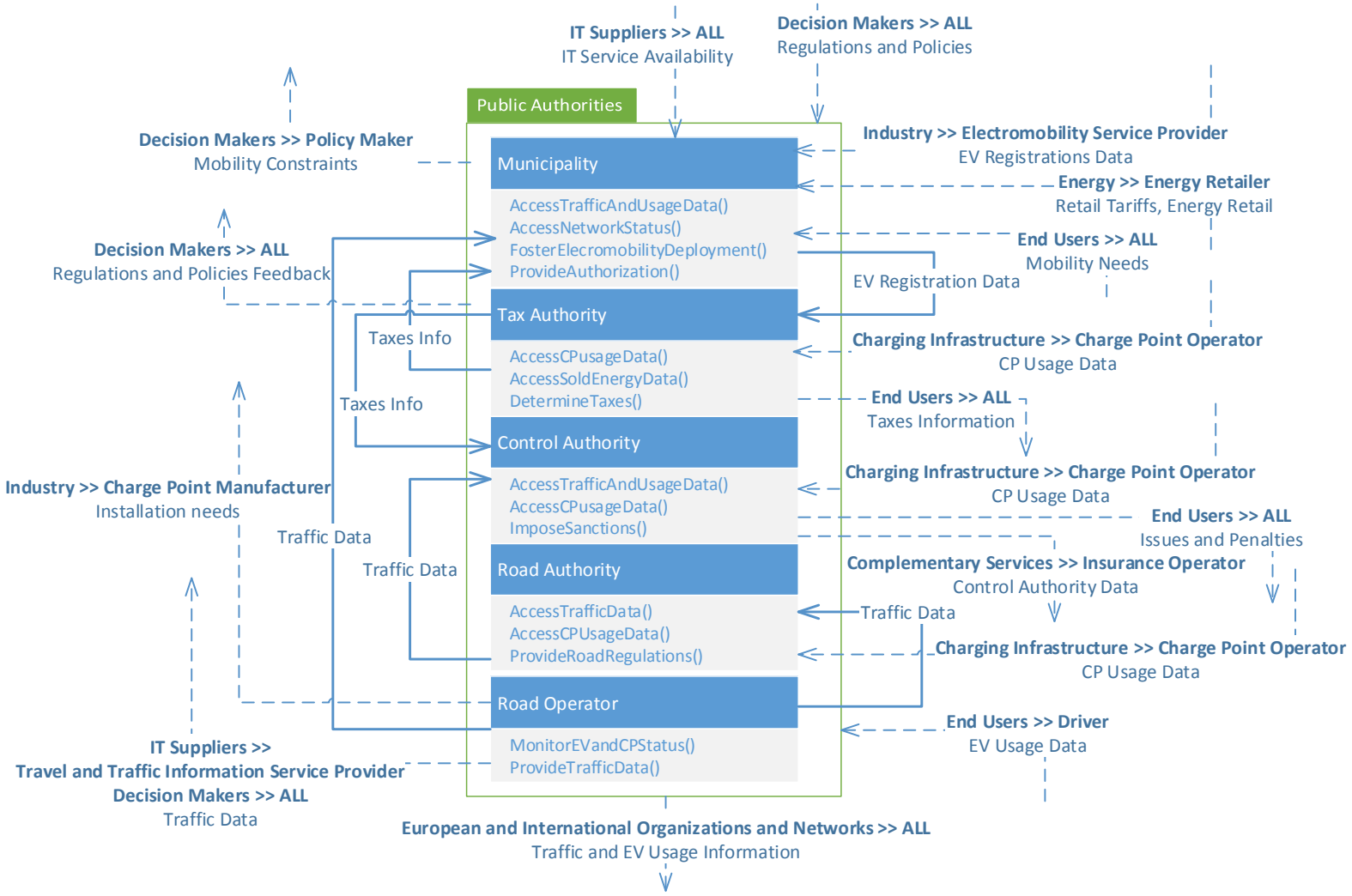


Figure 8: The Class Diagram of the Public Authorities actors

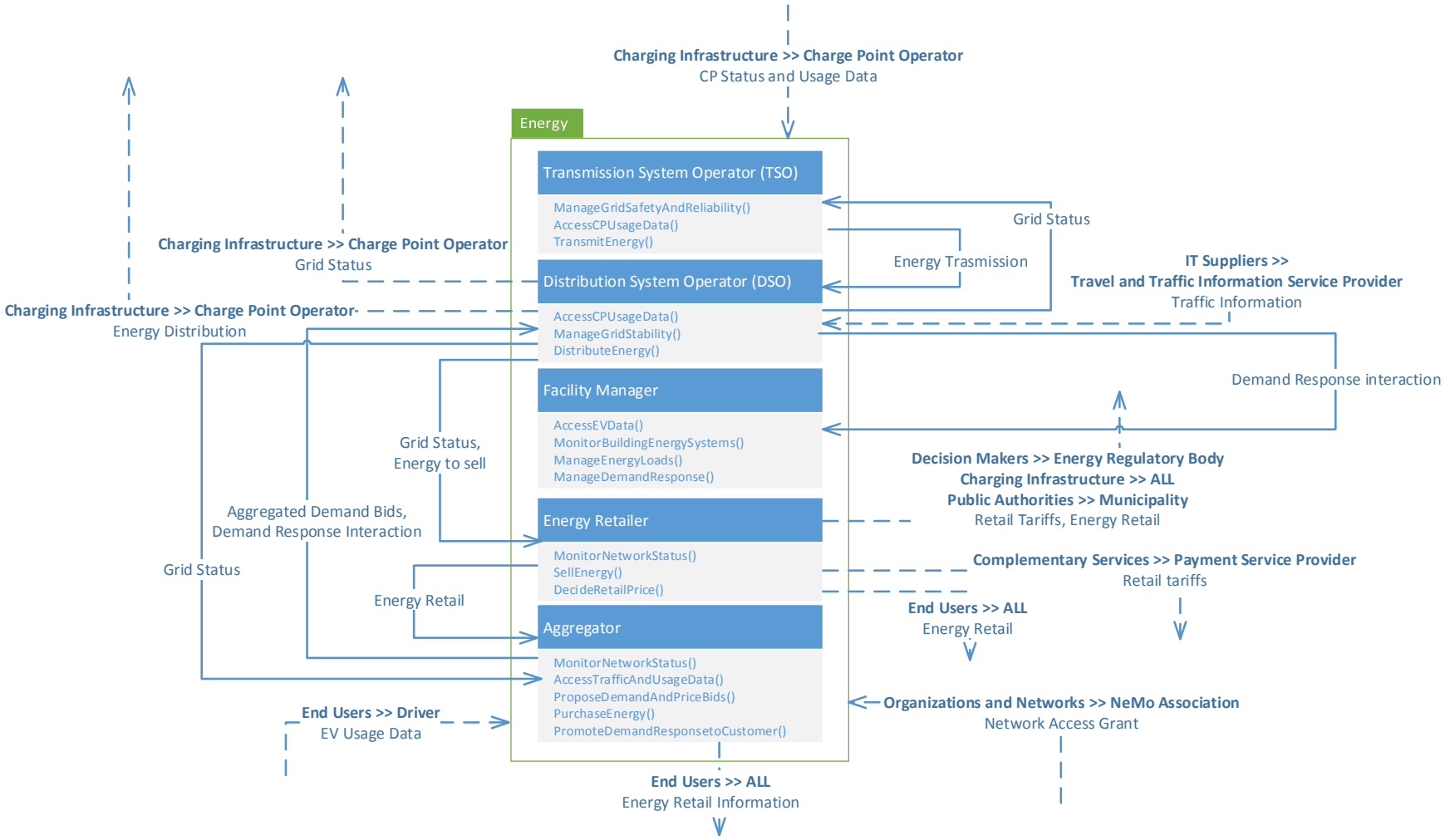


Figure 9: The Class Diagram of the Energy actors

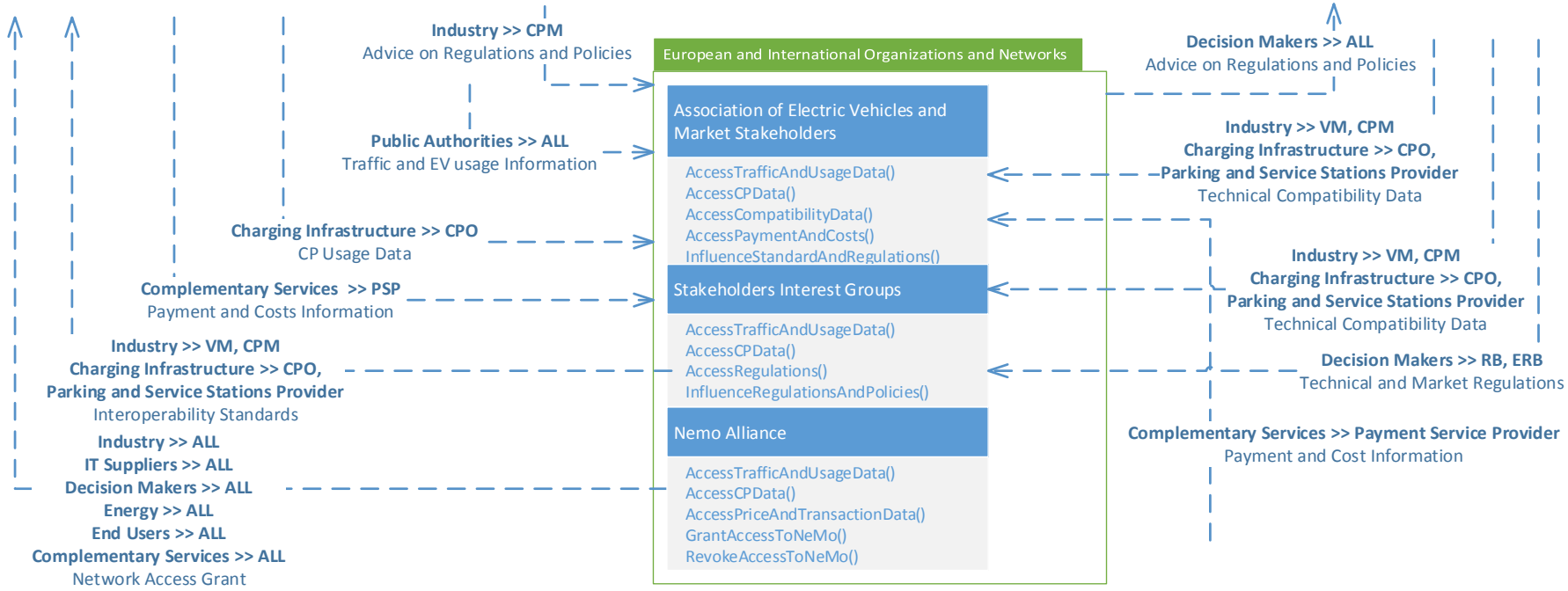


Figure 10: The Class Diagram of the European and International Organizations and Networks actors

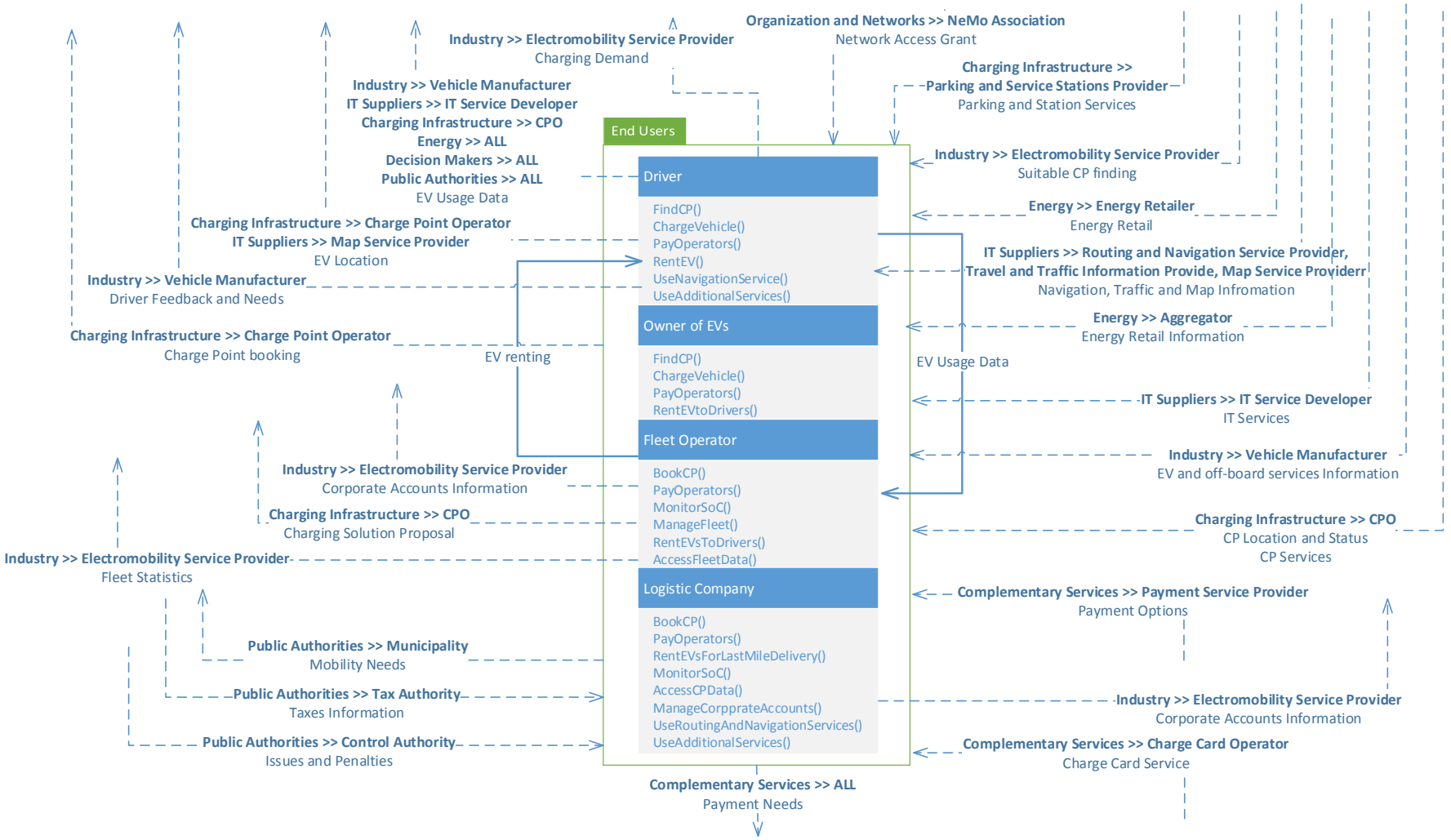


Figure 11: The Class Diagram of the End Users actors

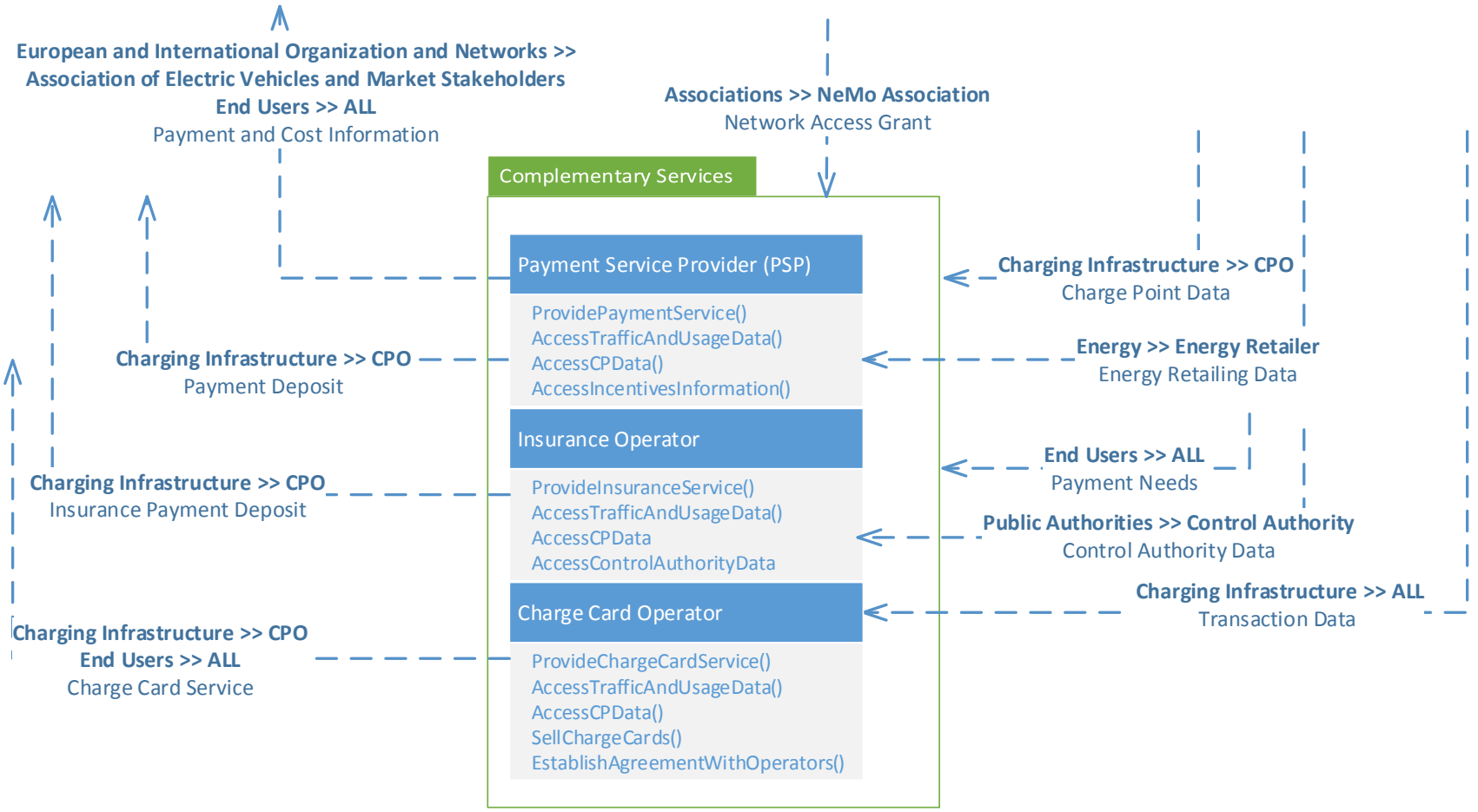


Figure 12: The Class Diagram of the Complementary Services actors

3. Business Scenarios

This chapter presents the Business Scenarios (BS) to be tested in the pilot sites. Each BS is described based on the template used in the Green eMotion project[1] and adapted to NeMo. The description of each BS has been carried out by sub-groups of NeMo partners who have proposed a set of scenarios to be tested. Based on the agreement among the sub-groups, the most significant scenarios have been selected by also paying attention to avoid overlaps among services proposed different BS.

The result of this work is a description of each BS, with its specific objectives, flow of events, actors involved, and improvements with reference to the current situation and what can be expected after the implementation of the NeMo Hyper-Network. The description of the BS includes a proposal for a set of KPIs aimed at the evaluation of the BS and test sites in which the proposed services will be tested.

3.1 Smart Charging (BS1)

3.1.1 Business Scenario short description

Smart Charging Business Scenario aims to provide services to EV driver and fleet manager to efficiently charge vehicles based on their preferences (cost, time, required SoC and place of recharge), charge point and renewable energy availability and respecting grid constraints.

On the other hand, Smart Charging aims to provide services to other stakeholders involved in the EV charge, like energy actors (DSO, retailers, and regulators) in order to optimise efficiency of energy grid, CPOs to optimise their charge point use rate or policies to avoid traffic congestion.

The battery's state-of-health (SOH) describes the current condition of the battery. SOH can be defined as the ratio of the maximum charge capacity of an aged battery to the maximum charge capacity when the battery was new. This value typically degrades over time and use of the system. SOH is used as a parameter for indication of the degree of performance degradation of a battery and for estimating the battery remaining lifetime.

The battery capacity can be determined by analysing the charging data measured at the charge point. Influencing factors for the SOH can be determined during vehicle operation and charging. This information can be communicated to the vehicle manufacturer via the NeMo Hyper-Network. This is specifically interesting for vehicles, which are not connected to an OEM server directly.

The SOH information can be used in order to determine an appropriate charging strategy for the vehicle and to adapt this strategy in hand. This may include respecting the available time for the charging process depending on the SoC level as well (compare previous use-case).

[1] www.greenemotion-project.eu

3.1.2 Objectives

The objectives of this Business Scenario are several and refer to the three sub-scenarios that are described more in detail in the next section related to main flow of events. The objectives are:

- Provide the best options to charge a vehicle or a fleet according to driver/owner needs (cost, time, required SoC, place of recharge, renewable energy rate, …), charge point availability, grid constraints, traffic congestion, …;
- Optimize waiting and charging time;
- Avoid grid congestion;
- Provide innovative grid balancing services (like Vehicle2Grid);
- Avoid traffic congestion;
- Maximize charge points use rate;
- Maximize renewable energies use;
- Improve battery life and performance;
- Improve range prediction;
- Improve battery charging strategy;
- Support battery maintenance prognosis.

3.1.3 Main Flow

In this paragraph, the main flow of events is described for each of the Sub-Scenarios which mainly consist of services addressed to the different actors.

Sub-Scenario A.1

This Sub-Scenario involves three main Actors: the EV driver, the Electromobility Service Provider (EMP) and the Charge Point Operator (CPO). The flow of events is the following:

1. The EV driver informs his/her EMP on all information about his/her mobility needs (needed range, when to charge, cable compatibility …) and desired charge location.
2. The EMP uses a NeMo service to convert these mobility needs into charge need (kWh needed, when, cable compatibility …).
3. The EMP uses a NeMo service to get better propositions from CPOs satisfying the needs of the EV driver.
4. The NeMo service requests CPOs having a charge point on the location desired by the user, giving them the charge need of the EV driver. CPOs answer for each charge point: the availability, the B2B tariff, *the rate of renewable energy,* max power available …
5. The NeMo service sorts the offers according to the level of fulfilment of the EV driver needs and according to the constraints from stakeholders like traffic congestion, and returns the sorted list to the EMP.
6. The EMP displays to the EV driver all the received information. The B2B price can have an impact on the B2C price proposed to the EV driver by the EMP.
7. The EV driver selects a charge point and books it through the EMP.

8. The EMP books the charge point to the CPO using a NeMo service.
9. The EV driver drives to the charge point following EMP routing.
10. The EV driver authenticates himself/herself to the charge point.

Sub-Scenario A.2

This Sub-Scenario implies an exchange of information between Distributor System Operators (DSO) and CPOs.

1. Energy actors (DSO, retailers) provide information to the CPOs using a NeMo service:
   - They set the price of energy regarding location (energy local network) and time based on DSO near-real time needs for grid balancing. Given the strategic and confidentiality status of these kinds of data, NeMo service will only receive punctual needs and no overall network asset and configuration will be shown to all the other stakeholders.
   - They inform the CPO on the max power available, possibly including renewable energy on an area (charge point, local network …).
   - Other stakeholders provide information to NeMo using a NeMo service. For instance, public authorities send information on traffic congestion.
2. The CPO, according to all constraints and information, takes a decision on the B2B tariff proposed to EMP he is in contract with.
3. The CPO provides tariffs and other information on each charge point to all concerned actors through a NeMo service.

Sub-Scenario B

This Sub-Scenario deals with the management of the electric grid and the charging infrastructure. The main flow of events is described as follows:

1. DSO sends constraints/opportunities (increase/decrease power, V2G …) to the CPO and Aggregator/retailers using a NeMo service.
2. Regarding charge needs of all EV plugged on its stations, the CPO pilots charge of EVs. The communication between stations and EVs can be done using 15118 protocols.
3. Aggregators evaluate near future balancing capabilities of the prosumers they manage (industrial fleets, car sharing, private EV), mainly evaluating their availability to provide additional services in terms of power curtailment or V2G.
4. When talking about V2G, NeMo platform should be able to take into consideration only the charging infrastructure and the EV designed and enabled for these kinds of grid services.

Sub-Scenario C

The last Sub-Scenario is related to the Battery Management:

1. The vehicle is connected to charge point and charging process starts.
2. The Charge Point measures the energy transferred to the vehicle until battery is fully charged.
3. A service estimates the remaining capacity based on last charging events.
4. The service communicates the remaining capacity to the vehicle manufacturer.
5. The vehicle starts driving.
6. The usage profile of the vehicle is measured (especially electric load distribution, electric battery currents).
7. The vehicle stops driving.
8. A service aggregates the measured usage data.

3.1.4 Actors Involved

There are several Actors involved in the three Sub-Scenarios. These Actors have been described in Chapter 2 and are detailed for each standard service in paragraph **Error! Reference source not found.**.

The Actors involved can be summarized as follows: DSO, CPO, EV Driver, EMP, Routing and Navigation Service provider, Energy Retailer, Aggregator, Payment service provider, Vehicle manufacturer, Fleet operator, Public authorities, eRoaming Platform Provider.

3.1.5 Current situation

This paragraph describes the current situation related to the EV charging. Based on this description, the improvements brought by NeMo are identified. Currently:

- A driver can only select among all available charge points. No public functionality exists to smart search for a charge point.
- Pilot recharge according to stakeholder's constraints is only done through some experimentation (Research projects; ...).
- No grid balancing services are currently implemented in a commercial and industrialized way in Europe (only Proof of Concept and testing facilities).
- Currently the SOH information is only available for diagnostic purposes within the vehicle. The information is not detected outside (e.g. by the charging system). Additionally, the maintenance information is only used within service applications.

3.1.6 What is expected after NeMo

There are several improvements that the Smart Charge Business Scenario is expected to bring. Some of them are listed below, while additional impacts will be surveyed and evaluated in the context of WP6 on verification and validation.

- The EV drivers will be able to choose the best charge option according to their preferences and respecting other constraints. The charge phase will be optimized by integrating renewable energies (if existing), minimizing charging and queuing time, and avoiding grid and traffic congestion.

- CPOs will be able to answer energy grid constraints/opportunities by piloting recharge of EVs plugged to their charge points.
- With reference to the Status of Health of Batteries, it will be possible to detect the SOH using the charging system. Additionally, it will be possible to optimise the potential for maintenance and charging strategy alignment, depending on battery and vehicle usage behaviour.

3.1.7 Needed services

The following table lists the standard services that have been identified for this Business Scenario and the corresponding Actors involved. The Use Cases described in Chapter 4 are identified based on these services.

Convert mobility needs to charge needs	EMP, VM, Routing and Navigation Service Provider, eRoaming Platform Provider
Search better charge offers (from CPO according to EV driver/fleet operator needs and stakeholders constraints)	EMP, CPO, eRoaming Platform Provider, DSO, Aggregator, Energy retailer
Search and find charge points	EMP, Routing and Navigation Service Provider, CPO, eRoaming Platform
Book a charge point	EMP, CPO, eRoaming Platform Provider
Change energy tariffs	CPO, DSO, Aggregator, Energy retailer, (eRoaming Platform Provider)
Provide information on energy (max power available, …)	CPO, DSO, Aggregator, Energy retailer, (eRoaming Platform Provider)
Provide information about traffic	Public authorities, Routing and Navigation Service Provider
Get information about traffic	All
Change information about an EVSE (B2B tariff of the charge, max power available, dynamic status)	CPO, EMP, eRoaming Platform Provider
Propose energy grid constraints/opportunities	CPO, DSO, Aggregator, Energy retailer, (eRoaming Platform Provider?)
Get information from vehicle	EMP, VM, eRoaming Platform Provider
Aggregate user/driving behaviour	VM, eRoaming Platform Provider
Monitor Vehicle status	VM

Access User preferences (e.g. from previous use-case time until drive-off for maintenance)	VM, EMP

3.1.8 Key Performance Indicators (KPIs)

The list of KPIs that will be needed to evaluate the Business Scenarios was derived according to the description of the requirements. The following list reports a preliminary selection of possible KPIs:

- Take up rate of the provided service
- Increase the number of EVs which are charging on the same Charge Point
- Increase usage rate of the CPO network
- Service usage after roll-out
- Battery SOH after completion of driving cycle and after charging
- Degrading speed of SOH

3.1.9 Possible Test Sites

The Smart Charging Business Scenario will be tested in two different pilot sites. For fleet customers, the test will be carried out in France or Austria depending on the customers. For private customers, the pilot will take place in Austria.

3.2 Itinerary Planning (BS2)

3.2.1 Business Scenario short description

The Itinerary Planning aims to assist an EV driver in planning and driving an itinerary. The Itinerary Planning should take into consideration static restrictions and information (such as route, vehicle's charge capacity, etc.) as well as dynamic restrictions and information (such as remaining charge, weather, charge price, driving profile, etc.) and security sensitive situations (such as running out of charge in a highway with no emergency stopping lane).

3.2.2 Objectives

The objectives of this Business Scenario are presented from the point of views of the involved Actors. For an EV driver the objective is to:

- Ensure that the driver can complete his/her itinerary
- Help a driver finding the cheapest option available for charging his/her EV and reaching his/her destination
- Reduce time spent queuing for a charge point
- Reduce time spent waiting for charging to complete
- Enhance user experience
- Help managing range anxiety

For a DSO:

- Avoid grid congestion

For a CPO:

- Maintain a high usage rate in a charge point

For Road Operators:

- Ensure the safety and efficiency of critical infrastructure such as highways and tunnels

3.2.3 Main Flow

The flow of events of this BS consists of three main phases, two of which take place before the trip and the third one which consists of recommendations provided to the driver before and during the trip.

Driver-itinerary planning:

- An EV driver plans driving to a destination that requires a long itinerary, possible across multiple countries.
- Through an HMI, the driver is presented with a proposed itinerary plan, including:
  - Availability of the necessary and compatible charge points along the route (available stations should be preferred)
  - Charge status throughout the itinerary, as estimated from a consumption model and dynamic and static information
  - Charging times in each charge point
  - Charging goals in each charge point (for example, it may not be necessary to charge an EV to 100% battery capacity to complete the next leg of the trip)

Suggested-itinerary planning:

- An EV driver gets a recommendation to drive to a charge point according to a trigger condition:
  - Charge level insufficient to complete a planned trip
  - Charge level insufficient to complete a predicted trip (i.e. even if the charge level is sufficient to arrive to the next planned destination but not to fulfil the predicted usage for the rest of the day)
  - Price spike detected: a spike in prices later in the day would make it cheaper for a driver to start charging now.
  - Proximity to a preferred charging place (for example, due to automatically learned user preference, cheaper charge point with dynamically updated price, grid constraints, etc.)
  - Point-of-no-return detection, after which a driver would not be able to reach a preferred charge point.
  - Manual request from a driver

- Dangerous situation detection (for example: a planned or predicted route nears the end-of-range in an area where stopping may be dangerous, such as a tunnel or a highway with no emergency stopping lane).

- A charge point should be recommended to the driver. There should be a rule set for charge stations preference:
  - Charging stations with a lower grid load should be preferred
  - Cheaper stations according to their dynamic B2C pricing should be preferred
  - Charging stations close to the planned route or final destination should be preferred
  - Charging stations with relevant POIs should be preferred (e.g. charging stations with nearby public transport, or a nearby place where the driver can wait while the EV is charged)
- If the driver accepts the recommendation, an itinerary to a charge point is presented through an HMI.

Security features:

- During the itinerary planning as well as during the driving phase, real time information on the current route will be presented, including battery status and detection of possible security critical situations, such as running out of battery while in a tunnel or in a highway with no emergency lane. When detecting security critical situations, recommendations will be available for a driver if appropriate (e.g.: turn heating off to reduce battery consumption).

3.2.4 Actors Involved

The Actors involved in this BS are: Drivers, CPO (Charge Point Operator), Navigation service provider, VM (Vehicle Manufacturer), Map service provider (B2B), eRoaming platform provider, Routing service provider, EMP, Cloud platform provider.

3.2.5 Current situation

Currently, independent services exist, with no integration solution available. No user preferences are usually considered as guiding principle to select a charge point.

3.2.6 What is expected after NeMo

After the implementation of this Business Scenario, the driver will be able to plan a full itinerary across multiple countries with an EV, drivers will be able to drive comfortably even without a route. A driver will know the minimum time required to recharge enough for his/her next trip leg.

3.2.7 Needed services

The services that will be needed in this Business Scenario are the following:

- Get charging points
- Navigation to CP

- User profiling
- Booking services
- Load forecasting
- Battery capacity calculation

3.2.8 KPIs

As previously stated, only a preliminary proposal can be made on KPIs aimed to evaluate the Itinerary Planning. Indicators may include the decreased distance of trips after service roll-out, the estimation accuracy of time of arrival which may improve and the waiting time which should decrease.

3.2.9 Possible Test Sites

The Itinerary Planning can be tested in all NeMo test sites, including the cross-country pilot.

3.3 Cross-provider border booking authorization and payment management (BS3)

3.3.1 Business Scenario short description

The BS3 aims to automate the different phases of EV charging so that an EV driver can have a seamless charging experience across different providers and networks: the different steps in the charging process (such as booking, authorization/authentication and payment) will be simplified or fully automated. This includes a seamless experience for a single CPO in the "base network" of a driver, as well as different CPOs outside of the driver's base location (for example, by performing a cross-border EV charge plan).

3.3.2 Objectives

The objective of this BS is to improve the interoperability, defined as the ability for an EV driver to seamlessly use the Electric Vehicle Charging Infrastructure wherever it is located, whichever EV the customer uses, whoever operates the Charging Point, whoever supplies the charging service and electricity. To ease accessing the charging infrastructure, every charge point operator shall have a business agreement with every Electromobility Service Provider.

- Roaming: Offer EV drivers the option to charge their vehicles at all charging stations – regardless of any contracts concluded with operators. The billing occurs subsequently via the customer's own contractual partner.
- Enhance user experience: Provide a fully seamless (preferably "touchless") user experience by enabling drivers to book and use a charge point from a centralized HMI while eliminating the required interaction with 3rd party systems, including the charge point itself.
- Simplify the operation of a charge point: Allow drivers to commence a charging process without the need to interact with a charge point directly, thereby simplifying the charge process.

- Optimize charge times.
- Increase EV/renewable market share.

3.3.3 Main Flow

The flow of events consists of the following steps:

- The EV Drivers provide their EMP account details through the HMI, so that seamless authentication and payment is enabled.
- Through an HMI, EV drivers check the real time B2C expected pricing and availability of compatible charge points.
    - This could be triggered as an extension of the Business Scenario 'Itinerary planning', where an EV driver will be notified of different low-charge conditions and offered a route to a suitable charge point.
    - Availability and expected pricing should account the time-slot to be booked (that is, when the driver will reach the charge point and when will the charge process start).
- Once a charge point is selected by a driver, the HMI attempts to automatically book a time slot in the charge point.
- Once a driver reaches the charge point, the authentication/unlocking process is seamless (preferably with no intervention from the user). For example, if the position of the vehicle is accurate enough, the vehicle itself is used as an authentication token by having the navigation system selecting the appropriate nearby charging station and requests it to be unlocked through a NeMo service.
    - The feasibility of alternative authentication mechanisms will be evaluated when a vehicle's position is not accurate enough (such as RFID, Bluetooth, etc.).
- Direct or ad-hoc payment is possible at any of the charge points where a driver is authorized/able to charge his/her vehicle. The payment is handled through the driver's account considering specific payment policies and charging conditions, such as roaming. CPOs provide standard payment options via payment providers registered in the NeMo network. In addition, popular payment providers such as PayPal and others should be supported by any European CPO.
- NeMo makes the clearing process between DSOs, CPOs and other intermediaries transparent to the end user by allowing any driver to operate any compatible charging station, including roaming in different countries.

3.3.4 Actors Involved

The Actors involved in the Business Scenario 3 are: VM (Vehicle Manufacturer), eRoaming platform provider, Navigation service provider, CPO (Charge point operator), Drivers, Banking service operator, Charge Card Operators (DKV, UTA), Tax authority

3.3.5 Current situation

Currently, some independent services exist. However, there is not an integrated solution available.

3.3.6 What is expected after NeMo

Thanks to this BS, the driver will be able to automatically and seamlessly authenticate to a charge point, pay for charge ad-hoc, and book a point in a cross-border trip.

3.3.7 Nemo services to be used

The NeMo services that will be deployed in this BS are:

- User preferences storage/retrieval
- Vehicle/driver consumption profile
- Bulk charge point list
- Charge point availability
- Charge point booking
- Charge rating/pricing
- Charge point remote unlock

3.3.8 Possible Test Sites

All test sites will be involved. Different aspects of the Business Scenario may be demonstrated in different test sites.

3.3.9 KPIs

The following is a preliminary list of KPIs, mainly related to the payments and service utilization:

- Service usage after roll-out
- Number of cross-provider payments
- Network utilization increase
- Number of electronic payments x user
- Number of total payments

3.4 NeMo Service Development Environment (BS4)

3.4.1 Business Scenario short description

The NeMo Service Development Environment develops and offers new IT services (OPEN Cloud Marketplace) to the participants (Business Partners) in the NeMo Hyper-Network.

Participants (Business Partners) in the Hyper-Network are able to exchange data and services such as IT service offerings. Stakeholders, like IT service providers offer their IT service in the network, either to all or selected participants (Hyper-Network Service).

3.4.2 Objectives

The objective of the Business Scenario is to develop, offer and contract IT services and interact with other NeMo business partners.

3.4.3 Main Flow

The flow of events is focused on the IT developer and on the participants in the NeMo Hyper-Network. An IT developer uses the Open Cloud Marketplace, Service Creation domain and develops a new IT service. An example of such an IT service (application) could be:

- Calculate route optimal route to closed charge point based on battery-capacity and remaining range (data read via VM connected car API), available charging stations etc.
- Payment service to pay at charging stations by various payment providers (PayPal, HiPay etc.).
- Calculate charging plan based on grid contains and energy forecast.

The IT service developer is able to offer this IT service to the participants in the NeMo Hyper-Network, like EMP, CPO, VM or others (B2B) and directly to end-customers (B2C).

The participants in the NeMo Hyper-Network are able to search and find the IT services.

The IT service developer and participants in the NeMo Hyper-Network close contracts for usage of the IT service.

The IT service developer is paid by the participants in the NeMo Hyper-Network using the IT service.

Business partners in the NeMo Hyper-Network, could be EMP, CPO, VM, DSO and other actors. The business partners could be either acting as:

- Offering Parties offering an IT service; **Service Providers**
- Accessing Parties using an IT service: **Service Requestors**

The Use Case description uses three entities which are relevant in this Business Scenario. Those are:

- Business Partner Registry (in the context of **Partner Management** within the NeMo Hyper-Network)
- Service Registry (in the context of **Service Management** within the NeMo Hyper-Network)
- Contract Registry (in the context of **Contract Management** within the NeMo Hyper-Network)

Within the Use Case of the Business Scenario the different parts of a service are divided into two terms:

- Service Specification: Comprises the service interface information and the semantic service description (functional and non-functional) of a service.
- Service Process & Content: Comprises the service implementation part and all documentation and files related to the respective service.

3.4.4 Actors Involved

The Actors to be involved are the following: ICT service developer, ICT cloud platform, CPO, EMP, VM, DSO.

3.4.5 Current situation

Currently, this service does not exist.

3.4.6 What is expected after NeMo

The objective is to put in place the services to be run in the NeMo Hyper-Network and make it operational.

3.4.7 Possible Test Sites

The BS4 will be tested in the Italian and German test sites. However, the connection to NeMo Hyper-Network will involve all test sites.

3.4.8 KPIs

Possible KPIs could be related to the improved attractiveness of EVs achieved through a seamless and ergonomic energy management cycle (spanning the entire cycle from re-charging spot selection/reservation to plug-out after re-charging).

3.5 Horizontal services: Actors monitoring and profiling, Service finder and optimizer, Service brokerage and Rating/Pricing services (BS5)

3.5.1 Business Scenario short description

The BS will develop and deploy horizontal NeMo Hyper-Network services for electromobility actors' monitoring and profiling, service finding and optimizer, service brokerage and rating/pricing services.

The horizontal services are developed on top of the NeMo Hyper-Network in order to provide new added-on functionality usable in the whole environment. These services will provide specific functionality to be used by different kind of actors such as end users (e.g. EV driver) and IT service developers. More specifically, the horizontal services differentiate from the "vertical" services described in the BS4 by the fact that are initially developed inside NeMo and provide a set of functionalities to be used by the different actors according to their interests.

The end users will be able to look for EM services according to their preferences, common locations, trips and other real-time information. The *service finder* finds the best services candidates and proposes alternatives for those that have a big demand. Additionally, the *service brokerage* proposes better prices if the end user finally chooses different but suitable alternatives not fully aligned with his/her preferences but better for the CPO or service provider.

From the IT service developer point of view, they will be able to implement new services taking advantage of the functionalities already provided by the horizontal services. The aim is to provide a common set of tools and functionalities to the IT service developers in order to speed up the implementation on the new services as described in the BS4.

3.5.2 Objectives

The objectives of this BS are several:

- Know the user profile and preferences.
- Monitor user activity to propose better alternatives according to their common trips, services, locations and preferences.
- Disseminate an integrated offer of available electromobility services.
- Balance electromobility services demand, avoiding service congestion and promoting the less used services.
- Implement pricing strategies to enhance policies to avoid traffic congestion or promote sustainable electromobility.
- Avoid grid congestion.
- Optimize available power.
- Optimize customer energy bill.
- Avoid waiting times.

3.5.3 Main horizontal services

Actors' monitoring and profiling: This service will focus on the activity of the different electromobility actors inside the Hyper-Network, so as to estimate their activity, interests and behaviour in the electromobility environment.

- *Actor Profiling* will be based on static information directly specified by the actor as well as on inferred information extracted dynamically from its tracked activity.
- *The Actor Monitoring* service will analyse the activity of the actor in order to infer valuable information, such as mobility patterns, common places, user interests, irregular activity, etc. with the objective of generating valuable personal and non-personal information to the service optimiser and reconciliation algorithms.

Service finder The service finder will be responsible to look for available services according to the actor request. This is mainly restricted to functional parameters. Thus describing which input and output a service should provide. This search can be specialized with further specification of a precondition and effect of the service.

Service optimizer: The optimiser will be able to consider, up to date context information like the actor profile, the environmental variables or the EV status, to provide the most appropriate services ranking. This is done based on the result of the service finder. The optimizer sorts the services found by the service finder e.g. due to their location, availability or other non-functional parameters.

Service brokerage: The service brokerage domain will fit the actors' requests to existing services of the Hyper-Network, reconciling concurrent requests from different actors for the

same electromobility service provided by the Hyper-Network. It will consider restriction factors such as number of resources, capacity, demand provision, local policies, etc. These reconciliation algorithms will be connected to the service optimiser, for looking up suitable services according to actors' profiles and other restrictions.

Rating/pricing services: The relevant work will focus on the pricing (static or dynamic) of the different services. In a dynamic context, the price of each type of service could be established by its offer/demand matrix but also by other external variables like local policies. Also, the rating and pricing services will be connected with the service brokerage and optimiser to move the demand of limited resources from most to less requested ones.

The rating and pricing services will allow end-users to access a centralized listing of prices for those services that meet their service request criteria. In close collaboration with the service finder and optimizer, each service available in the NeMo Hyper-Network will be able to utilize the rating and pricing services to make available an individualized price in accordance with the dynamic context in which a request was made, including the actor from which the request originates, the type of service requested, the requested time and date of service delivery/execution, etc. Considering the available dynamic price, different actors in the NeMo Hyper-Network will be able to pick the most suitable service according to the offered price and service characteristics.

3.5.4 Main Flow

In order to show the interaction among the actors and the horizontal services, two different Sub-Scenarios are introduced next:

Sub-Scenario 1: EV drivers looking for charge points

- An EV driver needs a charge point in a specific area (defined by his/her destination, current location or other). The service profiler sends a request according to their actor profile that takes into account his/her preferences, previous choices, trips, locations and other real-time information.
- The Service finder finds the best charging points and proposes alternatives for those that have a big demand.
- The Service brokerage manages the different requests for the most demanded charging points and balances them prioritizing, for example, those less demanded.
- In order to push the user to select specific charging points, the service pricing proposes better prices for suitable alternatives maybe not fully aligned with the user preferences but better for the grid, service provider or the policy maker.

Sub-Scenario 2: CPO developing an IT service for its charge points

- The IT service developer creates service availability and service booking for its charge points.
- For each CP, he details all the expected attributes so that they may be selected on the requests according to the user profiles.
- To test the service, he uses the service finder to look for the CPs in a specific area.

- In order to promote the users to book those less used, he configures the service pricing to modify the price of the service dynamically, reducing the price for those less used and increasing it for those more demanded.

3.5.5 Actors Involved

The Actors involved in this BS are: IT service developer, IT cloud platform, CPO, DSO, EMP, eRoaming platform provider, EV Driver.

3.5.6 Current situation

Currently, the service does not exist and it constitutes a main innovation brought by the NeMo project.

3.5.7 What is expected after NeMo

The objective is to make the new NeMo horizontal services operational.

3.5.8 Needed services

There are several services that will be part of the Horizontal BS. The list of the standard services is reported here below:

- User preferences and profile
- User monitor
- Service Search
- Service Discovery
- Manage service catalogue
- Service Selection
- Services Rating
- Service Brokerage
- Dynamic tariff management
- User profiling
- eRoaming
- Big Data analytics
- Get charging station and its attributes
- Real time status info
- Transferring IDs or service identification

3.5.9 KPIs

A preliminary list of KPIs is reported here below:

- Service utilization KPIs
- Profile data inferred by the actor monitor
- Booked services with dynamic pricing
- Brokerage for most demanded services
- Number of actors profiles
- Number of actors monitored

- Optimized found services per request

3.5.10 Possible Test Sites

All test sites will be involved in this BS.

3.6 eMobility Report (BS6)

3.6.1 Business Scenario short description

Many stakeholders could be interested in e-mobility reports to assess their business/policies, such as:

1. DSO: Grid load report, to anticipate grid impact, charge need by location to know the need of energy for each location (available renewable resources).
2. Public authorities: Charge point usage, mobility patterns, to check alternative fuel directive roll out, traffic information.
3. CPO: Charge point usage rate per type of location, to identify best locations for investment, charge needs per location (i.e. possibility to subscribe a low tariff with the DSO based on monthly usage rate of CPO), constraints for the DSO on the supply / available power supply (related to smart charging for instance increase / decrease the possibility to power supply).
4. VM: mobility patterns, to better specify batteries and engines, usage rate of infrastructure (same indicator as CPO but different purpose: assure the customer about the level of re-charging).

3.6.2 Objectives

The objectives of this Business Scenario are to:

- Improve public and private infrastructure investments
- Make infrastructure more profitable for players
- Reduce traffic congestion
- Promote environmental policies
- Enable charging infrastructure planning, enable smart charging
- Integrate information from different players

3.6.3 Main Flow

The BS6 is related to BS1 and BS7. Data collection is in BS6 and data forecast and smart charging is in BS1. What data has to be forecasted is decided in BS1. Forecasts can be used by different actors (i.e. DSO to fix prices) or to have a decision making system that automatically take the data and fix the prices based on all data (i.e. based on smart charge).

In case of dynamic pricing, the price can be fixed based on forecasts (i.e. based on energy costs).

The flow of events in this BS is mainly related to the need for information specified in BS1 and BS7.

Bridge charge need from customer (SOC, charging power) with Charge detailed record, driver profile (mileage, autonomy) and grid load to perform analysis.

Collect and aggregate data from vehicles, from CPOs and on grid loads.

Roaming platforms are the link between CPO and EMP because they are gathering information on the car and the grid.

The information on the grid constraints is provided to roaming platform or to the CPO. This has to be decided when the IT architecture is discussed and set up. In France, the energy provider, instead of being connected to each CPO, it is connected to the eRoaming platform which interconnects all actors. An overview where and how data are collected is reported in the following table:

which indicator	Usage rate of ePOI, Consumption of local grid, Driver usage
which actor	OEM, eRoaming platform, DSO
what data input	Mileage, SOC, Energy delivered CP / POD (Point of Delivery)
which BS will use the data	BS1, BS6, BS7

3.6.4 Actors Involved

The Actors involved in this BS are: CPO (Charge point operator), Stakeholders interest groups, VM (Vehicle Manufacturer), Policy maker, Parking and service stations provider and Regulatory bodies.

3.6.5 Current situation

Data could be available but no organism is trying to merge them to propose reports.

3.6.6 What is expected after NeMo

The energy distributors will have access to history of CPO charge sessions/events of EVs per geographical area, history of usage of specific CPs. In this way, a better planning of future charge infrastructure may be implemented. Additionally, smart charging algorithms will have access to real time metering data in order to perform smart charging

3.6.7 Needed services

The services that will be used by this Business Scenario are:

- Calculate User profile
- Get station and attributes (dynamic status, energy delivered)
- CDR (Charge Detail Record)

- Grid load (energy / power consumption)
- Big data
- Data collection
- Data analytics

3.6.8 KPIs

Possible KPIs to evaluate the BS are the evolution of occupation rate of public charge points and the maximum intensity peak evolution at the test site.

3.6.9 Possible Test Sites

The BS6 will be tested in France where constraints are known for the local grid (i.e. behind a point of delivery). If a DSO is involved, there will be no possibility to have grid constraints above the POD unless measurement device is installed meanwhile.

In Austria, it may be possible to get information on the local grid from the DSO.

Finally, the BS will be implemented in the cross-country pilot where every test site should make a report. However, it must be noticed that it may be difficult to have grid constraints for all countries so that a solution should be found to avoid ending up with partial reports.

3.7 Vehicle Preparation for Drive Off (BS7)

3.7.1 Business Scenario short description

If the vehicle knows its drive off time, it can prepare its systems for the specific moment of the drive off. Preparation of the systems could cover all energy consuming systems. Therefore, the cabin heating, thermal conditioning of the battery and/or final battery charging to 100% SoC shall be performed, while the vehicle is still connected to the charge point. As the energy for the preparation in such cases is not taken from the battery, such preparation measures do not have influence on the driving range. The more precise the drive off estimation can be performed, the more efficient the preparation strategies can be aligned.

A fleet owner having multiple vehicles from different OEMs can push the drive off information from its own system to the vehicles via NeMo's standardized interfaces.

A private owner might use an app (developed based on the NeMo service development from a third party) that predicts his personal mobility behaviour, derives the drive off time automatically and pushes this information to the target vehicle.

The vehicle might know the preferences of the driver in order to align the preconditioning to his/her specific preferences.

3.7.2 Objectives

The objectives of this BS are to:

- Improve available vehicle performance at drive off

- Improve energy consumption from battery
- Improve user comfort experience
- Improve battery life
- Improve battery dimensioning opportunities

3.7.3 Main Flow

The flow of events can be summarised as follows:

- Vehicle is connected to charge point and is charging
- Fleet owner system or other app predicts drive off time
- Drive off time is pushed to the vehicle
- Vehicle stops charging at 99% (depending on charge point capabilities and battery capacity)
- Short before the drive off, the vehicle starts heating up the cabin including defrosting of windshield if necessary, depending on the drivers' preferences (if available)
- Short before drive off, the vehicle puts its battery into the optimal temperature range (cooling and/or heating possible), if necessary
- Short before drive off, the vehicle charges to 100%
- Vehicle is disconnected from charge point and drives off

3.7.4 Actors Involved

The Actors involved in this BS are: Vehicle, Driver, Fleet Operator, VM (Vehicle Manufacturer)

3.7.5 Current situation

Currently, approaches for such applications are partial and simplified existing as OEM specific solution; no solution with a "closed" information base (multi levels of information) is currently existent, prediction mainly requires maintenance of interfaces/preferences by customer (e.g. shared and maintained agenda) or regular behavioural patterns. No vehicle transferable (cross-manufacturer, cross-system) approach exists, meaning that especially for shared vehicles or fleet vehicles such functionalities are completely missing.

3.7.6 What is expected after NeMo

Vehicles will be automatically prepared and their usability will be enhanced by a better and interoperable information basis. Prediction services may optimize the accuracy of the predicted time of drive off and thereby enhance the potentials for an overall efficient vehicle preparation, respecting as well drivers' preferences. Additionally, the prediction service can provide relevant influence factors for an adaptation of the charging strategy (compare follow-up use-case).

3.7.7 Needed services

These following services will be implemented in the BS7:

- User preferences
- User condition (current health and physical status of the driver)

- User activity (determination of prediction)
- Battery status and charge control
- Climate status and climate control
- Vehicle status

3.7.8 KPIs

A preliminary list of evaluation criteria is reported here below:

- service usage after roll-out
- number of cross-vehicle information flows
- battery SoC after completion of driving cycle with and without NeMo preparation
- available vehicle driving range at drive off
- available vehicle driving range at end of drive
- personal comfort experience of driver
- time delta between optimal vehicle condition state and drive off

3.7.9 Possible Test Site

This BS will be tested in France.

4. Use Cases

4.1 Use Cases

This chapter presents a description of the Use Cases (UC) defined as the specific functionalities of the services proposed by NeMo. The UC have been identified based on the BS by the participants in Task 1.4 Requirements following the discussion with each BS responsible. The template for the description of the UC has been adapted from Green eMotion project. Each UC include a reference to the BS they refer to, indicated in the field "Scope and Level". The UC template includes the description of the requirements which will be further analysed in Chapter 5.

4.1.1 Convert mobility needs into charge needs (UC01)

Scope and Level	BS1-Smart Charging		
Goal in context	The aim is to translate mobility needs of an EV driver or a fleet manager into charge needs. The mobility needs can be described as: I want to go to point B from point A, starting at date X. The charge needs is the amount of energy that should be transferred to the battery to complete the itinerary.		
Preconditions	The EV driver or fleet manager, through his EMP or NSP tools, provides his/her mobility needs.		
Successful outcome	The charge needs is returned.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return the charge needs	Error message describing the error	Service unavailable; impossible itinerary
Primary actors	EMP, VM, NSP, eRoaming Platform Provider, VM		
Secondary actors	-		
Main scenario	▪ The EMP/NSP tool requests the NeMo Hyper-Network for this service giving as inputs the mobility needs and VIN or other EV identifier. ▪ The NeMo Hyper-Network routes this call to an eRoaming Platform giving inputs parameters. ▪ (Optional): The eRoaming Platform requests an NSP to get energy in Joule needed to cover an itinerary. ▪ The eRoaming Platform requests the VM backend to get vehicle real time data (SOC, preconditioning activated?) ▪ Computations and aggregations are done in VM back end and/or in eRoaming Platform to give more information about the charge needs. ▪ The charge needs is returned to the requestor.		
Requirements	▪ #ID EMP_mob_need_01: Get end users' mobility need: The EMP should get information related to the end-user mobility need (where to go? When? From where? …) in order to calculate the "Charge need" and optimise the charging process. ▪ The mobility needs must describe at least an amount of km to run or, more precisely, a trajectory to run. ▪ EV owner must have given his/her approval in order to get personal information about his EV from VM backend.		

Scope and Level	BS1-Smart Charging
	▪ An eRoaming Platform must provide this service through the NeMo Hyper-Network. ▪ VM backend must be able to provide "real time" data of an EV. ▪ End to end time constraint: ○ If no request to the vehicle (data previously uploaded by the vehicle to the VM backend): The user must have an answer within the normal app usability response time ○ If request the vehicle (OEM backend requests vehicle to get real time data): The user can have an answer longer than the normal app usability response time.

4.1.2 Search better charge offers (UC02)

Scope and Level	BS1-Smart Charging		
Goal in context	The aim is to get better charge offers proposed by CPOs according to the charge needs of the requestor, a location/perimeter, and traffic and energy grid constraints.		
Preconditions	Charge need of the requestor is defined.		
Successful outcome	Better offers according to stakeholders interests.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return a successful response	Error message describing the error	Service unavailable
Primary actors	EMP, CPO, eRoaming Platform Provider, DSO, Aggregator, Energy retailer		
Secondary actors	-		
Main scenario	▪ The EMP requests the NeMo Hyper-Network for this service giving as inputs charge needs, where the EV is and a defined location/perimeter where he/she wants to charge. ▪ The NeMo Hyper-Network routes this call to an eRoaming Platform or directly to CPO giving inputs parameters. ▪ In case of eRoaming Platform, it requests each CPO which has at least one charge point in the given perimeter to get offers according to charge needs. ▪ Each CPO returns offers related to their own constraints, energy grid constraints/opportunities and traffic constraints. ▪ In case of eRoaming Platform, it sorts and filters offers according to priorities of requestor and returns results.		

Scope and Level	BS1-Smart Charging
Requirements	▪ Send Charge-needs: The EMP may send to the eRoaming Platform (final destination is CPO), the charge-need of an EV, in order (for the CPO) to optimise the charging process (taking into account the availability of energy - constraints and opportunities). ▪ Send Charge-needs: The EMP may send to the CPO, the charge-need of an EV, in order (for the CPO) to optimise the charging process (taking into account the availability of energy - constraints and opportunities). ▪ An eRoaming Platform or a CPO must provide this service through the NeMo Hyper-Network. ▪ The eRoaming Platform must be able to detect EVs in a defined perimeter. ▪ DSO/ Aggregator/Energy retailer must be able to send constraints/opportunities to CPOs. ▪ If needed by actors, traffic congestion must be taken into account by CPOs to make offers. ▪ End to end time constraint: The user must have an answer within the normal app usability response time.

4.1.3 Search and find charge points (UC03)

Scope and Level	BS1-Smart Charging		
Goal in context	The aim is to get a detailed list of charge points located on a requested perimeter. Results must be in accordance with criteria defined by the requestor (accessible charge points, cable capacity, min/max power available, B2C price of charge, ...)		
Preconditions	Criteria and perimeter are defined		
Successful outcome	All charge points which match with criteria, and located on the requested perimeter are returned.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return a successful response	Error message describing the error	Service unavailable; perimeter too large
Primary actors	All actors		
Secondary actors	-		
Main scenario	▪ All actors are able to request the NeMo Hyper-Network for this service giving all inputs (perimeter and criteria). ▪ The NeMo network routes this call to an eRoaming Platform or directly to a CPO to get detailed list of charge points regarding inputs. ▪ In case of eRoaming Platform, it is already connected to CPOs and having real time data about charge points, return the detailed list requested to the Search and find service to the initial requestor. ▪ In case of CPO, the CPO returns detailed list requested to the Search and find service to the initial requestor.		
Requirements	▪ #EMP_CP_stat_01: Get Charge Point static data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infrastructure element description change notification (creation modification deletion of a charging point for example), it will push the new description data to the EMP, if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs with description to the Driver. ▪ #EMP_CP_dyn_01: Get Charge Point dynamic data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infrastructure element status change notification (for example the notification that a charging point switched from “Occupied” to “available”), the eRoaming Platform will push the new status to the EMP if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs and availability status to the Driver. ▪ All actors must be able to request this service. ▪ An eRoaming Platform and/or a CPO must provide this service through the NeMo Hyper-Network. ▪ End to end time constraint: The user must have an answer within the normal app usability response time		

4.1.4 Plan itinerary with charge stops (UC 04)

Scope and Level	BS2-Itinerary Planning		
Goal in context	Display an itinerary with charge stops for an electric vehicle, optimizing different environment constraints.		
Preconditions	None		
Successful outcome	The driver can see (one or more) itineraries fulfilling their search criteria.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No itinerary found	No itinerary shown to a driver	One or more restriction criteria can't be fulfilled
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, Travel and traffic Information Service Provider, Map service provider, CPO		
Main scenario	▪ The driver inputs his/her starting position to the navigation system by manually selecting it or by having it automatically detected or deduced. ▪ The driver selects a destination (manually, out of a list of predetermined locations, by letting the navigation system predict a location, etc.) ▪ Optionally: a set of restrictions and preferences is provided to the navigation system (to be defined by extending this scenario) ▪ The navigation system creates one or more itineraries, which are displayed to the driver. These itineraries take into consideration the restrictions and preferences defined by the driver and the vehicle and include at least: ▪ Reachability area ▪ Suggested charge stops ▪ The charge point selection in this scenario may be based only in the proximity to the planned itinerary. This may be extended in other scenarios.		
Requirements	▪ Battery consumption parameters should be available to actors who need to determine energy consumption rates while driving. ▪ A navigation system should be able to detect, through an API with the vehicle, the battery charge status. ▪ A navigation service provider should be able to determine a reachability area based on a vehicle's current situation (e.g.: current battery charge and status, driving environment, temperature, etc.). ▪ A navigation service should be able to plan an itinerary between two points including charge stops in the route. ▪ A preference model should be available to influence the selection of charge points along the route (e.g.: prefer cheaper CPs, prefer closer CPs, prefer faster CPs, etc.) ▪ An EMP/eRoaming platform provider should be able to provide a list of charge points, including its relevant attributes such as charger types, in a given area. ▪ An eRoaming platform provider should be able to estimate the time it would take to recharge a vehicle's battery up to a certain percentage of total capacity. ▪ An eRoaming platform provider should be able to estimate the waiting time to use a charge point. ▪ A navigation service provider should be able to determine an estimated total trip time for an itinerary, when possible (e.g.: it may be impossible to determine the waiting time for a charge point)		

4.1.5 Change/switch/plan energy tariffs (UC05)

Scope and Level	BS1-Smart Charging		
Goal in context	This service is used by DSO/Aggregator/Energy retailer/TSO to switch energy tariff for CPO from level A to level B, or to plan energy tariff levels. Each tariff level could be fixed in an agreement signed by DSO/Aggregator/Energy retailer/TSO and CPO.		
Preconditions	Energy tariffs are set in the B2B contract between DSO/Aggregator/Energy retailer/TSO and CPO.		
Successful outcome	Energy tariff level is switched or planning is updated.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return a successful response	Error message describing the error	Service unavailable;
Primary actors	CPO, DSO, Aggregator, Energy retailer, TSO, eRoaming Platform Provider		
Secondary actors	-		
Main scenario	▪ The DSO/Aggregator/Energy retailer/TSO requests the NeMo network to set the planning of tariff levels or to switch from level A to level B. ▪ The NeMo network routes this call: o To the CPO directly which takes into account the change and answer to the requestor. ▪ or o To an eRoaming Platform which is already connected to CPOs. o The eRoaming Platform requests the CPO which takes into account the change and responds. o The eRoaming Platform responds to the requestor.		
Requirements	▪ CPO must agree to accept and take into account tariff information from the DSO/Aggregator/Energy retailer/TSO. A contract could be signed by the two parts to agree on the process: set a tariff planning or switch in real time the tariff level. ▪ Grid local perimeter/network must be defined in order to switch tariff level/set tariff planning on localised perimeters which are relevant for DSO/Aggregator/Energy retailer/TSO. ▪ End to end time constraint: The requestor must have an answer within the normal app usability response time.		

4.1.6 Provide traffic information (UC06)

Scope and Level	BS1-Smart Charging		
Goal in context	The aim is to provide information on traffic regarding a location. Information on traffic can be an indicator of the traffic congestion at this location (blue, green, red) or more specific information like traffic events or ETAs on itineraries at a specified time, average speed in real time compared to speed limits at a location. This information is used by IT suppliers, Public authorities and CPOs close to policies		
Preconditions	An NSP, MSP and/or Public authorities are able to provide these information.		
Successful outcome	The requester has information about traffic consistent with his request		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return a successful response	Error message describing the error	Service unavailable
Primary actors	CPO, IT Suppliers, Public authorities		
Secondary actors	-		
Main scenario	CPO, IT Suppliers or Public authorities pulls these information: ▪ The CPO, IT Suppliers or Public authorities requests the NeMo Hyper-Network for this service giving as inputs the location or search area for which traffic information is needed. ▪ The NeMo Hyper-Network routes this call to the IT Suppliers or Public authorities which answers with traffic information on the location requested.		
Requirements	▪ NSP and/or MSP and/or Public authorities must have this information and are able to provide it through services. ▪ End to end time constraint: The requestor must have an answer within the normal app usability response time.		

4.1.7 Push Charge Plan (UC07)

Scope and Level	BS1-Smart Charging		
Goal in context	The calculated charge-plan is being pushed (uploaded) from the NeMo Network to the Electric Vehicle (EV) via the VM.		
Preconditions	The charge-plan has been calculated based on users' needs and energy grid constrains by the EMP or any other 3rd Party providing this service to end-users.		
Successful outcome	The charge-plan is pushed to the EV in order to operate independently of any communication-problem to any server while charging.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Charge-plan could not be pushed to the EV via the VM	Error message is returned and no off-line operation for charging is possible	The communication link between the EV to the VM could not be established
Primary actors	End-User, EMP, VM		
Secondary actors	eRoaming Platform in case EMP, VM or CPOs are connected to		
Main scenario	1. The EMP requests the charge-plan for the EV based on users' needs and energy-grid constrains 2. Based on the technical capabilities of the EV the EMP sends the charge-plan via the VM to the EV. 3. The EMP sends the charge-plan via the VM 4. The VM pushes the charge-plan and returns a message back to the EMP if the charge-plan has been pushed to the EV successfully or not 5. The EMP returns an appropriate a message back to the end-user		
Requirements	▪ The EV must be connected to the VM (OEM) backend ▪ VMs must provide standard API to upload Charge Plans (e.g. ISO 20078 Extended Vehicle Standard) ▪ Secured service transactions between VM with EMP for billing and payment. End to end time constraint: The requestor can have an answer longer than the normal app usability response time.		

4.1.8 Propose energy grid constraints/opportunities (UC08)

Scope and Level	BS1-Smart Charging		
Goal in context	The aim is to provide a service to DSO/Aggregator/Energy retailer/TSO to submit energy constraints/opportunities to CPO in order to enhance grid congestion. A constraint/opportunity could be: increase or decrease power of X kW on this grid sub network or provide me Y kW using Veh2Grid capabilities.		
Preconditions	CPO and DSO/Aggregator/Energy retailer/TSO agree to use this feature (contract signed?)		
Successful outcome	CPO has accepted or not the constraint/opportunity		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return a successful response	Error message describing the error	Service unavailable
Primary actors	CPO, DSO, Aggregator, Energy retailer, eRoaming Platform		
Secondary actors	-		
Main scenario	▪ DSO/Aggregator/Energy retailer/TSO requests NeMo network to propose a constraint/opportunity to a CPO on a grid sub network. ▪ NeMo network routes the call: ○ Directly to the CPO which analyses, according to cars connected to its charge points network on this grid sub network, if he can manage with this proposal. ○ CPO answers yes or no (and give more information?) to the requestor ▪ or ○ To an eRoaming Platform which routes the request to the targeted CPO. ○ The CPO analyses, according to cars connected to its charge points network on this grid sub network, if he can manage with this proposal. ○ CPO answers yes or no (and give more information) to the eRoaming Platform which answers to the requestor.		
Requirements	▪ DSO/Aggregator/Energy retailer/TSO must agree with CPO to use this service. ▪ CPO and/or eRoaming Platform must provide this service through NeMo Hyper-Network. ▪ Charge infrastructure of the CPO must be able to be piloted by CPO to answer constraints of DSO/Aggregator/Energy retailer/TSO; or CPO must be able to push a charge plan to the vehicle (see UC7) End to end time constraint: The requestor can have an answer longer than the normal app usability response time.		

4.1.9 Push a CDR from EMP to VM (UC9)

Scope and Level	BS1-Smart Charging		
Goal in context	Provide a service which improves the battery durability		
Preconditions	Some information and requirements are needed: • History of driver's charging sessions • The CDR (Charge Details Record) is sent by the CPO at the end of the charging session • The EMP is able to communicate with the VM		
Successful outcome	The CDRs is sent to the OEM which analyses it to calculate a good steering strategy for battery durability		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The CDRs are not sent to the VM	The steering strategy is not calculated	No communication between EMP and VM
Primary actors	VM, EMP, CPO		
Secondary actors	eRoaming platform provider		
Main scenario	• The driver realizes several charging sessions • At the end of each charging session, the CPO sends the CDRs to the EMP and EMP sends it to the VM • The VM analyses the CDRs and calculates a steering strategy to improve the battery durability, for instance a SOC range to respect (20%-80%) or a maximum duration at 100% SOC to preserve cells chemistry		
Requirements	• The CPO shall be able to send CDR to the EMP • The EMP shall be able to send the CDR to the VM		

4.1.10 Push charging strategy from OEM to CPO (UC10)

Scope and Level	BS1-Smart Charging		
Goal in context	When the charging strategy to improve the battery durability is calculated, the VM communicates it to the CPO		
Preconditions	• Some information and requirements are needed • Charging strategy calculated by the VM • The CPO can receive the strategy and consider it as another parameter in the smart charging process		
Successful outcome	• The CPO takes into account battery constraints in the smart charging process		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**

Scope and Level	BS1-Smart Charging		
	CPO doesn't consider the battery durability as a constraint	The battery have a lower lifetime	The charging strategy is not sent to the CPO
Primary actors	VM, CPO		
Secondary actors	eRoaming platform provider		
Main scenario	• The VM have calculated the charging strategy thanks to the charging historical • The charging strategy is sent to the CPO from VM • The CPO takes into account the charging strategy in his algorithm of smart charging. He will have 3 constraints: electric grid, mobility need and battery constraints (limited SOC, power, etc.)		
Requirements	• The VM shall be able to send the charging strategy to the CPO • The CPO shall be able to integrate the battery constraints in his model of smart charging • The driver shall be able to choose between two modes: one which prioritizes the mobility needs (charging speed and full range), the other the battery durability		

4.1.11 Plan itinerary with safety features (e.g. tunnel detection) (UC11)

Scope and Level	BS2-Itinerary Planning		
Goal in context	BS2		
Preconditions	Extension of use case "Itinerary planning with charge stops" considering itineraries where the driver's, vehicle's or infrastructure's safety might not be ensured.		
Successful outcome	A basic itinerary as described in the base use case can be successfully planned. The driver follows the planned itinerary (that is: the vehicle follows the route planned by the navigation system).		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Itinerary becomes invalid	No itinerary shown to a driver / safety features become invalid or disabled	The vehicle deviates from the planned/predicted itinerary, invalidating it. The use case may be re-triggered on real time in this case.
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, Travel and traffic Information Service Provider, Map service provider, CPO		

Scope and Level	BS2-Itinerary Planning
Main scenario	With an itinerary planned, the navigation system detects the predicted reachability area. The navigation system gathers from other systems real-time information to improve the reachability area estimates, such as traffic, weather, etc. Using information from the Travel and traffic Information Service Provider and from the Map service provider, potential safety critical situations near the end of the reachability area are detected, such as: o Tunnels o Highways without an emergency lane The safety critical situations are displayed through the HMI. An alternative itinerary may be offered to avoid the safety critical situation. This new itinerary may avoid the situation if at all possible, or offer a charge-stop to ensure the vehicle won't run out of battery while in an unsafe situation.
Requirements	▪ Maps should allow the detection of security critical features, such as tunnels and highways with no emergency lanes. ▪ The navigation system's HMI should display safety critical situations, such as running out of charge near a tunnel. ▪ The itinerary should be available for a driver to depart on short notice (i.e. planning an itinerary shouldn't take more than 30 seconds)

4.1.12 Interoperability: Display charge point real-time availability (UC12)

Scope and Level	BS2-Itinerary Planning, BS3-Cross-provider/border booking authorization and payment management		
Goal in context	Improve the list of charge points displayed/offered to a driver by considering real-time restrictions such as availability and compatibility.		
Preconditions	▪ Charge points to be considered in this scenario offer a service to determine their availability on a specified time and date. ▪ A charge point is considered to be available when a customer may use it for charging and its associated parking spot is not occupied. ▪ The case where a CP is not booked (i.e. considered available) but its parking spot is occupied by a vehicle not charging is considered out of scope.		
Successful outcome	The driver can see a list of available charge points compatible with his/her dynamic (e.g.: time, place, charge needs) and static (e.g.: plug type) restrictions.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No availability service	Charge point won't display real time availability	Charge point doesn't provide an availability service which lets NeMo applications verify its availability on a certain time-slot.
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, Travel and traffic Information Service Provider, Map service provider, CPO		

Scope and Level	BS2-Itinerary Planning, BS3-Cross-provider/border booking authorization and payment management
Main scenario	▪ The navigation system retrieves a list of charge points relevant to the driver's needs: ○ Nearby the planned itinerary, if one is available. ○ Near the end of the reachability area, if no itinerary is available. ○ in the area of a displayed map, if no other information is available. ▪ The list of charge points is further refined considering static criteria, such as connection compatibility. ▪ The list of charge points is further refined considering dynamic criteria, such as real-time availability or by the expected arrival time (if applicable). ▪ An extension point of this scenario may be offered to present extra information and services to a driver, such as price estimation and charge point booking.
Requirements	▪ An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes, such as charger types, in a given area as POIs (point of interest) in a map. ▪ An EMP/eRoaming platform provider should be able to show the real-time availability of a charge point (optionally, the availability of its associated parking spot may be omitted when considering a charge point available). ▪ An EMP/eRoaming platform provider should be able to show the availability of a charge point in a future date for a certain period of time (optionally, the availability of its associated parking spot may be omitted when considering a charge point available). ▪ An eRoaming platform provider should be able to estimate the waiting time to use a charge point. ▪ A preference model should be available to influence the selection of charge points along the route (e.g.: prefer cheaper CPs, prefer closer CPs, prefer faster CPs, etc.) ▪ The HMI should be able to display a list of charge points as POIs (point of interest), with the list of their attributes and their availability. ▪ Charge point information should reflect the current status of the network to provide meaningful information to a driver.

4.1.13 Monitor reachability and dynamically suggest charge saving alternatives (UC13)

Scope and Level	BS2-Itinerary Planning		
Goal in context	Make an EV driver aware of the alternatives he/she can use to extend the effective range of his/her vehicle.		
Preconditions	An itinerary is planned or predicted.		
Successful outcome	The driver can evaluate different charge saving measures and their impact on the itinerary.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No alternatives available	No alternatives shown to user	There are no charge saving measures a driver could realistically undertake.
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, Travel and traffic Information Service Provider, Map service provider, CPO		
Main scenario	▪ The navigation system detects when the battery state is not enough to complete the planned/predicted itinerary. ▪ A list of charge-saving actions is created according to a dynamic context model, for example: ○ The effect of turning the AC or heating unit off ○ Choosing an alternate route when a traffic jam is detected ▪ The list of charge-saving actions and their impacts on the itinerary (e.g.: change in travel time) is displayed to the driver through an HMI.		
Requirements	▪ The navigation system should automatically detect when the battery status is not enough to complete a trip. ▪ The navigation system should automatically detect when dynamic conditions reduce the reachability area (e.g.: wind conditions require greater energy usage than anticipated). ▪ Charge saving measures should be proposed to the driver when the reachability area does not guarantee reaching the destination		

4.1.14 Optimize itineraries (UC14)

Scope and Level	BS2-Itinerary Planning		
Goal in context	In an EV itinerary with multiple charge stops, optimize the charge plan according to a user-defined goal.		
Preconditions	-		
Successful outcome	The driver can see (one or more) itineraries and the savings for each alternative.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No itinerary found	No itinerary shown to a driver	One or more restriction criteria can't be fulfilled
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, Travel and traffic Information Service Provider, Map service provider, CPO		
Main scenario	Offer different goals for itinerary optimization: ▪ Optimize an itinerary for the total travel time by, for example: ○ Prefer charge stations with a lower predicted waiting time. ○ Prefer charge stations which can be booked, to ensure the minimal waiting time. ○ Choose a route that is longer but has faster chargers. ○ Stop charging early: notify a driver when a partial charge might be enough to reach the next charge station (for example, in case the second charge station is faster than the first one). ▪ Optimize an itinerary for a cheaper plan: ○ Prefer charge points with (possibly predicted) lower expected charge. ○ Prefer charge points with a cheaper parking fee. ○ Avoid charge points for which prices can't be estimated. ▪ Optimize an itinerary to reduce waiting in charge points ○ Prefer charge points which can be booked, to ensure the minimal waiting time. ○ Choose faster chargers.		
Requirements	▪ The navigation system should provide alternative itineraries according to different (and customizable) criteria ▪ The driver should be able to select which optimization criteria to use when creating an itinerary (e.g.: optimize for cheaper routes, faster routes, etc.). These estimations should account for charging times, prices, etc. ▪ An EMP or eRoaming platform provider should be able to estimate the time it would take to recharge a vehicle's battery up to a certain percentage of its total capacity. ▪ An EMP or eRoaming platform provider should be able to estimate the waiting time to use a charge point. ▪ An EMP or eRoaming platform provider should be able to estimate the price a driver will pay for charging in a certain location during a certain timeslot. (NB: cross-border payment estimations may be left out of scope).		

4.1.15 Interoperability: Book a charge point (UC15)

Scope and Level	BS1-Smart Charging, BS2-Itinerary Planning, BS3-Cross-provider/border booking authorization and payment management		
Goal in context	Reserve a charge point so that drivers can reduce their waiting time (ideally, eliminate waiting times).		
Preconditions	▪ The CP has an API through which an external actor can reserve a timeslot for usage. ▪ When a CP has been booked, other drivers won't be able to make use of it. ▪ Reserving a CP implies reserving both the station for usage and its associated parking spot. ▪ Cases of vehicles parked in the associated parking spot of a charge point without having booked it won't be considered in this use case. ▪ Drivers will be assumed to stop using the charge point and vacate the associated parking spot after their reservation expires.		
Successful outcome	A charge point can be booked and is therefore reserved exclusively for the driver who requested the booking.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Charge point not available.	Reservation can't be booked. Usage of the charge point is not guaranteed.	Charge point wasn't available or no booking mechanism exists.
Primary actors	CPO		
Secondary actors	EMP, eRoaming Platform Provider, CPO		
Main scenario	▪ Through an HMI, the driver is able to select a charge point (displayed as a point-of-interest or POI) in a map. ▪ The HMI should make a list of actions available for the selected charge point, in which it should be possible to check for the charge point's availability in a specific timeslot [see use case "Interoperability: Display charge point real-time availability"]. ▪ Through an API, it should be possible to submit a reservation to the CPO operating the charge point selected by the driver, for the specified timeslot. ▪ The CPO should save this reservation request and make the charge point unavailable to any other driver. ▪ Once the driver approaches the charge point in the requested timeslot, the driver should be able to unlock the said charge point according to his/her reservation [see use case "Authenticate/unlock a charge point using a vehicle as an authentication token"] ▪ The charge point should cease to be available once the timeslot requested by the driver expires.		

Scope and Level	BS1-Smart Charging, BS2-Itinerary Planning, BS3-Cross-provider/border booking authorization and payment management
Requirements	▪ It should be possible for a CPO to reserve a timeslot in a charge point. ▪ An EMP or eRoaming platform provider should be able to request a charge point to be reserved through a service available in the Hyper-Network ▪ An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes such as charger types, in a given area as POIs (point of interest) in a map. ▪ An EMP/eRoaming platform provider should be able to show the availability of a charge point in a future date for a certain period of time (optionally, the availability of its associated parking spot may be omitted when considering a charge point available). ▪ An EMP/eRoaming platform provider should be able to show the real-time availability of a charge point (optionally, the availability of its associated parking spot may be omitted when considering a charge point available). ▪ It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver, provided that the actor has some kind of authorization or authentication token which proves that the driver has granted permission to this actor to do so. ▪ The reservation process should work on real time to ensure a booking can be timely confirmed.

4.1.16 Interoperability and roaming: Seamlessly manage payments (UC16)

Scope and Level	BS3-Cross-provider/border booking authorization and payment management		
Goal in context	Manage a driver's payments seamlessly, so that no interaction with a charge point is necessary after charging a vehicle.		
Preconditions	The navigation system can access the driver's EV account/credentials. The driver is charging his/her vehicle in a charge point.		
Successful outcome	A navigation system can handle payments after charging without the driver's explicit intervention.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No payment method available	Charge can't be initiated	The driver has no valid payment method/no payment method accepted by CPO
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, CPO		
Main scenario	▪ A driver is charging his/her vehicle in a charge point. ▪ Upon charging completion, or whenever the driver wishes to stop charging, he/she unplugs the charger. ▪ The navigation system, with some kind of knowledge of the driver's account with the EMP, settles any necessary payment through the EMP.		
Requirements	▪ It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver, provided that the actor has some kind of authorization or authentication token which proves that the driver has granted permission to this actor to do so. This should also include payments. ▪ A navigation system should be able to detect, through an API with the vehicle, the battery charge status. ▪ A navigation system should be able to detect, through an API with the vehicle, when the driver is preparing to drive-off after being parked. ▪ It should be possible to get a report of charge delivered from the CPO to other actors, possibly through an eRoaming platform provider, and the price to be paid by the driver. ▪ The vehicle's position should be updated in real time. The authentication and payment process shouldn't take more than a few seconds, so that a driver never has to wait on the payment process.		

4.1.17 Authenticate/unlock a charge point using a vehicle as an authentication token (UC17)

Scope and Level	BS3-Cross-provider/border booking authorization and payment management		
Goal in context	Seamlessly unlock a charge point with minimal driver interaction.		
Preconditions	▪ The vehicle is using a navigation device with sufficient positioning accuracy. ▪ The navigation system can access the driver's EV account/credentials. ▪ The charge point can be remotely unlocked. ▪ The charge point is compatible with the vehicle.		
Successful outcome	The driver can begin charging his/her vehicle without interacting with the charge point.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Positioning not accurate enough	Charging can't be seamlessly initiated	The positioning service is not accurate enough to uniquely identify a charge point.
Primary actors	EV Driver		
Secondary actors	EMP, eRoaming Platform Provider, Navigation service provider, CPO		
Main scenario	▪ An EV driver parks his/her vehicle in a spot with a charge point. ▪ The navigation system, by detecting the vehicle's parked state, its position and its proximity to a charge point, offers the driver an option to unlock the charge point. ▪ The navigation system, through an EMP, requests the appropriate charge point/plug to be unlocked. ▪ After notifying the driver of a successful charge point unlock via the HMI, the driver can begin charging his/her vehicle.		
Requirements	▪ It should be possible to unlock charge points using a common API available through the Hyper-Network. A driver should be able to immediately start charging his/her vehicle after it has been unlocked using this API. ▪ It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver; provided that the actor has some kind of authorization or authentication token which proves that the driver has granted permission to this actor to do so. ▪ A navigation system should be able to detect, through an API with the vehicle, when the driver has parked his/her vehicle. ▪ A navigation system should be able to detect, through an API with the vehicle, the battery charge status. ▪ An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes such as charger types, in a given area as POIs (point of interest) in a map. ▪ The navigation system should be able to detect the position of the vehicle with sufficient accuracy to determine when it is parked near a charge station. ▪ The vehicle's position should be updated in real time. The authentication and unlock process shouldn't take more than a few seconds, so that a driver never has to wait on the unlock process.		

4.1.18 Search and Select Service (UC18)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	Find the best fitting service regarding the search criteria of a service request.		
Preconditions	The searching entity is part of the NeMo Network Service. A semantic service description and the service have been deployed in the NeMo Hyper-Network.		
Successful outcome	The requesting partner receives a list of services ordered according to their relevance regarding the request.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No service is found	Display if the request has been badly formatted or if no service satisfying the search criteria are found.	Malformatted request or no services available.
Primary actors	IT Service Developer or other Business Partners, such as Service Providers		
Secondary actors	-		
Main scenario	There are two possible scenarios: ▪ A service developer provides a search request to find a fitting service for his service composition and integrates the selected service at design time. ▪ A service composition contains a request template and the Service Finder is invoked during runtime to find a fitting service to be executed. In both cases the Service Finder accesses the Service Store in order to get all available services.		
Requirements	The services as well as the request are semantically annotated with OWL-S and the domain model of the NeMo Hyper-Network.		

4.1.19 Call of Service (UC 19)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The Service Requester performs a call of a contracted service and receives a result / response.		
Preconditions	The Service Requester has a valid Service Contract (state = active) with the Service Provider.		
Successful outcome	The Service Requester receives the result of a service call from the contracted Service Provider.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Invalid request	Create an error response message and a log entry.	The service request is not compatible with the service request.
Primary actors	Service Requester		
Secondary actors	Service Provider		
Main scenario	1. The Service Requester makes a request based on the Service Interface Specification available in the "Service Management" database 2. Based on the "Service Management" entry the request is forwarded to the contracted Service Provider 3. The Service Provider fulfils the request OR returns an error message 4. A Service Transaction is created for accounting purpose 5. The response is send back to the Service Requester		
Requirements	▪ Data integrity must be guaranteed ▪ Secured service transactions between Service Provider and Service Requester for billing and payment.		

4.1.20 Search and Select Service Contracts (UC20)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	Find Service Contracts that satisfy search criteria of owned and contracted services.		
Preconditions	The actor searching for Service Contracts has the access right from the Business Partner (Service Provider or Service Requestor) to search for.		
Successful outcome	The business partner receives an ordered list that meets the search criteria. This can be: a. Valid or terminated Service Contracts b. Active or suspended Service Contracts c. Contracts which are expired or close to prolongation		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No Service Contracts found	Return and display an error message	There are no Service Contracts that meets the search criteria
Primary actors	Business Partner (Service Provider or Service Requestor)		
Secondary actors	-		
Main scenario	a. The actor (Business Partner) enters the search criteria b. The list generated contains valid Service Contracts that satisfy search criteria c. The actor is able to select one Service Contract from the list		
Requirements	The actor needs to have the access right (role) to perform the search and select contracts function		

4.1.21 Register Service (UC21)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The IT Service Developer should be able to register a developed service in the NeMo Hyper-Network.		
Preconditions	Such a service is not yet registered.		
Successful outcome	The service is registered in the distributed Services Registry (Service Management) and all assets of the service (Service description, ontologies, resources, documentation ...) are available within the NeMo Hyper-Network to all business partners in the network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service has not been published because of missing assets.	Error message describing the missing resource or asset.	Missing obligated asset or resource.
Primary actors	IT Service Developer or Service Providers, such as the Electro Mobility Service Provider (EMP)		
Secondary actors	-		
Main scenario	▪ The IT Service Developer or the EMP requests the registration of a Service ▪ Service assets are available for the NeMo Hyper-Network. ▪ The service is available to be referenced in a contract ▪ The service switches into the live cycle phase Published		
Requirements	The NeMo network must be built and based on a decentralized architecture. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. One service description might reference multiple equivalent services. The equivalent services are listed in the grounding (technical location of the service) of the service description. Models (e.g. Ontologies) used to describe services and service parameters are made available by the "Open Cloud Marketplace" to all partners.		

4.1.22 Start/Stop Service (UC22)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A Service can be temporarily disabled.		
Preconditions	1. A Service is selected 2. The Service is available and published		
Successful outcome	The status of the Service has changed "Started" or "Stopped"		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Status error	Return an error message	Service was not in status "Published"
Primary actors	Service Provider		
Secondary actors	-		
Main scenario	1. The Service Provider starts or stops an IT service 2. The status of the IT service changes (started or stopped) 3. Business partners do not need to be informed as the pending service transactions has to be completed before service gets restarted.		
Requirements	▪ Role based access that only authorized persons are allowed to start or stop an IT service ▪ Current and pending transactions must not be adversely affected as the service is offline (stop to start) for a very short time		

4.1.23 Delete Service (UC23)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A service no longer provided should be removable by the Service Provider. The service can only be deleted when there are no valid service contracts in place with other Service Requestors.		
Preconditions	The service was available and offered in the NeMo Hyper-Network No valid service contracts with other Service Requestors exist Service Contract is in "Inactive" state		
Successful outcome	The service is removed, all its assets are removed and there are no more contracts referencing the service.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service is not deleted because there are still contracts (service compositions) referencing the service or Service Contract is in "Active".	A message is displayed which contract or service composition is still using the service which should be deleted.	Use of service.
Primary actors	IT Service Developer or Service Provider		
Secondary actors	-		
Main scenario	The IT Service Developer selects the service to be deleted It will be checked if there are no valid service contracts with other Service Requestors in place OR the Service Contract is in "Inactive" state The service is being removed and no references for the service are found anymore. All assets (service description, documentation, resources...) referenced by the service are checked if they still are referenced by other services. If not, they are removed as well. The Service Registry (Service Management) is being updated that this service has been removed All service contracts regarding the service are being removed from the Contract Registry (Contract Management).		
Requirements	Smart contracts are verifiable, signed and encoded in ANY programming language. For each programming language a run-time container is used. The shared services metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Services Store", and accessible to all partners in the network. Service metadata includes semantic and QoS information about the services. The system should provide the means for automatic deletion of data, when the latter are no longer needed, in compliance with standards and data protection regulations in effect. One service description might reference multiple equivalent services. The equivalent services are listed in the grounding (technical location of the service) of the service description.		

4.1.24 Search and Select Business Partner (UC24)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A business partner in the NeMo Hyper-Network is able to search for any other partners of the NeMo Hyper-Network and view their profile information.		
Preconditions	All partners have entered their profile information. Each service is referenced to a business partner who is responsible as a service provider.		
Successful outcome	A list of partner profiles of the NeMo Hyper-Network is provided.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No partner has been found.	Empty list of partners	The search criteria are not fitting to the NeMo Hyper-Network partner profiles.
Primary actors	Business Partner		
Secondary actors	-		
Main scenario	1. A NeMo business partner enters his search criteria according to different criteria, such as role, domain, country, etc. 2. The Business Partner Registry (Partner Management) is being accessed with the search criteria 3. It returns a list of fitting profiles, which is being displayed to the requester. 4. The searching partner can select a specific profile and view details, such as provided services		
Requirements	▪ All partner profiles of the NeMo Hyper-Network are available within the Partner Management. ▪ There is a filter mechanism by the Partner Management to search according to specific parameters.		

4.1.25 Create Business Partner Account (UC25)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A new partner is registered in the NeMo Hyper-Network Partner Management.		
Preconditions	The NeMo Hyper-Network does not yet know this partner; the requester is authorized to become a partner in the NeMo Hyper-Network.		
Successful outcome	The Business Partner account is created and all profile information are described. The Business Partner can be found via the search by any other partner in the Hyper-Network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Account is not created because it already exists.	Account is not created	Account already exists. An account is unique and cannot be created twice.
	Unauthorized request	Account is not created	Requester is not authorized becoming a partner of the Hyper-Network.
Primary actors	NeMo Alliance, Business Partner		
Secondary actors	-		
Main scenario	▪ A Registration Request Form (company data and responsible person) is filled and submitted by a legal authorized representative of the Business Partner. ▪ The new profile is checked for completeness. ▪ NeMo Alliance checks and ensures that the legal entity has the rights to join the Hyper-Network ▪ NeMo Alliance grants access to the network for the new Business Partner. ▪ The profile is published into the decentralized Partner Management of the NeMo Hyper-Network by the NeMo Alliance ▪ All Business Partners in the NeMo Hyper-Network are informed about a new partner joining.		
Requirements	▪ The Business Partner is not yet registered. ▪ All profile information of the business partner has to be filled in during the registration. ▪ Participants in the network are known and trusted. ▪ The NeMo network must be built and based on a decentralized architecture. ▪ "NeMo Alliance" grants access to participants to access the NeMo network. ▪ Participants are legal entities. Such as EMP, CPO, etc. who wants to access the NeMo network and to exchange transaction records with the participants. ▪ The shared partner metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Business Partner Management", and accessible to all partner in the network.		

4.1.26 Change Business Partner Account (UC 26)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The account information should be updated if information about the Business Partner change.		
Preconditions	The Business Partner is a registered NeMo Hyper-Network partner. The partner's profile exists. The partner is the owner of the business account.		
Successful outcome	The account information is updated and propagated on the decentralized Partner Management of the NeMo Hyper-Network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No update is made	Error message is displayed showing the appropriate contract.	Account information conflicts with existing contracts.
Primary actors	Business Partner		
Secondary actors	-		
Main scenario	▪ NeMo Hyper-Network business partner updates account information. ▪ If the change on the account information is relevant for contracts, all contracts regarding this business partner are checked for inconsistency and contract partners are notified. ▪ Updated information is propagated into the decentralized Partner Management within the NeMo Hyper-Network.		
Requirements	There should be NO central system or any single-point in the network to support scalability and availability. "NeMo Alliance" grants access to participants to access the NeMo network.		

4.1.27 View Business Partner Account (UC27)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	Partners of the NeMo Hyper-Network should be able to view information about any other NeMo business partners. This information includes the business account as well as all provided services and QoS parameters of these services.		
Preconditions	All NeMo Business Partners have a business account.		
Successful outcome	The NeMo Hyper-Network partner account information is displayed and can be browsed e.g. the services contracts are listed and each service profile including QoS can be viewed.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Account is not available	Nothing to display	Account information is not available.
Primary actors	Business Partner		
Secondary actors	-		

Scope and Level	BS4-NeMo Service Development Environment
Main scenario	▪ A NeMo business partner has already searched and selected a profile of another NeMo business partner ▪ The partner profile is retrieved from the Partner Management and displayed including the related services contracts (Contract Management) and its respective conditions.
Requirements	The shared partner metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Business Partner Management", and accessible to all partners in the network.

4.1.28 Activate Business Partner Account (UC28)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The NeMo Alliance can activate a partner account in the NeMo Hyper-Network.		
Preconditions	The account does not yet exist.		
Successful outcome	The account is registered to Business Partner Management and distributed within the NeMo Hyper-Network. All account information is available to all partners in the NeMo Hyper-Network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The account is not created	Error message describing the conflict is displayed including	Conflicting account information.
Primary actors	NeMo Alliance		
Secondary actors	-		
Main scenario	▪ The NeMo Alliance receives a request for registration of a new Business Partner ▪ In case of approval: The account is published into the decentralized Partner Management of the NeMo Hyper-Network		
Requirements	The NeMo network must be built and based on a decentralized architecture. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. One service description might reference multiple equivalent services. The equivalent services are listed in the grounding (technical location of the service) of the service description. Models (e.g. Ontologies) used to describe services and service parameters are made available by the "Open Cloud Marketplace" to all partners.		

4.1.29 Deactivate Business Partner Account (UC29)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The business account should be no longer available within the NeMo Hyper-Network. This includes notifying all service users of the soon to be removed partner account. All contracts referencing this account should be removed as well.		
Preconditions	The account exists in the NeMo Hyper-Network.		
Successful outcome	The account is removed, all its assets are removed and there are no more contracts referencing the services of this account.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The account is not deleted because there are still contracts (service compositions) referencing the account.	A message is displayed which contract or service composition or contract is still using one of the services of the account which should be deleted.	Use of service.
Primary actors	Business Partner, like Service Provider (e.g. EMP)		
Secondary actors	NeMo Alliance		
Main scenario	The NeMo Hyper-Network Business Partner selects his account to be deleted The NeMo Hyper-Network does search for all references to this account (e.g. in contracts and service composition) The service consumers are notified by the NeMo Hyper-Network of the pending deletion. The account enters the life cycle "pending deletion". The account is removed after a deadline has been reached or no references for the service are found anymore. All referencing (service still using the deleted) services enter the life cycle phase "Pending update" All assets (Service description, Documentation, Resources....) referenced by the account are checked if they still are referenced by other services. If not, they are removed as well. All contracts regarding the account are suspended and their contract partners are notified.		
Requirements	Smart contracts are verifiable, signed and encoded in ANY programming language. For each programming language a run-time container is used. The shared services metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Services Store", and accessible to all partner in the network. Service metadata includes semantic and QoS information about the services. The system should provide the means for automatic deletion of data, when the latter are no longer needed, in compliance with standards and data protection regulations in effect. One service description might reference multiple equivalent services. The equivalent services are listed in the grounding (technical location of the service) of the service description.		

4.1.30 Create Service Contract Offering (UC 30)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	During the service registration process, a Service Contract Offering is being created from a Service Contract Template Framework by the Offering Party (Service Provider). The contract template does not contain pricing information.		
Preconditions	A Service Contract Template Framework is being provided by the NeMo Alliance.		
Successful outcome	A new Service Contract Offering is being created.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service Contract data is incomplete.	Display an error message.	Not all fields from the Service Contract template are entered.
Primary actors	Offering Party (Service Provider)		
Secondary actors	-		
Main scenario	▪ The Offering Party chooses all required sections and additionally optional sections from the Service Contract Template Framework ▪ The chosen sections configure a new Service Contract Offering ▪ The new Service Contract Offering is published within the NeMo network.		
Requirements	▪ Access: Role based authorization required ▪ Data security: Data integrity must be guaranteed; Data confidentiality must be ensured ▪ Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required ▪ Performance: Sub-seconds response time for service transactions ▪ Availability: 24x7 ▪ Scalability: Able to grow with market demand and increase of connected IT systems		

4.1.31 Create Service Process & Content (UC31)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The Service Creation domain of the Open Cloud Marketplace lets the developer specify the implementation (e.g. as a composition of existing services in form of a BPMN diagram) of a service. This is not a mandatory task.		
Preconditions	Service has been created before.		
Successful outcome	The new service process and all content that is being utilized within it has been created.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service Process or Content is incomplete	Error message is being displayed to the IT Service Developer	Parts of the required input is missing
Primary actors	IT Service Developer		
Secondary actors	-		
Main scenario	The IT Service Developer utilizes the tools from the Service Creation domain to define a new service process and content.		
Requirements	The shared partner metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Business Partner Management", and accessible to all partners in the network. Each eRoaming-Platform provider as participants in the network provides the IDs of the EMPs and CPOs as part of the partner metadata Participants in the network are known and trusted. Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.32 Download Service Process & Content (UC32)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The service implementation can be downloaded if necessary. This can only be done by the IT Service Developer who has developed this service or any other actor who has received the rights from the IT Service Developer.		
Preconditions	The service is owned by the IT Service Developer or the actor trying to download the data received the rights from IT Service Developer.		
Successful outcome	The service implementation (e.g. a .jar) is downloaded to the service developer's computer.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service not found	An error is displayed which shows that no service content of this service could be found.	The service has no implementation.
Primary actors	IT Service Developer or other Business Partner, like Service Provider		
Secondary actors	-		
Main scenario	▪ The IT Service Developer searches for a service ▪ The IT Service Developer selects a service of the returned list of services ▪ He/she triggers the download of this service process & content and stores it locally.		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.33 View Service Contract Details (UC 33)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	View the details of a Service Contract		
Preconditions	One out of a list of Service Contracts has been selected		
Successful outcome	The business partner views the details and history of one Service Contract; e.g. Terms and conditions, pricing, contract period		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	NA	NA	NA
Primary actors	Business Partner		
Secondary actors	-		

Scope and Level	BS4-NeMo Service Development Environment
Main scenario	▪ The Business Partner views the details of its own Service Contract details Variation: ▪ The Business Partner views the details of all Service Contracts details
Requirements	The Business Partner has the access right to perform the action.

4.1.34 Create Service Contract Termination Request (UC 34)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The termination of a bilateral Service Contract is a two-phase process. A Business Partner makes first a termination request on a Service Contract.		
Preconditions	A valid Service Contract exists between Business Partners and is selected.		
Successful outcome	A termination request is created and the other Business Partner is being notified.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Request is incomplete	Display an error message	Business Partner did not provide all data necessary
Primary actors	Business Partner (Offering Party of Accessing Party; Service Provider or Service Requestor)		
Secondary actors	-		
Main scenario	▪ The Business Partner fills the contract termination form ▪ A Service Contract history is created ▪ The other Business Partner is notified		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with market and increase of connected IT systems		

4.1.35 Confirm Service Contract Termination Request (UC35)

Scope and Level	BS4-NeMo Service Development Environment
Goal in context	A termination of a bilateral Service Contract is a two-step process. A Business Partner (Service Provider or Service Requester) accepts or rejects a termination request on a Service Contract made by the other Business Partner.
Preconditions	A Service Contract between the Offering Party (Service Provider) and the Accessing Party (Service Requester) exists. The Business Partner has been notified on a termination request on the Service Contract.

Scope and Level	BS4-NeMo Service Development Environment		
Successful outcome	The termination request on the Service Contract is accepted or rejected. The other Business Partner is notified.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service contract termination rejected	Display error message	Business Partner rejecting service contract termination request
Primary actors	Business Partner (Service Provider or Service Requester)		
Secondary actors	-		
Main scenario	▪ The Business Partner accepts or rejects the termination request on the Service Contract. ▪ If yes: o The Service Contract is terminated. o A Service Contract History Entry is created. o The Service Contract status is being set to "Inactive". ▪ The other Business Partner is notified.		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with the market and increase of connected IT systems.		

4.1.36 View Service Contract Template (UC 36)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The template of the Service Contract Framework is displayed to the Business Partner		
Preconditions	Contents of the Service Contract Framework are provided by a platform independent representation (e.g. XML) by the NeMo Alliance.		
Successful outcome	The Business Partner is able to view the contents with all available options of the Service Contract Framework		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	NA	NA	NA
Primary actors	Business Partner		
Secondary actors	-		
Main scenario	▪ The Business Partner chooses the details view of the contents with all available options of the Service Contract Framework ▪ The detail view is displayed.		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with the market and increase of connected IT systems.		

4.1.37 View Service Specification Details (UC 37)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	View the details of a service for the service developer to choose a fitting service for his implementation.		
Preconditions	▪ The service has a semantic description and a service interface specification. ▪ The Service has been found and selected by the IT Service Developer		
Successful outcome	The functional and non-functional description as well as the Service Interface is shown to the requester.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service not found	An error is displayed which shows that no description of this service could be found.	The service has no service description and interface published yet.
Primary actors	IT Service Developer or any Business Partner with respective rights (such as other Service Provider)		
Secondary actors	-		
Main scenario	The semantic description of the service as well as the service interfaces is displayed from the Service Registry (Service Management).		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.38 Create Service Contract (UC 38)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A Service Contract is created by accepting a Service Contract Offering by the Accessing Party (Service Requester).		
Preconditions	A Service Contract Offering is available for a service in the Contract Registry (Contract Management).		
Successful outcome	A new Service Contract is created.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service Contract incomplete	Display of an error message	Not all required sections from the Service Contract Offering are chosen
	Service Contract Offering rejected	Display a message and cancellation of the process	Accessing Party do not accept the Service Contract Offering
Primary actors	Accessing Party (Service Requester)		
Secondary actors	-		
Main scenario	▪ The Service Requester chooses a Service and reviews the Service Contract Offering with its options. ▪ The Service Requester chooses all required sections and may choose available optional sections and is able to store his selection. ▪ The Service Requester accepts the Service Contract Offering with his selection on options. ▪ The chosen sections configure a new Service Contract between the Service Provider and the Service Requester. ▪ The new created Service Contract is stored in the Contract Registry. ▪ The Service Provider will be notified as soon as the contract becomes active.		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with the market and increase of connected IT systems		

4.1.39 Enable / Disable Service (UC39)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	An IT Service Developer can enable and disable services. An enabled service is available to the members of the NeMo Hyper-Network. A disabled service cannot be executed.		
Preconditions	The service is properly deployed in the Hyper-Network.		
Successful outcome	The service is either disabled or enabled.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service is in use	An error is displayed which shows that the service is in use and cannot be disabled	The service stays enabled and the use of the service is displayed for the service provider to be able to contact this user.
Primary actors	IT Service Developer / Service Provider		
Secondary actors	Service Requestors using this respective services		
Main scenario	▪ The IT Service Developer or Service Provider sends a request for enabling / disabling a service belonging to him. ▪ In case of disabling a service, it will be checked that there are no valid service contracts (state "inactive") s if the service is still in use by any other service or contract. ▪ All of the business partners are being notified of the disablement of the service by updating the state in the Service Registry (Service Management) ▪ The service state is updated either to "Enabled" or "Disabled".		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.40 Download Service Specification (UC40)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The service specification includes the semantic service description as well as the service interfaces description (e.g. WSDL). This data shall be downloadable.		
Preconditions	The service is published and documentation has been uploaded to the NeMo Hyper-Network.		
Successful outcome	The IT Service Developer has access to the service documentation.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service not found	An error is displayed which shows that no description of this service could be found.	The service has no service specification published yet.
Primary actors	IT Service Developer or other Business Partners, such as Service Provider		
Secondary actors	-		
Main scenario	▪ The IT Service Developer has found and selected the service from the Service Registry ▪ The IT Service Developer downloads this specification to his local computer		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.41 Upload Service Process & Content (UC41)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The service implementation and additional files, such as documentation, can be uploaded by the IT Service Developer creating a service.		
Preconditions	The service has been registered in the NeMo Hyper-Network IT Service Developer has the rights to upload the service process & content		
Successful outcome	The service process and content is uploaded to the Service Registry and distributed within the NeMo Hyper-Network		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service process cannot be uploaded	Error message being displayed	Service has not been registered
Primary actors	IT Service Developer or Service Provider		
Secondary actors	-		

Scope and Level	BS4-NeMo Service Development Environment
Main scenario	▪ The IT Service Developer uploads the implementation and additional documentation to the Service Registry
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.

4.1.42 Upload Service Specification (UC42)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The service specification includes the semantic service description as well as the service interfaces description (e.g. WSDL). This service specification is publicly available for the business partners in the NeMo Hyper-Network.		
Preconditions	The service is published and documentation has been uploaded and distributed within the NeMo Hyper-Network via the Service Registry (Service Management).		
Successful outcome	The Service Specification has been uploaded and distributed in the Service Registry in the NeMo Hyper-Network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service not found	An error is displayed which shows that no description of this service could be found.	The service has no service description published yet
Primary actors	IT Service Developer or Service Provider		
Secondary actors	-		
Main scenario	▪ The IT Service Developer uploads the semantic description and the service interface to the Services Registry ▪ These are then linked to the previously uploaded Service Prices and Content in the Service Registry		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.43 Suspend Resume Service Contract by Service Requestor (UC43)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A Service Requester suspends or resumes a Service Contract, i.e. the Service Contract itself is not changed or terminated, but indicated as not to use during a service call.		
Preconditions	A Service Contract between the Service Provider and the Service Requester exists and is selected.		
Successful outcome	A Service Contract is suspended or resumed by the Service Requester.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service contract is not suspended or resumed	Error message is displayed	Service contract not in state active
Primary actors	Accessing Party (Service Requestor)		
Secondary actors	-		
Main scenario	▪ The Service Requestor marks the Service Contract as suspended, respectively for use. ▪ A Service Contract History Entry is created in the NeMo Hyper-Network Contract Registry (Contract Management)		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with the market and increase of connected IT systems		

4.1.44 Publish a Service (UC44)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	A service is published into the NeMo Hyper-Network, when its specification, its implementation, its contract details and its description is uploaded to the NeMo Hyper-Network. The publishing of the service enables the service for use of the other partners.		
Preconditions	The service is published and documentation has been uploaded, a contract has been specified for this service and a service description is uploaded to the NeMo Hyper-Network.		
Successful outcome	All service developers have access to the service.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service is a duplicate	An error is displayed which shows the duplicate service	The service is already deployed.
Primary actors	IT Service Developer or Service Provider		
Secondary actors	-		
Main scenario	▪ An uploaded service gets a unique ID (e.g. a URI) ▪ The service is linked to all its documentation, description, contracts, implementation and meta information. ▪ The service becomes available in the distributed Service Registry (Service Management). ▪ The service is set to enabled / published.		
Requirements	Each EMP as participant in the network provides the unique ID (eMAID) as part of the partner metadata. Each CPO as participant in the network provides the unique ID (<Country Code><Spot Operator ID>) as part of the partner metadata Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network.		

4.1.45 Change Service Contract Offering (UC 45)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	An existing Service Contract Offering can be changed from the template of the Service Contract Framework by the Service Provider. Existing Service Contracts based on the Service Contract Offering are not affected.		
Preconditions	▪ Contents of the Service Contract Framework are provided to the Contracts database by a static platform-independent representation (e.g. XML). ▪ A Service Contract Offering is selected		
Successful outcome	An existing Service Contract Offering is changed.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service Contract Offering incomplete	Display of an error message	Not all required sections from the Service Contract Framework are chosen
Primary actors	Offering Party (Service Provider)		
Secondary actors	-		
Main scenario	▪ The Offering Party (Service Provider) changes optional sections from the Service Contract Framework. ▪ The chosen sections configure a changed Service Contract Offering. ▪ The changed Service Contract Offering is stored.		
Requirements	1. Access: Role based authorization required 2. Data security: Data integrity must be guaranteed; Data confidentiality must be ensured 3. Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required 4. Performance: Sub-seconds response time for service transactions 5. Availability: 24x7 6. Scalability: Able to grow with the market and increase of connected IT systems		

4.1.46 Create Service Specification (UC 46)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	The service contains a semantic service description comprising functional and non-functional parameters. Additionally, the service also contains a concrete service interface.		
Preconditions	The IT Service Developer has the right to describe the service and its QoS aspects.		
Successful outcome	A requesting party, or a service developer can develop a service specification using the Service Creation domain of the Open Cloud Marketplace.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service specification cannot be created	Error message is being displayed	Service specification was incomplete
Primary actors	IT Service Developer		
Secondary actors			
Main scenario	▪ The IT Service Developer opens a tool, which enables him to specify the functional as well as the non-functional details of a service. ▪ These can be technical aspects or quality parameters that are being guaranteed.		
Requirements	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network. The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance. Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.		

4.1.47 Create Service Example Request and Response (UC47)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	In order to evaluate a service before integrating it into a service composition, an example invocation message and response should be provided.		
Preconditions	The service for which the example request is built is deployed and running in the NeMo Hyper-Network.		
Successful outcome	The service request can be invoked and the response is delivered to the invoking party. No cost is created and the IT Service Developer can test his connection, session and implementation within the NeMo Hyper-Network.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service request and response fail to fit together. This means the request does not foster the specified response message.	An error message is displayed stating the reason of failure.	A malformed request or a not running service.
Primary actors	IT Service Developer		
Secondary actors	-		
Main scenario	▪ The service developer publishes a new service and uploads an example request and response message via the Upload Service Process & Content ▪ The Service Registry is being updated with the example ▪ All business partners in the NeMo Hyper-Network are being informed		
Requirements	▪ The service which will be tested needs to be available. ▪ The request triggers the specified response. ▪ The service developer owns the service which is tested.		

4.1.48 Call Service Example Request (UC48)

Scope and Level	BS4-NeMo Service Development Environment		
Goal in context	An IT Service Developer wants to try-out / test a service before integrating it into his service composition.		
Preconditions	The service can be invoked with the parameters of the example request. The message of the response can then be parsed and the result can be extracted.		
Successful outcome	The service request can be invoked and the response is delivered to the invoking party.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service invocation failed due to malformed request or failure in the service e.g. unavailability	An error message is displayed stating the reason of failure.	A malformed request or a not running service.
Primary actors	IT Service Developer		
Secondary actors	-		
Main scenario	1. The service developer finds a new service and selects the example request from its documentation. 2. The service developer wants to test the found service with the example response to see if the service is useful. 3. The service developer looks up the service request message and fills in the example parameters. 4. The service responds with the response message which can be checked against the example. 5. The service developer can now integrate this response into his service composition.		
Requirements	▪ The service which will be tested needs to be available. ▪ The request triggers the specified response. ▪ The service developer owns the service which is tested.		

4.1.49 Monitor actor activity (UC49)

Scope and Level	BS5-Horizontal Services		
Goal in context	Track the actor activity with the NeMo services as well as its interaction with the other actors. For the case of the EV driver, the monitor tracks also the user's location in order to identify frequently visited locations and mobility patterns.		
Preconditions	The actor has to be registered with an actor profile and should have agreed to the privacy consent form. The actor has not deactivated the monitor/tracking functionality.		
Successful outcome	The actor's activity and locations are registered.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Monitor deactivated	The activity is not monitored	The actor has not an actor profile, has not agreed to the privacy consent form or he/she has temporally deactivated the tracking functionality
Primary actors	Driver		
Secondary actors	CPO, EMP		
Main scenario	▪ An actor interacts with the NeMo Hyper-Network looking for services, checking its characteristics, booking or using them. ▪ The monitor registers all these interactions looking for preferences patterns to incorporate in its profile. ▪ In the case of the EV driver, its locations are also registered and analysed looking for common places and mobility patterns.		
Requirements	▪ The Actor is informed about the information that will be tracked according to the privacy and security procedure ▪ The actor accepts to be tracked. He/she can withdraw consent any time ▪ The actor can deactivate the tracker in specific situations ▪ The monitor registers the user's locations ▪ The monitor deduces common actor/user locations according to the registered positions ▪ The monitor registers the user activity and its interaction with the services or other actors in the network ▪ The monitor deduces the electromobility services with which the actor uses/consumes/interacts frequently ▪ The monitor updates the actor profile with the information deduced from the actor's activity ▪ The monitor registers the actor activity in real-time. The analysis for the monitored information could be in real-time (if needed to provide instant feedback or user support) or in an elapsed time, if instant feedback is not needed.		

4.1.50 Update actor profile (UC 50)

Scope and Level	BS5-Horizontal services		
Goal in context	The actor accesses its profile, updates it with some modifications and saves them.		
Preconditions	The actor has an actor profile.		
Successful outcome	The modifications have been saved properly.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Modifications not saved	An error message informs that the modifications have not been saved. The profile does not change.	The actor has not permissions to update the profile
Primary actors	End user		
Secondary actors	EMP, CPO, Business partners		
Main scenario	• The actor accesses its user profile • The actor makes some modifications • The actor saves them		
Requirements	▪ The actor has a profile ▪ The actor can access his/her profile ▪ The actor can update his/her profile ▪ The actor has been informed about the personal information the profile stores and has consented that they can be managed according to the privacy procedures ▪ The actor can withdraw his/her consent that the profile can manage his/her personal information. In that case, the personal information related to the actor has to be removed ▪ Updates on the actors profile must be visible for other processes immediately		

4.1.51 Read actor profile (UC 51)

Scope and Level	BS5-Horizontal services		
Goal in context	The actor accesses its profile, checks/reads it and closes it without saving.		
Preconditions	The actor has an actor profile.		
Successful outcome	The actor has checked the information registered on its user profile		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The user profile does not open or does not show any information	An error message informs that the profile information is not accessible	The actor has not permissions to see the actor profile
Primary actors	EV driver, End user		
Secondary actors	EMP, CPO, Business partners		
Main scenario	▪ The actor accesses its user profile ▪ The actors check its stored preferences and other profile attributes ▪ The actor closes the profile without saving any changes		
Requirements	▪ The actor has a profile ▪ The actor can access his profile ▪ Actors profile information must be quickly accessible ▪ Actors profile must be up-to-date before being loaded		

4.1.52 Find an electromobility service (UC 52)

Scope and Level	BS5-Horizontal services		
Goal in context	A service developer or a dynamic service composition has the possibility to search for a service given a semantic service request (e.g. OWL-S). The service finder proposes a list of fitting services rated according to the functional aspects of the request.		
Preconditions	The services are registered in the Hyper-Network		
Successful outcome	The requesting party receives a list of services with the appropriate functionalities		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No service has been found	An error message informs the requester that no service satisfying the request can be found.	Mal formatted request or semantically so different that no service is similar to the request.
Primary actors	Service Developer		
Secondary actors	Service Optimizer		
Main scenario	▪ The requester sends an OWL-S service profile as request to the service optimizer ▪ The service optimizer hands over the functional requirements as a request to the service finder ▪ The service finder gets all available services from the service registry ▪ The service finder ranks the services according to their similarity to the request (functional properties of the service e.g. input or output) ▪ The service optimizer gets a list of ranked services.		
Requirements	▪ The services have to be registered in the service registry of the Hyper-Network including a semantic description of the service ▪ The services have to be described based upon the common information model of the NeMo Hyper-Network ▪ The request is specified as an (incomplete) semantic service description ▪ The service description may include an (optional) price for fulfilling a request or contracting a service. The specified price may be constant (the same for all requests and types of service) or a service-specific call-back which will dynamically calculate a price for a specific service profile.		

4.1.53 Optimize Search for electromobility services (UC 53)

Scope and Level	BS5-Horizontal services		
Goal in context	The service optimizer ranks services with respect to non-functional or Quality-of-Service (QoS) aspects. The requesting party specifies a set of non-functional parameters (price, location, etc.) and the optimizer optimizes the feasible services with respect to them.		
Preconditions	The services are described with non-functional parameters.		
Successful outcome	The requesting party receives a ranked list of services that fit from a functional as well as from a non-functional point of view.		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No service with this QoS parameters has been found	An error message informs the requester that no service similar to this service can be found.	Mal formatted request or the requested QoS are not described in the service's QoS
Primary actors	Service Developer		
Secondary actors	Service Optimizer		
Main scenario	▪ The requester sends a search template containing QoS parameters ▪ The Service Optimizer triggers the Service Finder to filter all services according to functional aspects and receives a list of those ▪ The Service Optimizer requests the user profile of the requesting actor from the Profiling Service ▪ The Service Optimizer requests the pricing for the services from the Pricing Service ▪ The Service Optimizer ranks the functionally suitable services regarding the profile, the pricing and other non-functional aspects described in the service request ▪ The requester gets a list of ranked services		
Requirements	▪ The Service Optimizer has access to a service finder component in order to get all services that fit from a functional point of view ▪ The Service Optimizer has access to Pricing and Profiling Services ▪ The service descriptions do contain non-functional parameters ▪ The request does contain non-functional parameters as search criteria		

4.1.54 Broker requests for electromobility services (UC 54)

Scope and Level	BS5-Horizontal Services		
Goal in context	Fit the actors' requests to existing services of the Hyper-Network, reconciling concurrent requests from different actors for the same electromobility service provided by the Hyper-Network.		
Preconditions	Different actors demand identical electromobility services. Those services have a limited offer they can afford that is less than that requested.		
Successful outcome	The service requests have been brokeraged and properly served		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Service not possible to broker	Service Request rejected	There are not suitable alternatives for the service requested or the demand is too high to broker all the requests
Primary actors	End users, EV Driver		
Secondary actors	EMP, CPO, Business partners		
Main scenario	▪ Different actors request similar electromobility services at the same time and location. ▪ The Service brokerage manages the different requests for the most demanded services and balances them prioritizing, for example, those less demanded. ▪ For the requests that could not be served, the brokerage proposes different but suitable alternatives		
Requirements	▪ The broker identifies concurrent request to the same service ▪ The broker accesses the actor profile to identify preferences or constraints for the services requested in order to produce suitable alternatives ▪ The broker proposes alternatives for concurrent requests to same services ▪ The user can express his/her opinion or vote for the alternatives proposed ▪ The broker uses the actor feedbacks to improve future brokerages ▪ Brokerage is active for concurrent requests ▪ Brokerage response must be fast enough to not overload the acceptable waiting times		

4.1.55 Check the price for an electromobility service (UC 55)

Scope and Level	BS5-Horizontal services		
Goal in context	A service developer has the possibility to receive a quote for contracting a specific service, given a semantic service request (e.g. OWL-S). The service finder proposes a list of fitting services (See use case "Find an electromobility service"). Each of these services includes a mechanism through which prices can be requested.		
Preconditions	The services are available to the service finder.		
Successful outcome	The requesting party receives a list of similar services from a functional point of view		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No price is available.	Actors can't access a service price.	If no price is available (e.g. it doesn't make sense to offer a quote for the service, or the service is provided for free).
Primary actors	Service Developer		
Secondary actors	Service Optimizer, Service Finder		
Main scenario	▪ The requesting actor gets a list of services from the service finder. ▪ Each of these services offering prices shall have a call-back through which quotes are offered. ▪ The requesting actor may, at any time, invoke this call-back with a service profile. ▪ Once invoked, the call-back shall produce a quote for the specified service profile. This quote should include any additional terms needed for the usage of this service (e.g.: validity period, indication of being an estimative price if the final price can't be known at the time of invocation, etc.).		
Requirements	▪ The services need to be available to the service finder. ▪ The services may specify no-price (e.g.: if the service is offered for free). ▪ The services may specify a validity period for a quote, after which said quote becomes invalid. ▪ The services may specify a quote as estimative, if the final price can't be known in advance. ▪ The services published and available to the service finder shall include an endpoint that may be invoked with a service profile to obtain a quote.		

4.1.56 Offer a promotion for an electromobility service (UC56)

Scope and Level	BS5-Horizontal services		
Goal in context	A service provider may push special offers and promotion notifications to end users.		
Preconditions	The services are available to the service finder.		
Successful outcome	The requesting party receives a list of similar services from a functional point of view		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	End user not reachable	No notification can be pushed, therefore no offers are available for this particular user	An end user somehow becomes unreachable (e.g.: by dropping of the network)
Primary actors	Service Developer		
Secondary actors	Service Finder		
Main scenario	▪ The requesting actor gets a list of services from the service finder. ▪ Services offered in this list may, optionally, allow the calling actor to subscribe to a notifications channel by providing an endpoint to which the actor can submit a call-back. ▪ The service provider may, at any time, push notifications through this channel, offering new services, prices or promotions (e.g.: by storing a previous service profile requested by an actor, the service provider may offer new prices if they become available). ▪ The calling actor should be able to unsubscribe from the notification channel via a time-out mechanism or by utilizing an explicit unsubscription endpoint made available by the service provider.		
Requirements	▪ The services need to be available to the service finder. ▪ The services may specify in their service description an endpoint through which callers may subscribe to a notification channel. ▪ The notification channel should provide an unsubscription mechanism. ▪ To ensure relevance on the offers and promotions, a real-time channel must be available to process the notifications.		

4.1.57 Get grid load report (UC 57)

Scope and Level	BS1-Smart Charging, BS6-eMobility Report		
Goal in context	Provide a visualization tool for DSOs and CPOs, so that they may anticipate the needed financial investment in the grid / charging stations		
Preconditions	Data collection from CPOs and DSO about charging sessions and electrical grid should include: ▪ Occupation rate per delivery point (number of charges per day) ▪ Total energy delivered by the charge points ▪ Instantaneous power consumption during the day ▪ Data from the DSO about the energy / power consumption at a local scale (% power consumed on total available power)		
Successful outcome	A map of the electrical grid showing in different colours the areas according to the EV load		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The map is not available	Nothing is displayed on the map and an error message is send	Service unavailable
Primary actors	CPOs, DSO, eRoaming platform		
Secondary actors	-		
Main scenario	▪ The DSO or CPO requires the Grid load report to the NeMo Hyper-Network setting a geographical area (or a CPO network) ▪ The NeMo Hyper-Network gathers the data from the selected CPOs (for example based on the CDRs) and data from the concerned DSO (electrical consumption of the electrical grid) ▪ An algorithm calculates the impact of EVs on the local grid (curve of energy/power during the day, % of total consumption)		
Requirements	▪ The actor should get data from the DSO and the CPO (including CDR, energy consumption, EV impact in power limit on local grid) ▪ The NeMo Hyper-Network shall host the calculation which permits to complete the report ▪ The actor shall select the concerned CPOs and DSOs by the report localisation		

4.1.58 Display charge points occupation rate on Web (UC 58)

Scope and Level	BS1-Smart Charging, BS6-eMobility Report		
Goal in context	Provide reassurance to any customer willing to buy an EV on the capability to easily charge in real life		
Preconditions	Charge points have to be connected to interoperability platform and pushing their status (dynamic data) of eRoaming platform		
Successful outcome	A map with real data of occupation rate of charge points per day/time		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Not enough charge points connected and poor data update	No service or service not trustful for customers	Poor quality involvement of CPOs
Primary actors	CPO		
Secondary actors	eRoaming Platform		
Main scenario	eRoaming Platform analyses data from the CPO 's eRoaming platform in real-time		
Requirements	▪ CPOs shall push status change of charge points with harmonized delay ▪ The NeMo Hyper-Network shall host the calculation which will output a graphic of the occupation rate ▪ This graphic shall be displayed on a website, showing the real time availability of each charge point		

4.1.59 Analyse user usage (UC 59)

Scope and Level	BS5-Horizontal services, BS1-Smart Charging, BS6-eMobility Report		
Goal in context	Get a clear vision about the customers' usage of EVs to anticipate the needed evolution in the charge points network (location, power) and on vehicles (battery size,..)		
Preconditions	Data have to be sent by the vehicles		
Successful outcome	A visualization tool showing the usage of the EVs by the customer (for example mileage, number of charge sessions, charges on public charge points, ...)		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	The service doesn't return any visualization	Error message describing the error	Poor quality of data sending by EVs
Primary actors	VM,		
Secondary actors	-		
Main scenario	▪ The actor requests to the NeMo Hyper-Network the selected type of user for which he wants to conduct an analysis ▪ The NeMo Hyper-Network collects the data for this user type and calculates the usage ▪ A usage profile is given to the actor showing several indicators (for example mileage per day, number of charge sessions by day, ratio private/public charging)		
Requirements	▪ VM have to send data about EV usage (charging and driving details) to the eRoaming Platform ▪ EMP should send the CDR for public charges ▪ Private data shall be aggregated (VIN) to respect privacy laws. (list of VINs per type of user shall be sent by VM)		

4.1.60 Personal Mobility Plan (UC60)

Scope and Level	BS7-Vehicle Preparation For Drive Off		
Goal in context	Derive a mobility plan of a customer (in order to align the charging and operation strategy).		
Preconditions	Data on mobility request has to be collected from vehicle and user		
Successful outcome	A clearly defined plan on when the customer will required mobility services (starting times, amount of mobility and comfort); along with a degree of probability		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No personal mobility plan	Bad estimation on mobility demand	Undefined mobility behaviour/demand from customer
Primary actors	Driver		
Secondary actors	Fleet operator, Owner of EVs		
Main scenario	▪ The actor provides a preview information on when to use mobility services to the NeMo Hyper-Network ▪ The NeMo Hyper-Network collects the user information, stores this information and calculates the mobility demand profile of the customer ▪ The calculation of the profile considers correlative information on specific mobility patterns in order to determine probability conditions for certain demands		
Requirements	▪ Driver needs to be able to communicate his/her mobility demands to the network (either directly or indirect through app-services) ▪ An abstract mobility pattern (“mobility memory”) needs to be available ▪ Data privacy on the mobility demand needs to be ensured ▪ Mobility demand information must be available before drive off ▪ Mobility demand should be available before charging process ▪ Mobility memory should cover one year of history		

4.1.61 Derive EV Mobility Plan (UC61)

Scope and Level	BS7-Vehicle Preparation For Drive Off		
Goal in context	Derive a mobility plan for an EV (in order to align the charging and operation strategy).		
Preconditions	Data on mobility request has to be collected from vehicle and user(s) and individual mobility plans need to be available; probably in combination with customer data from fleet operator		
Successful outcome	A clearly defined plan on when to be ready for drive off (starting times), and probably a degree of probability level, derived by the combination of EV users		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No vehicle mobility plan	Bad estimation on drive off time	Undefined mobility behaviour/demand or bad fusion
Primary actors	IT Cloud Platform		
Secondary actors	Fleet operator, Owner of EVs		
Main scenario	▪ The actor collects the drivers who are probable to use EVs ▪ The actor analyses the abstracted mobility behaviour of the identified drivers ▪ The actor derives a combined mobility demand plan for the EVs		
Requirements	▪ Access to mobility behaviour needs to be available ▪ Deriving mobility plan needs to be possible without detailed personal information (person abstraction and limited look-ahead horizon) ▪ Mobility demand information must be available before drive off ▪ Mobility demand should be available before charging process ▪ Mobility memory should cover one year of history		

4.1.62 Estimate Energy Demand and Charging Strategy (UC62)

Scope and Level	BS7-Vehicle Preparation For Drive Off		
Goal in context	Estimate the energy demand and define an optimized charging strategy for a vehicle (based on mobility plan and EV state)		
Preconditions	Mobility plan for the EV exists and can be accessed Vehicle profile exists		
Successful outcome	A constraint plan on the charging timing strategy		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	No charging strategy constraints derived	No adaptive charging strategy can be selected	No or imprecise mobility demand
Primary actors	EV, CPO		
Secondary actors	Driver, EV		
Main scenario	▪ The actor collects the mobility demand of the vehicle ▪ The actor determines suitable charging strategy requirements for the active and probable follow-up charging cycles (for optimization), considering the usage demands and the SoH of the vehicles battery ▪ The actor selects the charging strategy ▪ The actor modifies the charging strategy upon changing demand		
Requirements	▪ Access to mobility aggregated mobility demand ▪ Constraints for optimization strategy for charging need to be available (e.g. SoH) ▪ Mobility demand profile for EV should be available immediately before charging process start ▪ Changing mobility demand needs to be available upon notice and before drive off time		

4.1.63 React on Short Notice Demands (UC63)

Scope and Level	BS7-Vehicle Preparation For Drive Off		
Goal in context	Mobility plan change or react to Short-Notice Demand		
Preconditions	Driver identifies mobility demand not aligning with “normal” mobility plan		
Successful outcome	Short notice demand is respected during the vehicle preparation phase		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Short notice demand not respected	Vehicle not prepared	Missing mobility demand information or no sufficient look ahead time horizon
Primary actors	Driver		
Secondary actors	CPO		
Main scenario	▪ The actor provides information based on which a mobility demand not aligned to his/her stored mobility profile can be identified ▪ The NeMo Hyper-Network forwards this anomaly as regards the mobility demand to the CPO ▪ The NeMo Hyper-Network forwards this anomaly in mobility demand to the user profile in order to update and refine the driver’s personal mobility plan		
Requirements	▪ Driver needs to be able to communicate his/her mobility demands to the network (either directly or indirect through app-services) ▪ An abstract mobility pattern (“mobility memory”) needs to be available ▪ Privacy of data in the mobility demand needs to be ensured		

4.1.64 Pre-heating of the car (UC 64)

Scope and Level	BS7-Vehicle Preparation For Drive Off		
Goal in context	Provide a service which permits the driver to have the car at an appropriate temperature		
Preconditions	The following data and information and requirements are needed ▪ Time of departure, as indicated by the driver to the EMP ▪ The status of the vehicle, it should be plugged in order to preserve the battery during the pre-heating		
Successful outcome	The car is at a good temperature when the driver enters it		
Failure outcome	**Failure**	**Outcome**	**Condition leading to outcome**
	Too high/low temperature	Bad customer experience	Car not plugged / Time of departure unknown
Primary actors	VM, EMP		
Secondary actors	-		
Main scenario	▪ The driver plugs his/her car and sets a time of departure in the EMP application ▪ The charging process is launched and completed ▪ Some minutes before the departure time, an algorithm (from EMP) calculates the needed duration for pre heating (according to external temperature sent by the car) using the energy from the grid ▪ When the driver enters the vehicle, the cabin interior and the battery are at a good temperature in order to use less energy during driving		
Requirements	▪ The driver shall be able to set his/her time of departure ▪ The car status (plugged/parked) and external temperature shall be sent to the EMP ▪ A pre heating signal shall be sent to the car from the EMP with a duration and a temperature ▪ The EMP shall be able to calculate the pre heating duration based on external temperature and EV status		

5. Current plans of the NeMo Test Sites

5.1 French Test Site

The objective of the Test Site France is to demonstrate the feasibility of several NeMo services. Indeed, the French test site will focus on services concerning smart charging, battery durability and monitoring of EVs impact on the ecosystem.

At the present moment, the French test site is equipped with charge points (pools) connected to an eRoaming Platform, a pool taxi for employee's moves, a supervision system for the charging stations and a building energy management system. Although fully operational, that equipment is disjointed and do not yet include the services and the connection to the NeMo Hyper-Network.

The French test site location will be mainly the technical centre of Renault, based in Parisian suburbs: 1 avenue du Golf 78280 Guyancourt. As will be difficult to test in Renault offices some of the services, such as preparation of cars before drive off or management of the battery durability, a customer fleet will be probably deployed. This fleet may be located somewhere else than Renault technical centre.

Some difficulties may interfere with the demonstration of NeMo services. Indeed, it may be difficult to install new charge points on the technical centre of Renault. Another issue is the accessibility of the charge points. Some of the charging points are not accessible to public customers but are reserved for employees. Moreover, due to a high number of electric vehicles on the site, the charge points have a high occupation rate, which may make difficult the demonstration of some services.

The Test Site in France will involve mainly RENAULT and GIREVE. RENAULT because the main test site will be located at the technical centre, GIREVE because charging points are connected to the eRoaming Platform, and several services to be tested will use its system to exchange data. As some of these data are considered as private by national authorities, it will be easier to handle data management considering national players.

At the present time, it is envisaged that the Test Site in France will cover the following Business Scenarios and corresponding Use Cases:

- Business Scenario 1: Smart Charging
  - UC1 - Convert mobility needs into charge needs
  - UC2 - Search better charge offers
    (from CPO according to EV driver/fleet operator needs and stakeholders constraints)
  - UC3 - Search and find charge points
  - UC15 - Interoperability: Book a charge point
  - UC5 - Change/switch energy tariffs

- UC7 - Push Charge Plan
  - UC8 - Propose energy grid constraints/opportunities
  - UC9 - Push a CDR from EMP to OEM
  - UC10 - Push charging strategy from OEM to CPO
- Business Scenario 6: eMobility Report
  - UC57: Get Grid load report
  - UC58: Display charge points occupation rate on Web
  - UC59: Analyse user usages
- Business Scenario 7: Preparation for drive off
  - UC64: Pre-heating of the car

5.2 German Test Site

The objective of the Test Site in Germany is to implement and demonstrate the development of the NeMo Project adapted and customized to the requirements of the participating partners. The Test Site in Germany will consider and balance the demonstration activities both in physical and virtual environment.

At the present moment, each of the participating partners have own demonstration facilities focused on own services and implementations. Although fully operational, those demonstrations are disjointed and do not yet include the services and the connection to the NeMo Hyper-Network. Such technologies include:

- Hubject eRoaming Platform – which is fully operational for commercial activities including information about more the then 44, 000 charging points. Furthermore, Hubject is planning to establish its own Test site, with own EVSE during the course of the project.
- TomTom Navigation Services – The source for the TOMTOM Traffic products are GPS probes data that are collected from personal navigation devices and users who have opted to share travel time information on an anonymous basis. These are largely from connected TOMTOM navigation devices (PND), TOMTOM Business Solutions devices, TOMTOM embedded navigation devices in cars and TOMTOM navigation applications on Smartphone (e.g. iPhone)
- GT-ARC – Four charge points with four parking sensors, behind a Gate, with a charging infrastructure backend running in a virtualization environment including a server cluster and Storage Area Network (SAN). This is augmented by a Network infrastructure offering public IPv4/IPv6 addresses with a fast uplink, including firewall, network virtualization for dedicated VLANs, and VPN services (IPsec and OpenVPN).
- IBM – IBM is utilizing cloud infrastructure to develop, deploy and run the IT services. This is based on Infrastructure-as-a-Service (IaaS) offering called IBM Softlayer. In addition, IBM is planning to use Platform-as-a-Service (PaaS) offering called IBM Bluemix. This contains the entire required infrastructure to deploy and run any IT services. The IT centre for IBM Softlayer and IBM Bluemix are located in Europe.

The Test Site in Germany will be implemented in both virtual and physical environments. Based on the implemented business scenarios and use cases, the Test Site in Germany will include:

- Demonstration Lab including different types of charging stations connected to the eRoaming platforms utilizing the Pan-European Inter-Roaming protocol.
- Demonstration virtual lab utilizing the existing partner networks of the eRoaming provider.
- Hackathon, both physical and virtual, implementing demonstration services for the NeMo Hyper-Network

At the present time, it is envisaged that the Test Site in Germany will cover the following business scenarios and corresponding use cases:

- Business Scenario 2: Itinerary Planning
- Business Scenario 4: NeMo Service Development Environment
- Business Scenario 5: Horizontal Services

5.3 Spanish Test Site

The site will focus on the public taxis scenario, with the main objective to optimise the user experience of the taxi drivers by providing them personalised electromobility services, according to their history, and extend it to all the electromobility actors.

The site will cover the following Business Scenarios:

- Business Scenario 1: Smart charging
- Business Scenario 5: Horizontal Services

5.4 Austrian Test Site

For the demonstration of the different business scenarios VERBUND and SMATRICS will run the demonstration on two different test sites. For the demonstration of Business Scenario 1 the test site will be located in a company/ commercial environment which is exclusively usable for company fleet cars. From a broad variety of different potential SMATRICS B2B customers the partner company will be selected in the course of the NeMo project. The demonstration of Business Scenarios 3 and 5 involves also publically accessible sites of SMATRICS' existing charging network (currently approximately 400 charge points).

The objectives of the Test Site Austria are the following:

- Customer services
  - Live status of charge spots
  - Navigation to charging location
  - Ad hoc access & payment

- Smart charging services
  - Through firmware configuration of the charge controller the equipment is capable of providing various smart charging functionalities such as charge profiles and session scheduling. By setting the available grid connection power

and the minimum & maximum output current per charge point enhanced load balancing algorithms steer the charge events. During NeMo project also user feedback such as prioritisation of charge sessions and desired battery state of charge will be integrated into the load balancing optimization (either via HMIs or via vehicle). Additionally, fleet management optimization will be taken into consideration (parking time, car fluctuation, usage patterns). In a further step the integration of charging infrastructure into an overall local master energy management system e.g. building automation will be shown using UDP, KNX or other relevant market standard protocols.

- Operations
  - Monitoring
  - Customer support (1st to 3rd level)
- Evaluation and reporting
  - The data generated during the test site demonstration will be recorded, aggregated and analysed. Data include e.g. average charge duration, energy delivered and the daily distribution of charges.

Existing technology and infrastructure which may be considered in the tests are:

- Approximately 400 publically accessible charge points of which 220 are fast charging spots (43 kW AC / 50 kW DC)
- Full operational interface to E-Clearing and Hubject roaming platforms
- AC as well as DC fast charging stations
- Online monitoring and maintenance of charging equipment
- Local load management controller

Actors Involved are:

- From NeMo consortium: Renault, Gireve. Hubject, IBM, TOMTOM, VERBUND
- B2B: commercial fleet operators, company fleets

NeMo technologies and expected benefits derived from the use of the selected NeMo modules/components are:

- Interoperability and roaming through the Pan European roaming protocol (Business Scenario 3)
- Value added horizontal services and data provision for users through the NeMo Hyper-Network (as described in Business Scenario 5)
- B2B offerings between EMP, CPO, NSP and OEMs
- Map and navigation services
- Payment & access (ad hoc access, remote start& stop, direct payment via payment service provider)
- Flexible tariffing according to power availability, daytime etc. (see Business Scenario 5)
- Advanced communication between vehicle and CPO: charge plans, 15118
- As a vertically integrated eMobility provider, SMATRICS will test advanced B2B service offerings between both roles, CPO and EMP.

The Austrian site will cover the following business scenarios:

- Business Scenario 1: Smart Charging
- Business Scenario 3: Cross-provider/border booking, authorization and payment management / Cross-country test site
- Business Scenario 5: Horizontal services: User profile, preferences and monitoring, Service Finder brokerage and pricing

5.5 Italian Test Site

The Italian test site is mainly composed of the A32 toll motorway. The A32 highway starts, in connection with the Turin by-pass, from the town of Rivoli and ends in Bardonecchia at the entrance of the Frejus Tunnel. It is almost 73 km long and lies within the European motorway corridor 5 (Kiev-Lisbon).

The A32 highway is subdivided in 18 km of tunnels, 19 km of bridges and 36 km of embankments.
There are two rest areas, one in each direction, located in proximity of the 55th km, near the toll station of Salbertrand, where petrol and food are available.

The Italian test site is hosted by SITAF S.p.A. that manages the A32 Motorway and the Italian-French T4-Frejus Tunnel, together with its French counterpart (SFTRF).

There are 2 toll stations (Avigliana – Salbertrand). The Control Centre for the traffic monitoring of A32 Motorway is located in Susa, while a Control Centre for Frejus Tunnel is situated in Bardonecchia inside SITAF's offices.

Tecnositaf is fully owned by Sitaf and has the responsibility of all maintenance of the motorway.

The part of the motorway involved in NeMo project starts near Km 0 and runs for almost 10 km after the exit of Avigliana Ovest. It includes a highway tunnel of almost 1 km.

The Italian vehicle demonstration will start from CRF premises in Orbassano; the site includes standard and fast charging stations for EVs and OEM backend for EV connection, a demonstrator EV with communication capability towards the CRF OEM backend and RSU. According to vehicle manufacturer security policy, the EV is allowed to communicate in bi-directional way only with the OEM backend in secure way. The Charge Points have Type 2 and Type 3a sockets and a maximum power of 22 kW.

Along the A32 motorway there are the following installations:

- Variable Message Signs
- Weather Stations
- SOS Stations
- TVCC and AID (Automatic Incident Detection) in the critical sections
- WI-FI Network

Type of NeMo components/services to be deployed in the network to carry out the tests are:

- Safe and secure communication of EV and NeMo Hyper-Network
- Management of all the Charge Points
- Architectural approach with ExVE and Neutral server for EV third party services

The site will focus on the architecture consisting of the OEM backend connected through secure interface to the NeMo Hyper-network. This architecture will be verified and demonstrated by an EV travelling between Orbassano and Susa through A32 highway and sending its SoC to the OEM backend. At the present time, it is envisaged that the Italian Test Site will cover the following business scenarios and corresponding use case:

- Business Scenario 2: Itinerary Planning
- Business Scenario 4: NeMo Service Development Environment

Expected benefits derived from the use of the selected NeMo modules/components are:

- Using services available in the Hyper-Network, there will be alerts to the driver and information about CP availability
- Reliable infrastructure for the deployment of services to OEM EVs

5.6 Cross-country Test Activities

At this moment travelling over Europe implies using different roaming platforms, different payment systems, different authentication systems, etc. Therefore, cross-border testing activities will be carried out in NeMo, to analyse the impacts for the final electric vehicle user.

A first test drive will be organized during the first project year to evaluate the initial situation and calculate the related KPIs, to serve as baseline. Then, a final test drive will be done at the end of the project to demonstrate the NeMo benefits.

As five test sites are participating in NeMo, the final test drive will go through these five countries at least (France, Germany, Austria, Italy and Spain). Only a part of these countries will be crossed during the first test drive as the original purpose is to evaluate the current situation and extrapolate the results using Hubject, Gireve, TomTom and Renault data.

The objective of the cross country test is to demonstrate the NeMo benefits in different countries and the seamless experience and easy service for electric vehicle users. It will be reached by comparing the first test drive results and the final ones regarding e-roaming possibilities, services like payment or booking, checking the availability and the quality of the services in place for each charging station.

The organization of these test drives will be based on a close collaboration between several participating partners:

- Renault
- Hubject eRoaming Platform
- Gireve eRoaming Platform
- TomTom Navigation Services
- IBM
- Ajuntament de Barcelona
- Mosaic
- Verbund
- IDIADA

At the present time, it is envisaged that the cross-country test site will cover only the Business Scenario 3 related to Cross-provider border booking authorization and payment management. However, the success of the related activities strongly depends on the implementation of the other business scenarios:

- Business Scenario 2: Itinerary Planning
- Business Scenario 4: NeMo Service Development Environment
- Business Scenario 5: Horizontal services: Actors monitoring and profiling, Service finder and optimizer, Service brokerage and Rating/Pricing services

6.Requirements Analysis

The following chapters list and describe the high-level requirements in tables. The requirements have been collected from all partners, based on the actors' needs and the use cases selected for NeMo. All the tables share the same structure which is described in the following illustration.

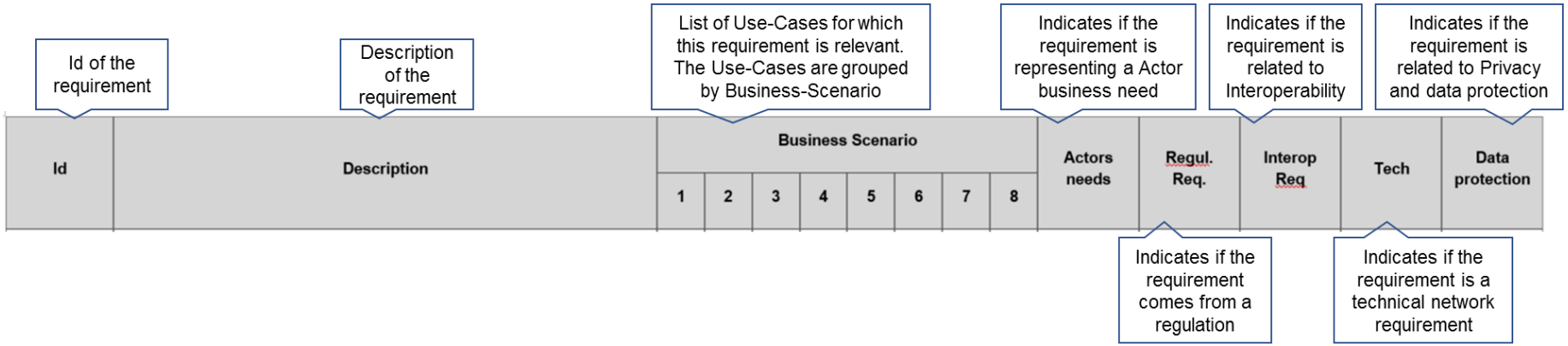


Figure 13: Template for the high-level requirements collection

6.1 Actors' requirements for the Hyper-Network

The actor's needs and main functionalities are formulated as high level requirements. The following table lists and describes these "Actors Requirements".

Table 1: Actors' requirements

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
		1	2	3	4	5	6	7					
Roaming-G_Auth_001	Local Authorisation: The CPO shall be able to request the NeMo Hyper-Network in order to get the Authorisation (or the non-authorisation) for delivering the service to the end-User who identified himself on a Charging Station								eRoaming Pltf				x
Roaming-G_Auth_002	Remote Authorisation: The EMP shall be able to request the NeMo Hyper-Network in order to activate (start) a charge session on a given EVSE, and for a given end user, and get an acknowledgment (or not) from the CPO that manages this EVSE								eRoaming Pltf				x
Roaming-G_Data_001	The CPO shall be able to "upload" to the NeMo Hyper-Network, the status changes of any EVSE it manages. This status shall describe if the EVSE is "Out of order", "In service", "Free", "occupied"," Booked" … This status upload feature has to be designed in order to ease "real-time individual upload"								eRoaming Pltf				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
Roaming-G_Data_002	The CPO shall be able to "upload" to the Nemo Hyper-Network, the EVSEs descriptions changes for any EVSE it manages. The EVSE description is a set of attributes listed and described in the NeMo Common Information Model. This description upload feature has to be designed in order to ease "real-time individual upload"								eRoaming Pltf				
Roaming-G_Data_003	The EVSE-Description shall be managed by the eRoaming Platforms and shall be stored, at least, on the eRoaming Platform to which the CPO is functionally and "contractually" connected								eRoaming Pltf				
Roaming-G_Data_004	The EMP and NSP (Navigation Services Provider) shall be able to "download" from NeMo Hyper-Network, the EVSEs descriptions of any EVSE managed by a CPO with which it is in contract. The EVSE description is a set of attributes listed and described in the NeMo Common Information Model								eRoaming Pltf				
Roaming-G_CDR_001	The CPO shall be able to "upload" to the NeMo Hyper-Network the Charge-Detail-Record (CDR) of each session, operated on an EVSE it manages (the CPO has to be the CPO of the EVSE). This CDR upload feature has to be designed in order to ease "real-time individual upload".								eRoaming Pltf				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
Roaming-G_CDR_002	The EMP shall be able to "download" from NeMo Hyper-Network the Charge-Detail-Record (CDR) of each session operated for a user which is a customer of this EMP. This CDR download feature has to be designed in order to ease "real-time push" to the EMP, and must be compliant with asynchronous pull mechanism								eRoaming Pltf				
Roaming-G_CDR_003	The Charge-detail Record shall be managed by the eRoaming Platforms and shall be stored, at least, on the eRoaming Platform to which the EMP is functionally and "contractually" connected								eRoaming Pltf				
Roaming-G_CDR_004	The Charge-detail Record shall contain the a set of attributes listed and described in the NeMo Common Information Model								eRoaming Pltf				
eRoaming-H_001	All Participants must agree on a standard to exchange charge point POI information: This standard should be the "NeMo Interface definition for POI static data"								eRoaming Pltf				
eRoaming-H_002	All participants must agree on a standard to exchange dynamic charge point information (e.g. availability and occupation status): This standard should be the "NeMo Interface definition for POI dynamic data"								eRoaming Pltf				
eRoaming-H_003	Dynamic data exchange must work in real-time: CPO shall upload their dynamic data changes in real time and EMP should be able to get this information immediately, using								eRoaming Pltf				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	the "NeMo Interface definition for POI data"												
eRoaming-H_004	Routing (navigation services) providers must be able to change routing based on real time information regarding availability/occupation status								eRoaming Pltf				
eRoaming-H_005	All charge stations must offer remote accessibility								eRoaming Pltf				x
eRoaming-H_006	All CPOs must share, with their corresponding EMPs via a platform, pricing information based on time or kwh meters contained in the CDR (at least duration and energy)								eRoaming Pltf				
eRoaming-H_007	No entity within the process chain should rely on exchange of whitelists of RFID Token (UIDs)								eRoaming Pltf				
eRoaming-H_008	All Roaming platforms should implement a push service to update dependent third parties about poi changes								eRoaming Pltf				
eRoaming-H_009	All Roaming platforms should implement a push service to update dependent third parties about dynamic POI changes								eRoaming Pltf				
eRoaming-H_010	All participants must agree on a standard to exchange charge detail records (CDR) This standard should be the "NeMo Interface definition for CDR"								eRoaming Pltf				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
eRoaming-H_011	Charge detail records must be traceable end to end								eRoaming Pltf				
eRoaming-H_012	CDR clearing must be handled between CPO and EMP								eRoaming Pltf				
eRoaming-H_013	All CPOs must offer remote access to their charging stations								eRoaming Pltf				
eRoaming-H_014	All charge stations must offer a unique identifier at the station on plug level EVSE level (charging Point)								eRoaming Pltf				
eRoaming-H_015	Charge stations should offer the possibility to read EVSE IDs by QR codes.								eRoaming Pltf				
eRoaming-H_020	QR code identifiers must follow a standard This standard should be the "NeMo Interface definition for QR codes"								eRoaming Pltf				
eRoaming-H_016	Charge stations should offer the possibility to read EVSE IDs by NFC								eRoaming Pltf				
eRoaming-H_017	All charge stations must be accessible by RFID								eRoaming Pltf				
eRoaming-H_0019	All charge stations must accept the following RFID badges types: Mifare Classic, Ultralight, Desfire								eRoaming Pltf				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
VM_EV_perf_01	In-vehicle data access: The Vehicle Manufacturer needs to access in-vehicle data from EV Owner in order to increase EV performance								Vehicle Manufacturer (VM)				x
VM_EV_charg_01	Allow EV charge seamless: The VM needs to ensure easy and seamless e-mobility through easy access to charge point								Vehicle Manufacturer (VM)				
VM_EV_serv_01	Get EV services availability: The VM needs to receive information on available EV services by the EMP, IT Suppliers, Charging Infrastructure and Complementary Services.								Vehicle Manufacturer (VM)				
VM_EV_serv_02	Provide EV off board applications: To allow the EV services utilization and data exchanging, the VM has to design and develop off board applications in the VM back end.								Vehicle Manufacturer (VM)				
VM_EV_serv_03	Provide HMI application: The End Users need to use HMI applications to access to EV services that are designed and developed by VMs.								Vehicle Manufacturer (VM) /End Users				
CPM_CPO_serv_01	The CPM must get the CPOs' needs in order to improve services and increase sale of charge points								Charge point Manufacturer (CPM)				
CPM_CPO_serv_02	The CPM needs to access charge points data from CPOs in order to increase charge point performance.								Charge point Manufactu				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
									rer (CPM)				
CPM_CPO_API_01	The CPM must communicate with CPO to provide API and technical support.								Charge point Manufacturer (CPM)				
EMP_Loc_auth_01	Local authorization: The EMP needs to manage local Authorisation Requests by interacting with the Driver, the CPO back-end and the eRoaming Platform Provider.								Electro Mobility service Provider (EMP)				x
EMP_Rem_auth_01	Remote authorization: The EMP needs to manage local Authorisation Requests by interacting with the Driver, the CPO back-end and the eRoaming Platform Provider.								Electro Mobility service Provider (EMP)				
EMP_CDR_01	Charge-Detail-Records: The EMP needs to get Charge-Detail-Records by the eRoaming Platform Provider and eventually to send detail to the End Users.								Electro Mobility service Provider (EMP)				
EMP_CP_stat_01	Get Charge Point static data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infra element description change notification (creation modification deletion), it will push the new description data to the EMP, if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs with								Electro Mobility service Provider (EMP)				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	description to the Driver.												
EMP_CP_dyn_01	Get Charge Point dynamic data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infra element status change notification, the eRoaming Platform will push the new status to the EMP if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs and availability status to the Driver.								Electro Mobility service Provider (EMP)				
EMP_CP_dyn_02	Get In-progress Charge Session data: During the charge session, the EMP may receive event notification related to an in-progress charge-session by the CPO (via eRoaming Platform Provider) such as "the charge started"; "the charge stopped", "the delivered power has been modified from 22kW to 3kW ...).								Electro Mobility service Provider (EMP)				
EMP_mob_need_01	Get End Users' mobility need: The EMP should get information related to the End-User mobility need (where to go? When? From where? …) in order to calculate the "Charge need" and optimise the charging process								Electro Mobility service Provider (EMP)				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
EMP_EV_status_01	Get End Users' vehicle status: The EMP should get information related to the End-User vehicle in order to calculate the "Charge need" and optimise the charging process. EMP will request eRoaming Platform (which will request the Vehicle Manufacturer (VM)). The state of charge, the autonomy and the "reachability" of its destination should be displayed to the Driver.								Electro Mobility service Provider (EMP)				
EMP_charg_need_01	Send Charge-needs: The EMP may send to the eRoaming Platform (final destination is CPO), the charge-need of an EV, in order (for the CPO) to optimise the charging process (taking into account the availability of energy - constraints and opportunities).								Electro Mobility service Provider (EMP)				
EMP_B2B_01	Get about B2B tariff and prices changes: The eRoaming Platform Provider sends notifications about tariffs changes to the EMP (due to energy cost changes ...) that come from CPO.								Electro Mobility service Provider (EMP)				
EMP_CP_book_01	Get and send CP reservation: The EMP may receive charge points reservations from End Users and send it to CPO.								Electro Mobility service Provider (EMP)				

6.2 Functional Requirements

The Business scenarios and the Use-cases require some high level functionality of the NeMo network and/or of some NeMo internal sub-systems. These functionalities are formulated as high level functional requirements. The following table lists and describes these “Functional Requirements”.

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
		1	2	3	4	5	6	7					
UC-01-001	#ID EMP_mob_need_01: Get End Users' mobility need: The EMP should get information related to the End-User mobility need (where to go? When? From where? …) in order to calculate the "Charge need" and optimise the charging process.	1							x				
UC-01-002	The mobility needs must describe at least an amount of km to run or, more precisely, a trajectory to run.	1							x				
UC-01-003	EV owner must have given his/her consent in order to transfer personal information about his/her EV from the OEM backend.	1								x			x
UC-01-004	An eRoaming Platform must provide this service through the NeMo Hyper-Network.	1							x				
UC-01-005	VM backend must be able to provide "real time" data of an EV.	1							x				
UC-01-006	End to end time constraint: ->If no request to the vehicle (data previously uploaded by the vehicle to the OEM back end) : The user must have an	1							x				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	answer within the normal app usability response time ->If request the vehicle (OEM back end requests vehicle to get real time data) : The user can have an answer longer than the normal app usability response time.												
UC-02-001	#EMP_charg_need_01: Send Charge-needs: The EMP may send to the eRoaming Platform (final destination is CPO), the charge-need of an EV, in order (for the CPO) to optimise the charging process (taking into account the availability of energy - constraints and opportunities).	2							x				
UC-02-002	Send Charge-needs: The EMP may send to the CPO, the charge-need of an EV, in order (for the CPO) to optimise the charging process (taking into account the availability of energy, any constraints and opportunities).	2							x				
UC-02-003	An eRoaming Platform or a CPO must provide this service through the NeMo Hyper-Network.	2							x				
UC-02-004	The eRoaming Platform must be able to detect EVSEs in a defined perimeter.	2											
UC-02-005	DSO/ Aggregator/Energy retailer must be able to send constraints/opportunities to CPOs.	2							x				
UC-02-006	If needed by actors, traffic congestion must be taken into account by CPOs to make offers.	2							x				
UC-02-007	End to end time constraint: The user must have an answer	2											

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	within the normal app usability response time.												
UC-03-001	#EMP_CP_stat_01: Get Charge Point static data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infra element description change notification (creation modification deletion), it will push the new description data to the EMP, if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs with description to the Driver.	3							x				
UC-03-002	#EMP_CP_dyn_01: Get Charge Point dynamic data: The EMP needs to receive dynamic information about Charge Points by the CPO (via eRoaming Platform Provider). Any time the eRoaming Platform Provider receives an EV charging infra element status change notification, the eRoaming Platform will push the new status to the EMP if the element is managed by a CPO with which the EMP is in contract. The EMP displays a map of CPs and availability status to the Driver.	3							x				
UC-03-003	All actors must be able to request the search and find service.	3							x				
UC-03-004	An eRoaming Platform and/or a CPO must provide the search and find service through the NeMo Hyper-Network.	3							x				
UC-03-005	End to end time constraint :The user must have an answer within the normal app usability response time	3											

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-04-001	Battery consumption parameters should be available to actors who need to determine energy consumption rates while driving.		4						x				
UC-04-002	A navigation system should be able to detect, through an API with the vehicle, the battery charge status.		4						x				
UC-04-003	A navigation service provider should be able to determine a reachability area based on a vehicle's current situation (e.g.: current battery charge and status, driving environment, temperature, etc.).		4						x				
UC-04-004	A navigation service should be able to plan an itinerary between two points including charge stops in the route.		4						x				
UC-04-006	An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes such as charger types, in a given area.		4						x				
UC-04-007	An eRoaming platform provider should be able to estimate the time it would take to recharge a vehicle's battery up to a certain percentage of total capacity.		4						x				
UC-04-009	A navigation service provider should be able to determine an estimated total trip time for an itinerary, when possible (e.g.: it may be impossible to determine the waiting time for a charge point)		4						x				
UC-05-001	CPO must agree to accept and take into account tariff information from the DSO/Aggregator/Energy	5							x				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	retailer/TSO. A contract could be signed by the two parts to agree on the process: set a tariff planning or switch in real time the tariff level.												
UC-05-002	Grid local perimeter/network must be defined in order to switch tariff level/set tariff planning on localised perimeters which are relevant for DSO/Aggregator/Energy retailer/TSO.	5							x				
UC-05-003	End to end time constraint: The requestor must have an answer within the normal app usability response time.	5, 6							x				
UC-06-001	NSP and/or MSP and/or Public authorities must have traffic information and be able to provide it through services.	6								x			
UC-07-001	The EV must be connected to the VM (OEM) backend VMs must provide standard API to upload Charge Plans (e.g. ISO 20078 Extended Vehicle Standard)	7											
UC-07-002	Secured service transactions between VM with EMP for billing and payment.	7											x
UC-07-003	End to end time constraint: The requestor can have an answer longer than the normal app usability response time.	7, 8											
UC-08-001	DSO/Aggregator/Energy retailer/TSO must agree with CPO to use energy constraints service.	8							x				
UC-08-002	CPO and/or eRoaming Platform must provide energy constraints service through NeMo Hyper-Network.	8							x				
UC-08-003	Charge infrastructure of the CPO must be able to be piloted by CPO to answer constraints of DSO/Aggregator/Energy retailer/TSO; or CPO must be able to push a charge plan to the vehicle.	8											

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-09-001	The CPO shall be able to send CDR to the EMP	9											
UC-09-002	The EMP shall be able to send the CDR to the VM	9											
UC-10-001	The VM shall be able to send the charging strategy to the CPO	10											
UC-10-002	The CPO shall be able to integrate the battery constraints in his model of smart charging	10											
UC-10-003	The driver shall be able to choose between two modes: one which prioritizes the mobility needs (charging speed and full range), the other the battery durability	10							x				
UC-11-001	Maps should allow the detection of security critical features, such as tunnels and highways with no emergency lanes.		11										
UC-11-002	The navigation system's HMI should display safety critical situations, such as running out of charge near a tunnel.		11										
UC-11-003	The itinerary should be available for a driver to depart on short notice (i.e. planning an itinerary shouldn't take more than 30 seconds).		11										
UC-12-001	An EMP/eRoaming platform provider should be able to provide a list of charge points, including its relevant attributes such as charger types, in a given area as POIs (point of interest) in a map.		12						x		x		
UC-12-002	An EMP/eRoaming platform provider should be able to		12						x		x		

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	show the real-time availability of a charge point (optionally, the availability of its associated parking spot may be omitted when considering a charge point available).												
UC-12-003	An EMP/eRoaming platform provider should be able to show the availability of a charge point in a future date for a certain period of time (optionally, the availability of its associated parking spot may be omitted when considering a charge point available).		12						x		x		
UC-12-004	An eRoaming platform provider should be able to estimate the waiting time to use a charge point.		12						x		x		
UC-12-005	A preference model should be available to influence the selection of charge points along the route (e.g.: prefer cheaper CPs, prefer closer CPs, prefer faster CPs, etc.)		12						x		x		
UC-12-006	The HMI should be able to display a list of charge points as POIs (point of interest), with the list of their attributes and their availability.		12						x		x		
UC-12-007	Charge point information should reflect the current status of the network to provide meaningful information to a driver.		12						x		x		
UC-13-001	The navigation system should automatically detect when the battery charge is not enough to complete a trip.		13										
UC-13-002	The navigation system should automatically detect when dynamic conditions reduce the reachability area (e.g.:		13										

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	wind conditions require greater energy usage than anticipated).												
UC-13-003	Charge saving measures should be proposed to the driver when the reachability area reduces far enough that reaching the destination is not guaranteed (e.g.: in cold weather, suggest turning heating off to reduce battery consumption)		13										
UC-14-001	The navigation system should provide alternative itineraries according to different (and customizable) criteria		14						x				
UC-14-002	The driver should be able to select which optimization criteria to use when creating an itinerary (e.g.: optimize for cheaper routes, faster routes, etc.). These estimations should account for charging times, prices, etc.		14						x				
UC-14-003	An EMP or eRoaming platform provider should be able to estimate the time it would take to recharge a vehicle's battery up to a certain percentage of total capacity.		14										
UC-14-004	An EMP or eRoaming platform provider should be able to estimate the waiting time to use a charge point.		14										
UC-14-005	An EMP or eRoaming platform provider should be able to estimate the price a driver will pay for charging in a certain location during a certain timeslot. (NB: cross-border payment estimations may be left out of scope).		14						x				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-15-001	It should be possible for a CPO to reserve a timeslot in a charge point.	15	15	15					x		x		x
UC-15-002	An EMP or eRoaming platform provider should be able to request a charge point to be reserved through a service available in the Hyper-Network	15	15	15					x		x		x
UC-15-003	An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes such as charger types, in a given area as POIs (point of interest) in a map.	15	15	15							x		x
UC-15-004	An EMP/eRoaming platform provider should be able to show the availability of a charge point in a future date for a certain period of time (optionally, the availability of its associated parking spot may be omitted when considering a charge point available).	15	15	15							x		x
UC-15-005	An EMP/eRoaming platform provider should be able to show the real-time availability of a charge point (optionally, the availability of its associated parking spot may be omitted when considering a charge point available).	15	15	15							x		
UC-15-006	It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver, provided that the actor has some kind of authorization or xauthentication token which proves that the driver has granted permission to this actor to do so.	15	15	15									
UC-15-007	The reservation process should work on real time to	15	15	15							x		

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	ensure a booking can be timely confirmed.												
UC-15-008	It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver, provided that the actor has some kind of authorization or authentication token which proves that the driver has granted permission to this actor to do so. This should also include payments.			16							x		
UC-16-001	A navigation system should be able to detect, through an API with the vehicle, the battery charge status.			16									
UC-16-002	A navigation system should be able to detect, through an API with the vehicle, when the driver is preparing to drive-off after being parked.			16									
UC-16-003	It should be possible to get a report of charge delivered from the CPO to other actors, possibly through an eRoaming platform provider, and the price to be paid by the driver.			16							x		
UC-16-004	The vehicle's position should be updated in real time. The authentication and payment process shouldn't take more than a few seconds, so that a driver never has to wait on the payment process.			16									
UC-17-001	It should be possible to unlock charge points using a common API available through the Hyper-Network. A driver should be able to immediately start charging his vehicle after it has been unlocked using this API.			17							x		x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-17-002	It should be possible for any actor to interact with an EMP or eRoaming platform provider on behalf of a driver; provided that the actor has some kind of authorization or authentication token which proves that the driver has granted permission to this actor to do so.			17							x		
UC-17-003	A navigation system should be able to detect, through an API with the vehicle, when the driver has parked his/her vehicle.			17									
UC-17-004	A navigation system should be able to detect, through an API with the vehicle, the battery charge status.			17									
UC-17-005	An EMP/eRoaming platform provider should be able to provide a list of charge points, including their relevant attributes such as charger types, in a given area as POIs (point of interest) in a map.			17							x		
UC-17-006	The navigation system should be able to detect the position of the vehicle with sufficient accuracy to determine when it is parked near a charge station.			17					x				
UC-17-007	The vehicle's position should be updated in real time. The authentication and unlock process shouldn't take more than a few seconds, so that a driver never has to wait on the unlock process.			17					x				
UC-18-001	The services as well as the request are semantically annotated with OWL-S and the domain model of the				18								x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	NeMo Hyper-Network.												
UC-19-001	Data integrity must be guaranteed				19								x
UC-19-002	Secured service transactions between Service Provider and Service Requester for billing and payment.				19								x
UC-20-001	The actor needs to have the access right (role) to perform the search and select contracts function				20				x				x
UC-22-001	Role based access that only authorized persons are allowed to start or stop an IT service				22								x
UC-22-002	Current and pending transactions must not be adversely affected as a service is offline (stop to start) for a very short time				22								x
UC-24-001	All partner profiles of the NeMo Hyper-Network are available within the Partner Management.				24								x
UC-24-002	There is a filter mechanism by the Partner Management to search according to specific parameters.				24								
UC-25-001	The Business Partner is not yet registered.				25								
UC-25-002	All profile information of the business partner has to be filled in during the registration.				25								
UC-30-001	Access: Role based authorization required				30 34								x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
					35 36 38 43 45								
UC-30-002	Data security: Data integrity must be guaranteed; Data confidentiality must be ensured				30 34 35 36 38 43 45								x
UC-30-003	Data privacy: Sensitive data is only stored if needed and authorized; Data export policy required				30 34 35 36 38 43 45								x
UC-30-004	Performance: Sub-seconds response time for service transactions				30 34 35 36 38 43 45								
UC-30-005	Availability: 24x7				30 34 35 36 38 43 45								

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-30-006	Scalability: Able to grow with market demand and increase of connected IT systems				30 34 35 36 38 43 45								
UC-33-001	The Business Partner has the access right to perform the View Service Contract Details action.				33								x
UC-47-001	The service which will be tested needs to be available.				47 48								
UC-47-002	The request triggers the specified response.				47 48								
UC-47-003	The service developer owns the service which is tested.				47 48								
UC-49-001	The Actor is informed about the information that will be tracked according to the privacy and security conditions.					49							
UC-49-002	The actor accepts to be tracked. The actors can withdraw in any moment their consent					49							
UC-49-003	The actor can deactivate the tracker in specific situations					49							
UC-49-004	The monitor registers the user positions					49							
UC-49-005	The monitor deduces common actor/user locations according the registered positions					49							

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-49-006	The monitor registers the user's activity and its interaction with the services or other actors in the network					49							
UC-49-007	The monitor deduces the electromobility services with which the actor uses/consumes/interacts frequently					49							x
UC-49-008	The monitor updates the actor profile with the information deduced from the actor's activity					49							x
UC-49-009	The monitor registers the actor activity in real-time. The analysis for the monitored information could be in real-time (if needed to provide instant feedback or user support) or in an elapsed time, if instant feedback is not needed.					49							x
UC-50-001	The actor has a profile					50 51							x
UC-50-002	The actor can access his/her profile					50 51							x
UC-50-003	The actor can update his/her profile					50							x
UC-50-004	The actor has been informed about the personal information the profile stores and has agreed that they can be managed according to the privacy procedures					50							x
UC-50-005	The actor can withdraw his/her consent that the profile manages his/her personal information. In that case, the personal information related to the actor has to be					50							x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	removed												
UC-50-006	Updates on the actors profile must be visible for other processes immediately					50							x
UC-51-001	Actors profile information must be quickly accessible					51							
UC-51-002	Actors profile must be up-to-date before being loaded					51							
UC-52-001	The services have to be registered in the service registry including a semantic description of the service					52							
UC-52-002	The services have to be described based upon the common information model of the NeMo Hyper-Network					52							
UC-52-003	The request is specified as an (incomplete) semantic service description					52							
UC-52-004	The service description may include an (optional) price for fulfilling a request or contracting a service. The specified price may be constant (the same for all requests and types of service) or a service-specific callback which will dynamically calculate a price for a specific service profile.					52							
UC-53-001	The Service Optimizer has access to a service finder component in order to get all services that fit a request from a functional point of view					53							
UC-53-002	The Service Optimizer has access to Pricing and Profiling					53							

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	Services												
UC-53-003	The service descriptions do contain non-functional parameters					53							
UC-53-004	The request does contain non-functional parameters as search criteria					53							
UC-54-001	The broker identifies concurrent request to the same service					54							
UC-54-002	The broker accesses the actor profile to identify preferences or constraints for the services requested in order to produce suitable alternatives					54							
UC-54-003	The broker proposes alternatives for concurrent requests to same services					54							
UC-54-004	The user can express his/her opinion or vote for the alternatives proposed					54							
UC-54-005	The broker uses the actor feedbacks to improve further brokerages					54							
UC-54-006	Brokerage is active for concurrent requests					54							
UC-54-007	Brokerage response must be fast enough to not overload the acceptable waiting times					54							

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-55-001	The services need to be available to the service finder.					55 56							
UC-55-002	The services may specify no-price (e.g.: if the service is offered for free).					55							
UC-55-003	The services may specify a validity period for a quote, after which said quote becomes invalid.					55							
UC-55-004	The services may specify a quote as estimative, if the final price can't be known in advance.					55							
UC-55-005	The services published and available to the service finder shall include an endpoint that may be invoked with a service profile to obtain a quote.					55							
UC-56-002	The services may specify in their service description an endpoint through which callers may subscribe to a notification channel.					56							
UC-56-003	The notification channel should provide an un-subscription mechanism.					56							
UC-56-004	To ensure relevance on the offers and promotions, a real-time channel must be available to process the notifications.					56							
UC-57-001	The actor should get data from the DSO and the CPO (including CDR, energy consumption, EV impact in power	57					57		x				

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	limit on local grid)												
UC-57-002	The NeMo Hyper-Network shall host the calculation which permits to complete the report	57					57		x				x
UC-57-003	The actor shall select the concerned CPOs and DSOs by the report localisation	57					57						
UC-58-001	CPOs shall push status change of charge points with harmonized delay	58					58						
UC-58-002	The NeMo Hyper-Network shall host the calculation which will output a graphic of the occupation rate	58					58						
UC-58-003	This graphic shall be displayed on a website, showing the real time availability of each charge point	58					58						
UC-59-001	VM have to send data about EV usage (charging and driving details) to the eRoaming Platform	59				59	59		x				
UC-59-002	EMP to send the CDR for public charges	59				59	59		x				x
UC-59-003	Private data shall be aggregated (VIN) to respect privacy laws. (list of VINs per type of user shall be sent by VM)	59				59	59						x
UC-60-001	Driver needs to be able to communicate his/her mobility demands to the network (either directly or indirect through app-services)							60					
UC-60-002	An abstract mobility pattern (“mobility memory”) needs to be available							60					

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
UC-60-003	Data privacy on the mobility demand needs to be ensured							60					
UC-60-004	Mobility demand information must be available before drive off							60 61					
UC-60-005	Mobility demand should be available before charging process							60					
UC-60-006	Mobility memory should cover one year of history							60 61					
UC-61-001	Access of mobility behaviour needs to be available							61					
UC-61-002	Deriving mobility plan needs to be possible without detailed personal information (person abstraction and limited look ahead horizon)							61					
UC-61-002	Mobility demand should be available before charging process							61					
UC-62-001	Access to mobility aggregated mobility demand							62					
UC-62-002	Constraints for optimization strategy for charging need to be available (e.g. SoH)							62					
UC-62-003	Mobility demand profile for EV should be available immediately before charging process start							62	x				
UC-62-004	Changing mobility demand needs to be available upon							62					

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	notice and before drive off time												
UC-63-001	Driver needs to be able to communicate his/her mobility demands to the network (either directly or indirect through app-services)							63	x				
UC-63-002	An abstract mobility pattern (“mobility memory”) needs to be available							63					
UC-63-003	Data privacy on the mobility demand needs to be ensured							63					x
UC-64-001	The driver shall be able to set his/her time of departure							64					x
UC-64-001	The car status (plugged/parked) and external temperature shall be sent to the EMP							64					x
UC-64-001	A pre heating signal shall be sent to the car from the EMP with a duration and a temperature							64					x
UC-64-001	The EMP shall be able to calculate the pre heating duration based on external temperature and EV status							64					x

6.3 System behaviour and elementary functions

The following table lists and describes the “System requirements”

Table 2: System requirements

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
		1	2	3	4	5	6	7					
TR-01	The NeMo network must be built and based on a decentralized architecture.				2 1 2 5 2 8							x	
TR-02	There should be NO central system or any single-point in the network to support scalability and availability.				2 6							x	
TR-03	The record of transactions in the NeMo network must be shared with all participants connected. Those records are stored in form of a "distributed ledger" by each participant.											x	
TR-04	Each participant provides a transaction record about own partner metadata, including "Role", "URL", "Services", etc. shared with all participants in the network.											x	
TR-05	Access to transactions records must be permitted. Business participants only see appropriate transactions of the shared system of record.											x	x
TR-06	Transactions in the network need to be authenticated. Cryptography is central to these transactions.											x	x
TR-07	Participants in the network are known and trusted.				2 5 3 1							x	x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
TR-08	Transactions could include "business rules" (smart contracts) and are executed as part of the transactions.											x	
TR-09	Smart contracts are verifiable, signed and encoded in ANY programming language. For each programming language a run-time container is used.				2 3 2 8							x	x
TR-10	Participants in the network can close contracts in a P2P manner.											x	
TR-11	"NeMo Alliance" grants access to participants to access the NeMo network.				2 5 2 6				x			x	
TR-12	Participants are legal entities such as EMP, CPO, etc. who wants to access the NeMo network and to exchange transaction records with the participants.								x			x	
TR-13	The shared partner metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Business Partner Management", and accessible to all partners in the network.				2 5 2 7 2 9 3 1							x	
TR-14	Each EMP as participant in the network provides the unique ID (eMAID) as part of the partner metadata.				4 4							x	
TR-15	Each CPO as participant in the network provides the unique ID (<Country Code><Spot Operator ID>) as part of the partner metadata				4 4							x	
TR-16	Each eRoaming-Platform provider as participants in				3							x	

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	the network provides the IDs of the EMPs and CPOs as part of the partner metadata				1								
TR-17	Participants in the NeMo network share transaction records about the offered and available "services".											x	
TR-18	Each partner in the NeMo network shares transaction records about "services" metadata and need to specify the details about the services offered								x			x	
TR-19	The shared services metadata is contributing to the "Open Cloud Marketplace" functional building block: "Business Services", "Services Store", and accessible to all partners in the network. Service metadata includes semantic and QoS information about the services.				2 3							x	
TR-20	Availability of a service is specified by the "Open Cloud Marketplace" and is accessible to all partners in the network.				3 1 3 2 3 7 3 9 4 0 4 1 4 2 4 4 4 6							x	

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
TR-21	The semantic description of a service is held separately to the service itself. Service provider and service description provider do not necessarily have to be the same instance.				2 1 2 8 3 1 3 2 3 7 3 9 4 0 4 1 4 2 4 4 4 6							x	
TR-22	Quality of service (QoS) parameters about a service held separately to the service itself. Service provider and the service evaluator do not have to be the same instance.				3 1 3 2 3 7 3 9 4 0 4 1							x	

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
					4 2 4 4 4 6								
TR-23	One service description might reference multiple equivalent services. The equivalent services are listed in the grounding (technical location of the service) of the service description.				2 1 2 3 2 8 2 9							x	
TR-24	Access restrictions are provided by the service description. This could be formulated in a precondition of the service.											x	
TR-25	Models (e.g. Ontologies) used to describe services and service parameters are made available by the "Open Cloud Marketplace" to all partners.				2 1 2 8							x	
TR-26	The system should provide the means for identifying the purpose of each data request and processing											x	x
TR-27	The system should provide mechanisms for specifying the compatibility between collection and processing purposes											x	x
TR-28	The system should provide the means for defining prevention rules regarding incompatible purposes.											x	x
TR-29	It should be ensured that the data collected and processed are necessary, adequate and											x	x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
	proportional, in particular with respect to processing activities, roles of entities involved and purposes.												
TR-30	The system should enable formal data categorization and the definition of different levels of data accuracy and granularity.											x	x
TR-31	The system should provide technical features that allow interoperation, as well as negotiation between domains, i.e. different charging networks, for the effective protection of privacy as well as business information confidentiality											x	x
TR-32	The system should ensure accountability and traceability through logging and auditing of transactions among partners, as well as all data access and exchange											x	x
TR-33	The system should support the automatic transformation of data, where needed, so as to comply with the necessity, adequacy and proportionality principles											x	
TR-34	The system should provide the means for automatic update of data, when the latter are outdated or incorrect.											x	x
TR-35	The system should provide the means for automatic deletion of data, when the latter are no longer needed, in compliance with standards and data protection regulations in effect.				2 3 2 9							x	x
TR-36	The system should provide the means for historical data being archived according to specified policies.											x	x
TR-37	The system should allow the configuration of data retention duration.											x	x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
TR-38	The system should provide appropriate mechanisms for anonymization, pseudonymization, selective disclosure, etc.											x	x
TR-39	The system should support the definition of fine-grained access and usage control policies, taking into account all applicable parameters, such as the role of the entity being the data recipient, the semantics of the data themselves, the underlying purpose of foreseen processing and the data confidentiality level.											x	x
TR-40	Privacy preferences and consent of data subjects should be reflected on the contracts “closed” among participants, and be considered in all data processed and exchanged											x	x
TR-41	Authentication mechanisms are required at any node of the system to allow the access just to the authorized users or modules.								x			x	x
TR-42	Communication should be encrypted to avoid eavesdropping and data tampering.											x	x
TR-43	Communication between nodes should keep the integrity of information.											x	x
TR-44	The overall operation of the system should be based on the semantics of all entities involved											x	
TR-45	The overall operation of the system should be based on a common reference information model for the semantic definition of all entities involved											x	
TR-46	The system must provide for the transparent transformation from custom/proprietary platform-specific data models into the common information model, and vice versa.											x	

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
TR-47	A participant can act, under specific conditions, on behalf of other participants (i.e. FCA backend on behalf of all the FCA EV drivers/owner.											x	
TR-48	The OEM backend, acting as representative of EV drivers, can make anonymous their id and all the data that could be used to find owner.											x	
TR-49	Each service description should contain also the list and/or relationship of all the services used.											x	
TR-51	Each service description should contain also the instructions for the data owner to see his/her data and, if needed, to delete them.											x	x
TR-52	It should be possible to test a service in a safe and isolated environment (sandbox).											x	x
TR-53	It should be possible to see all the info about any service supplier.											x	
TR-54	It would be nice to choose among different service usage profile especially during the initial testing of the service (like Amazon AWS services).											x	
TR-55	Connected IT systems to the NeMo network needs to be located in Europe to ensure "Data Protection Regulation" of personal data. In order to ensure this the "NeMo Alliance" has to grant access only to legal entities which do comply to European Data Protection regulation (TR-11).											x	x

6.4 Data protection and User privacy requirements

The NeMo Hyper-Network shall comply with Data protection and User privacy requirements. The following table lists and describes these "Data protection and user privacy requirements".

Table 3: Data protection and user privacy requirements

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
		1	2	3	4	5	6	7					
R-PRI-001	Sensible data like personal information is managed in compliance with the EU and local legislations												x
R-PRI-002	Operations on data creation, access and change must be authenticated and authorised according to the user permissions												x
R-PRI-003	It is provided a traceability of the operations performed by the users on sensible data												x
R-PRI-004	User profiles are stored in a server following the best-practice for privacy and personal data handling												x
R-PRI-005	Data integrity is guaranteed												x
R-PRI-015	Data shall be classified, at least in 3 parts: Confidential, Sensitive and Normal												
R-PRI-006	Confidential or sensitive data like personal data could not be accessed and/or stored without justification. Information is only distributed within users that need to deal with it												x
R-PRI-007	Personal data is anonymized or aliased when the personal information is not needed												x

Id	Description	Business Scenario							Actors needs	Regul. Req.	Interop Req	Tech	Data protection
R-PRI-008	An agreement for storing, transferring or accessing confidential data is signed by the person who "owns" the data												x
R-PRI-009	A data transfers policy is established. All data transfers must be compliant with that policy.												x
R-PRI-010	Sensible data transfer must be secured. Data communications transfer are encrypted												x
R-PRI-011	the user is aware of the extra functionalities provided in Exchange of identification himself/herself and will agree through a consent form												x
R-PRI-012	the basic functionalities are guaranteed even in case of no identification												x
R-PRI-013	the user can at any moment withdraw from the consent agreement												x
R-PRI-014	The system allows the users to cancel their account at any moment. Any personal information related to cancelled accounts will be removed or anonymised or aliased												x

7. Conclusions

This deliverable outlines the actors involved in the electromobility sector, especially those that will interact with the NeMo Hyper-Network. Their interrelations have been visually presented thanks to the UML diagram built based on an incidence matrix. A definition of the actors has been also provided to create a common language within NeMo project. The deliverable has also presented seven Business Scenarios that will be tested in the project test sites. During the preparation and identification of these Business Scenarios, the objective was to describe different services among which the pilot sites participants could select. Some of these services already exist while others are innovative. The proposed services will be provided thanks to NeMo Hyper-Network with the aim to create a seamless electromobility service. Particular attention has been paid during the services identification phase so that each Business Scenario proposes a different service. The identified Business Scenarios may be updated during the project execution when the preparation for the execution of the pilots becomes operative. Based on the Business Scenarios, detailed Use Cases have been identified. These describe more in detail which are the specific services that will be tested in the sites. The Use Cases include a flowchart of events and a high level description of the requirements which is summarised in the last section of the document, completely dedicated to the requirements analysis.

The deliverable is the basis for the future work in NeMo. Based on the requirements, WP2 will extract specifications from the technical point of view. The Use Cases will be the basis for the definition of the Common Information Model in WP3. Finally, the requirements will support the identification of the KPIs aimed at evaluating the NeMo Hyper-Network services in light of the test sites implementation to be carried out in WP6 which will start in a later stage (M18).

As explained, the deliverable presents a preliminary work on the Business Scenarios to be tested in pilot sites based on which Use Cases and requirements have been identified. As such, the services and the scenarios may be refined during the course of the NeMo project.

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